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Peripheral Nerve Disorders

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SUMMARY PEARLS

- The Seddon and Sunderland classification systems characterize focal nerve lesions based on the extent of damage to axons and surrounding structures. The pattern (length dependent, single nerve, nerve root) and fiber type (motor, sensory, both) guide the differential diagnosis.
- Electrodiagnostic testing can confirm, localize, and characterize both focal and generalized neuropathies and establish severity, chronicity, and often prognosis. Neuromuscular ultrasound complements electrodiagnosis in the assessment of focal and generalized neuropathies by evaluating structure.
- Depending on the nature of the lesion, recovery can occur through a combination of remyelination, axonal regrowth, and collateral sprouting. The recovery process can be serially assessed through electrodiagnostic study.
- Common compressive mononeuropathies include carpal tunnel syndrome, ulnar neuropathy at the elbow and wrist, radial neuropathy at the radial groove, fibular neuropathy at the knee, lateral femoral cutaneous neuropathy, femoral neuropathy, and sciatic neuropathy.
- Brachial plexopathies and regional upper limb nerve disorders can be recognized through characteristic sensory and motor deficits that span more than one nerve or root but derive from a common portion of the plexus. Upper brachial plexopathies present with weakness of shoulder movement and elbow flexion and sensory impairment along the lateral upper limb.
- Lower brachial plexopathies present with hand weakness and medial upper limb sensory impairment.
- Lumbosacral plexopathies comprise lesions of the lumbar or sacral plexuses, with different etiologies and risk factors driving each.
- Generalized peripheral polyneuropathies are characterized by pattern and subtype with most peripheral polyneuropathies being chronic, length dependent, symmetric, and axonal, involving both sensory and motor fascicles. Deviations from this pattern should signal consideration of atypical neuropathies that may be treatable.
- Inflammatory neuropathies, including Guillain-Barré syndrome (acute presentation) and chronic inflammatory demyelinating polyradiculoneuropathy with subacute or rapid and chronic presentation, are treatable disorders that are important to recognize.
- Hereditary neuropathies classically present with slowly progressive symptoms, foot deformities, and a predominance of negative over positive symptoms and signs. Those representing disorders of myelin present with uniform slowing of conduction velocities without conduction blocks.
- Complications of neuropathies include skin breakdown of the feet, pain, and functional impairment, including impaired balance and falls.

Introduction

Disorders affecting the peripheral nervous system may reflect either a focal injury to one or more neural structures or a generalized neuropathy often associated with other systemic disorders. Focal neuropathies are usually the result of localized compression, traction, or trauma to a specific nerve, whereas generalized neuropathies are typically caused by systemic processes that are often toxic, metabolic, or immunologic in nature. With electrophysiologic expertise and familiarity with musculoskeletal syndromes, physiatrists are positioned to accurately diagnose peripheral nerve disorders and provide rational advice regarding prognosis and therapy. An understanding of the anatomic course of nerves and points of vulnerability, pathophysiology of nerve disorders, and mechanisms of reinnervation enables accurate diagnosis, prognostication, and

treatment. This chapter introduces principles of nerve anatomy and function, pathophysiology, recovery, and prognostication; presents common focal peripheral neurologic lesions (focal upper limb, focal lower limb, regional nerve disorders, and lower abdominal neuropathies); describes generalized peripheral polyneuropathies; and addresses complications of peripheral polyneuropathies and paradigms in treatment and rehabilitation.

Anatomy and Principles of Lesion Localization

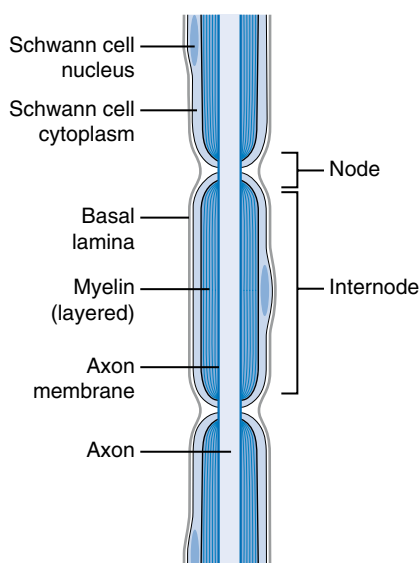
The peripheral nervous system is made up of 12 cranial nerves and 31 spinal nerves that innervate specific sensory distributions (dermatomes) and muscle groups (myotomes). In the cervical and

lumbosacral regions, the spinal nerves intermingle to create plexuses, which then form individual peripheral nerves, many of which branch further distally in the limb. A peripheral nerve branch may carry motor fascicles, sensory fascicles, or both. Thus when assessing nerve lesions, the distribution of motor and sensory abnormalities can aid in localization of the lesion within the peripheral nervous system to one or more nerves, portions of the plexus, or roots. For example, sensation to the fifth digit of the hand is supplied by nerve fascicles of the ulnar nerve that, more proximally, are part of the medial cord and lower trunk of the brachial plexus and the C8 nerve root. Thus a lesion of any of these neural structures can cause fifth-digit numbness. Assessment of sensory and motor function in other body regions sharing some, but not all, of the innervation pathway can help precisely localize the lesion.

The Role of Myelin in Impulse Conduction

The basic neural structure is the neuron and its associated axon, or nerve fiber. Schwann cells, the glial cells of the peripheral nervous system, encircle one or multiple axons, providing a myelin sheath and contributing to speed of action potential conduction along an axon. Unmyelinated nerve fibers are groups of axons bound by a single Schwann cell. These axons, lacking individual myelination, conduct action potentials relatively slowly, at a rate of about 1 to 2 m/sec. An example of unmyelinated axons are C fibers, which transmit nociceptive input from cutaneous and visceral sources. The experience of stubbing one's toe and feeling the pain hit moments later, on a delay, reflects the slow transmission of the C fibers.

In contrast, myelinated axons each are enwrapped by a series of Schwann cells along their lengths, separated by small sections of uncovered axon known as nodes of Ranvier (Fig. 42.1).⁷⁹ The nodes of Ranvier contain high concentrations of voltage-gated sodium channels. Once depolarized, the action potential of a myelinated axon “jumps” from node to node in the process of saltatory conduction.



• **Fig. 42.1** Myelinated axon invested by Schwann cells separated by nodes of Ranvier. (Modified from Jobe MT, Martinez, SF. Peripheral nerve injuries. In: Canale ST, Beaty JH, eds. *Campbell's Operative Orthopaedics*. 12th ed. Mosby; 2013:3062-3123.e6.)

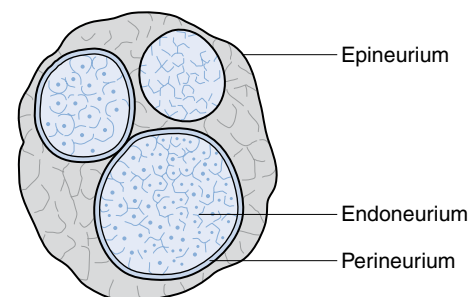
tory conduction. Impulse propagation by saltatory conduction is far faster (50 to 60 m/sec) than that of unmyelinated axons (1 to 2 m/sec).

Connective Tissue Structures and Extent of Nerve Injury

Each nerve fiber and its associated Schwann cells are surrounded by an endoneurium, which supports and nourishes axons via an endoneurial capillary system. Axons are grouped together in fascicles that are enclosed in a perineurium (although individual axons can cross from one fascicle to another along the course of a nerve). The fascicles are bundled together within the epineurium, which sheaths and nourishes the entire peripheral nerve (Fig. 42.2).¹⁰⁵

Focal nerve injuries are classified by the extent of damage to the axons and supporting structures. The main classification schemes are the Seddon system (Table 42.1)¹⁴⁶ and the Sunderland system (Table 42.2).¹⁶¹ In the Seddon system, *neurapraxia* refers to conduction failure without axonal loss, as seen with focal demyelination significant enough to cause conduction block. *Axonotmesis* refers to axonal damage with resultant degeneration of the distal component of the axon (wallerian degeneration), while the supporting connective tissue structures, including outer epineurium, remain intact. *Neurotmesis*, the most extensive category of nerve injury, refers to complete disruption of the axons and connective tissue structures, as with complete nerve transection.

The Sunderland system further delineates the degree of connective tissue damage by specific layer (see Table 42.2). First-degree injury corresponds to Seddon neurapraxia, and the higher degrees



• **Fig. 42.2** Anatomy of a nerve. Each axon is surrounded and associated Schwann cell enclosed by endoneurium. Multiple axons grouped in fascicles surrounded by perineurium. Fascicles are enclosed in epineurium to form the peripheral nerve. (Modified from Miller RD. *Anatomy of the peripheral nerve*. In: Miller RD, ed. *Miller's Anesthesia*. 6th ed. Elsevier; 2005.)

TABLE 42.1 Seddon Classification of Nerve Injury

Nerve Injury	Description
Neurapraxia	Transient conduction block without axon damage
Axonotmesis	Axon damage with wallerian degeneration, preservation of supporting structures
Neurotmesis	Complete nerve transection

Adapted from Seddon HJ. A classification of nerve injuries. *Br Med J*. 1942;2(4260):237-239.

TABLE 42.2 Sunderland Classification of Nerve Injury

Nerve Injury	Description
First degree	Transient conduction block without axon damage
Second degree	Axon damage with wallerian degeneration, preservation of supporting structures (endoneurium, perineurium, and epineurium)
Third degree	Intrafascicular lesion, damage to axons and endoneurium, with preservation of perineurium and epineurium
Fourth degree	Interfascicular lesion, damage to axons, endoneurium, and perineurium, with epineurium intact
Fifth degree	Complete nerve transection

Adapted from Sunderland S. The anatomy and physiology of nerve injury. *Muscle Nerve*. 1990;13(9):771-784.

represent axonal injury with varying degrees of damage to connective tissue structures. Higher degrees of injury carry poorer prognoses for spontaneous recovery, with neurotmesis and/or fifth-degree injury having no capacity for spontaneous axonal regrowth, due to lack of connective tissue pathways to the distal muscle or cutaneous territory. Notably, electrodiagnostic testing (EDX) can evaluate axonal continuity but cannot evaluate integrity of the supportive tissue structures.

Nerve Injury, Recovery, and Electrodiagnostic Assessment

EDX is particularly helpful in confirming and localizing focal neuropathies. Additionally, it provides critical information about the severity and chronicity of the injury and can be useful in determining the prognosis of the injury. The expected evolution of electrophysiologic abnormalities differs for primarily neurapraxic vs. axonal lesions (Fig. 42.3).

Until about 1 week postinjury, neurapraxia and axonal loss are electrophysiologically indistinguishable. In purely neurapraxic injuries, in which there is conduction block without axonal damage,

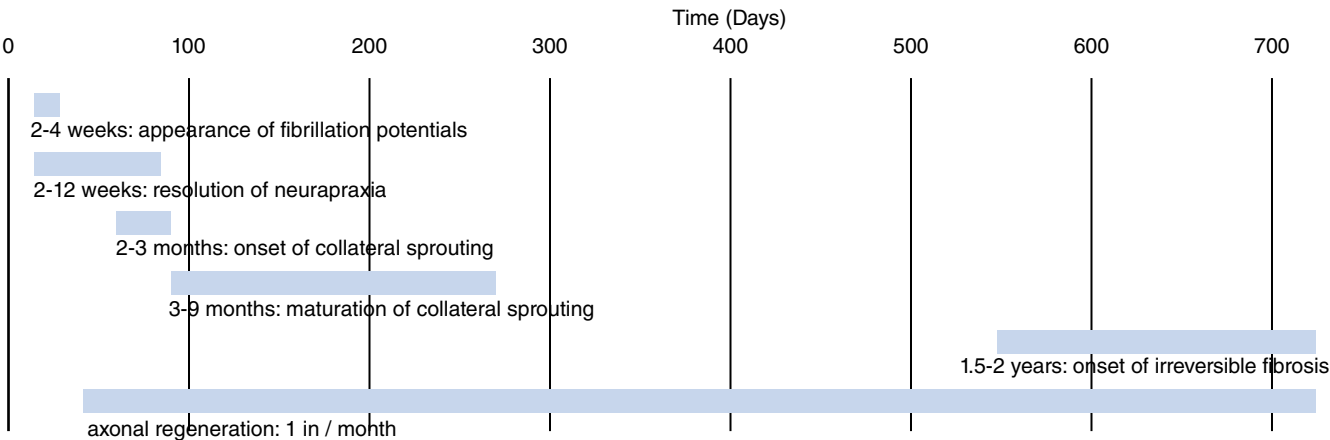
the compound muscle action potential (CMAP) and conduction velocity (CV) will be normal when stimulating distal to the lesion, as the motor axon remains intact, while stimulation proximal to the lesion will yield a CMAP with reduced amplitude (partial neurapraxia) or no response (complete neurapraxia). Needle electromyography (EMG) will show absent or decreased motor unit action potential (MUAP) recruitment commensurate with the proportion of axons blocked. Positive waves and fibrillation potentials are not seen, nor are changes in MUAP morphology because there is no motor axon loss.

In an axonal (rather than neurapraxic) lesion, the same pattern is observed in the initial days postinjury. This phase is known as *hyperacute axonal loss*, wherein the axonal segment distal to the lesion has not yet undergone wallerian degeneration. Specifically, sensory nerve action potential (SNAP) and CMAP amplitudes are normal when stimulating the nerve distal to the lesion and abnormal when stimulating proximally. On EMG, motor unit recruitment is decreased commensurate with the quantity of axons lost.

Wallerian degeneration spreads distally from the site of the lesion and occurs over a period of about 7 days for motor axons and about 10 days for sensory axons. Once these events have transpired, the relevant SNAP and CMAP amplitudes are low regardless of stimulation site. Muscle membrane instability, manifesting as denervation potentials (fibrillations and positive sharp waves), becomes evident around 3 weeks after injury, with abnormalities first noted proximally (closer to the lesion) and progressing distally over time.¹³⁹ It is important to note that fibrillation potentials may also be seen in direct muscle trauma, which can complicate localization of nerve injury.

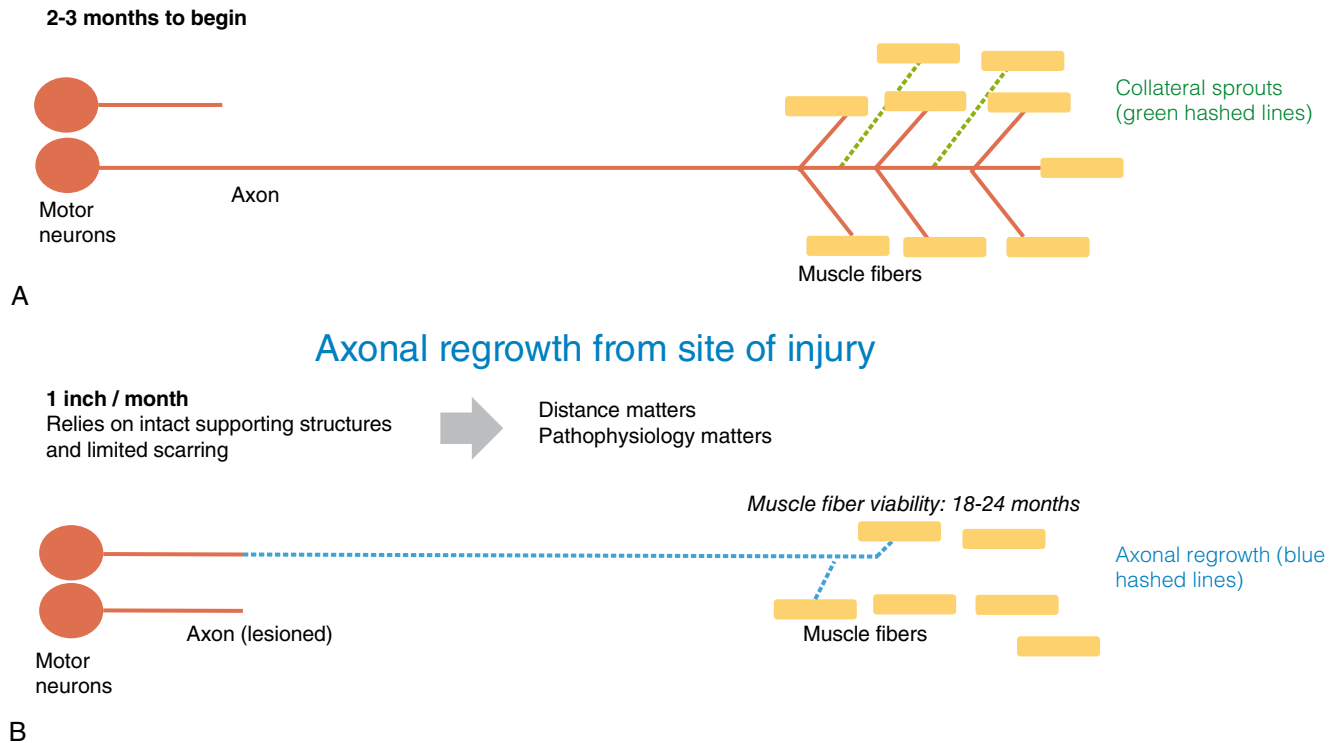
Recovery of focal nerve injury occurs by three mechanisms depending on the nature and severity of injury: resolution of neurapraxia, collateral sprouting, and axonal regeneration.¹³⁹ Resolution of neurapraxia occurs rapidly, with remyelination usually complete by 3 months, although more severe or chronic lesions may take longer. In contrast, reinnervation after axonal loss relies on collateral sprouting, axonal regrowth, or a combination of both mechanisms. These mechanisms are illustrated in Fig. 42.4 and described here.

Collateral sprouting is seen in partial axonal lesions, wherein intact axons sprout new terminal nerve branches to reinnervate “orphaned” muscle fibers (see Fig. 42.4A). An axon can reinnervate approximately up to five times its original number of muscle



• Fig. 42.3 Chronology of findings post nerve injury. (Modified from Hearn SL, Stino AM, Howard IM, et al. Serial electrodiagnostic testing: utility and indications in adult neurological disorders. *Muscle Nerve*. 2024;69(6):670-681.)

Collateral sprouting from nearby axons



• **Fig. 42.4** Illustration of collateral sprouting (A) and axonal regrowth (B), mechanisms of recovery after axonal loss. Note how collateral sprouting can occur quickly but depends on intact neighboring axons. Axonal regrowth is slower and relies on intact connective tissue structures.

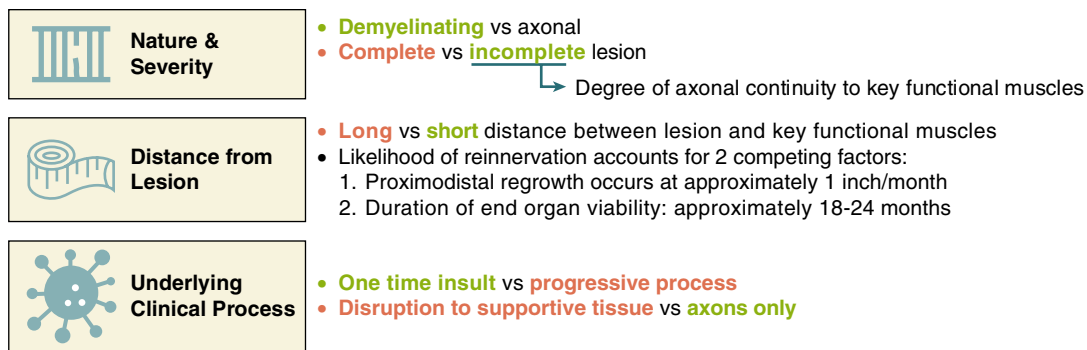
fibers, aligning with the observation that injuries with loss of less than about 70% to 80% of axons typically show good recovery with conservative management.¹⁵ Collateral sprouting takes about 2 to 3 months to manifest on EMG, and it is recognized through morphologic changes of the MUAPs. With early reinnervation, MUAPs show increased polyphasia, duration, and motor unit instability caused by desynchrony of the muscle fiber action potentials within the MUAP in the setting of immature neuromuscular junctions and more slowly conducting terminal nerve branches with newly forming myelin. As these new components mature, their electrophysiologic signal becomes assimilated into the primary MUAP, yielding waveforms with increased duration and amplitude (less polyphasia). Very remote and mature collateral sprouting manifests as triphasic MUAPs with significantly increased amplitude and only modestly increased duration.

Axonal regrowth represents the third mechanism of recovery. In partial axonal lesions, it occurs in conjunction with collateral sprouting, especially where there has been extensive axonal loss. Complete lesions with no axonal continuity to muscle rely entirely on axonal regrowth for reinnervation. The proximal nerve stump begins to sprout within 24 to 48 hours after injury and grows distally (see Fig. 42.4B). Recovery depends on degree of injury (including damage to supporting connective tissue structures), intervening scar tissue, and distance from target muscle. Regeneration occurs at a rate of 1 to 5 mm/day and progresses in a proximal to distal fashion, with the earliest signs seen in muscles just distal to the site of injury.¹⁶² On EMG, these initial MUAPs show low amplitude, increased polyphasia, and decreased recruitment.

Prognostication

Rendering a prognosis after peripheral nerve lesions is complex and depends on an array of factors. Purely neurapraxic nerve injuries have very good prognosis, with recovery in a few months, if the cause of compression is eliminated, although conduction slowing may persist on nerve conduction studies. Recovery of mixed demyelinating and axonal injury is usually biphasic, with demyelinating components resolving in a few months and axonal recovery proceeding more slowly. The prognosis of axonal nerve lesions depends on the degree of loss and can be estimated by the reduction of CMAP amplitude compared with the opposite side. In studies of facial nerve lesions, those with CMAP amplitude >30% of the opposite side had a very good prognosis, while those with less than 10% had a poor outcome.¹⁵³

Prognosis is more challenging for cases of severe axonal loss, with extent of axonal loss exceeding about 70%, including for complete lesions with no evidence of axonal continuity. Complete lesions rely on axonal regrowth (rather than collateral sprouting or remyelination). Capacity for axonal regrowth, in turn, depends on viability of connective tissue scaffolding, which is not directly assessed with electrophysiologic testing. On the one extreme, complete neurotmesis has no capacity to recover without surgical intervention; two severed ends of the nerve will retract, and there is no capacity for regrowth across a gap. When such lesions can be readily identified clinically (e.g., open wound with nerve laceration) or by imaging, surgical repair is typically indicated. In other cases, EDX testing may determine that a lesion is complete with no axonal continuity beyond the site of injury, but no information



• **Fig. 42.5** Factors impacting prognosis of peripheral nerve lesions. *Red text:* negative prognostic factors. *Green text:* positive prognostic factors.

can be derived about the viability of the supporting connective tissue elements. In such cases, repeat EDX may be considered to assess for evidence of early reinnervation by axonal regrowth. Serial EDX may thus guide surgical decision making, when performed at intervals up until the time of a surgical decision (typically 6 to 9 months postinjury, to allow sufficient time for axonal regrowth after the surgical intervention).⁶⁶ Because the distal muscle fibers and connective tissue structures remain viable for reinnervation for only about 18 to 24 months postdenervation, capacity for reinnervation is time limited and distance dependent.

Factors influencing prognosis of nerve lesions are depicted in Fig. 42.5. Aside from the neuroanatomic and neurophysiologic factors discussed earlier, assessment of prognosis should account for the underlying clinical process: whether the lesion represents an isolated event vs. a progressive process, and whether the pathophysiologic mechanism likely disrupts supportive tissue structures (e.g., laceration) vs. axon alone (e.g., immune-mediated neuropathy).

Ultrasound in Diagnosis of Peripheral Nerve Injury

The role of ultrasound in the diagnosis of peripheral nerve injuries has been growing. Ultrasound can complement EDX in diagnosing focal nerve injury and generalized neuropathies. In the setting of acute trauma, such as motor vehicle accidents, ultrasound may be able to localize sites of nerve trauma that are not accessible to nerve conduction study. As signs of denervation are not seen in muscle until wallerian degeneration occurs 3 to 4 weeks after trauma, needle EMG cannot identify axonal injury in the early postinjury phase, but ultrasound can demonstrate structural findings such as focal nerve compression or even complete nerve discontinuity, which may guide the immediate treatment approach. In some cases, the morphology of the injured nerve on ultrasound may also help predict prognosis for recovery. Coraci et al. showed that enlargement of the fibular nerve greater than 300% of normal on ultrasound had no recovery of function.²⁶ In nontraumatic peripheral nerve injuries, ultrasound can provide additional structural information. Ultrasound may find abnormality when EDX studies are normal and can be helpful for assessment after surgical intervention as EDX abnormalities can persist even after clinical improvement. The role of ultrasound in individual focal neuropathies will be discussed further.⁶⁷

Neuromuscular ultrasound may also assist in the diagnosis of generalized peripheral neuropathies. Those with acquired peripheral neuropathies, such as chronic inflammatory demyelinating polyra-

diculoneuropathy (CIDP), may have multifocal enlargement of peripheral nerves, fascicular enlargement, and increased nerve vascularity. Hereditary demyelinating neuropathies can be distinguished from acquired conditions, as hereditary demyelinating disorders will have diffuse and symmetric nerve enlargement. Ultrasound is also relevant in the diagnosis of multifocal motor neuropathy and will show regional nerve enlargement. This finding can help differentiate this condition from motor neuron disease, in which nerves will have normal or decreased cross-sectional area.¹¹

Focal Peripheral Nerve Injuries

Traumatic focal nerve injuries are usually caused by laceration, compression, or traction of a specific nerve, often in a location of anatomic vulnerability (superficial location, adjacency to bone, passage through a tight canal, change in trajectory along a long course).⁶⁵ They can occur after repetitive motion, direct forceful trauma, or prolonged compression caused by positioning or use of bracing or assistive devices; they can arise from a single injury or from repeated biomechanical stress over time. An understanding of the anatomic distribution of specific nerves, their course and areas of mechanical vulnerability, and the specific muscles and sensory distribution they supply is critical in localizing the patient's lesion and formulating strategies for treatment.

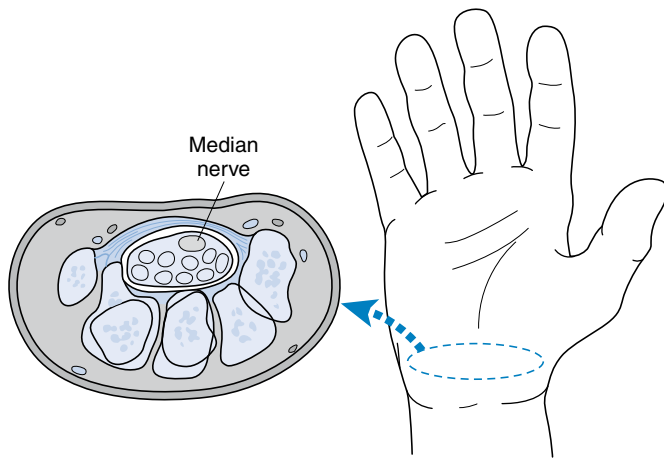
Notably, focal neuropathies often accompany multitraumatic events and may be initially overlooked in the setting of catastrophic traumatic injury. Life-threatening injury, major trauma to bone and tissue, and altered mental status may delay recognition of a serious nerve injury that later impedes expected functional progress. Up to 10% to 34% of patients admitted to rehabilitation units with traumatic brain injury have peripheral nerve injuries.²⁸ It is therefore important to be vigilant for these injuries in the appropriate clinical setting.

The following sections address common focal nerve lesions of the upper and lower limbs.

Focal Neuropathies of the Upper Limb

Median Mononeuropathies

The medial and lateral cords of the brachial plexus form the median nerve just distal to the level of the axilla. The nerve courses parallel to the intermuscular septum and then passes under the bicipital aponeurosis before entering the antecubital fossa medial to the biceps tendon and anterior to the brachialis muscles. In 1% to 2% of the population, there is a small bone spur,

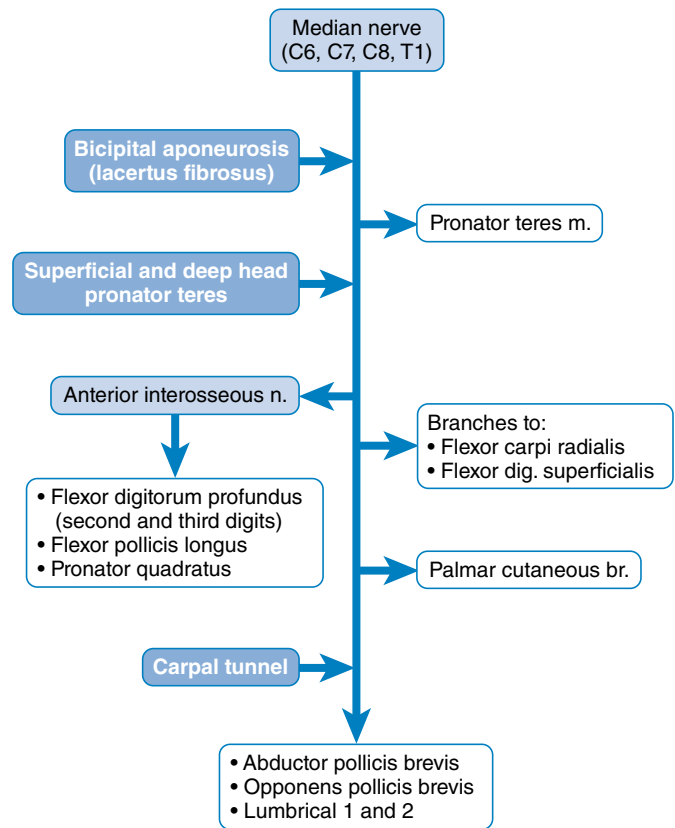


• **Fig. 42.6** The carpal canal is formed by the carpal bones and the flexor retinaculum. The contents include the median nerve, flexor pollicis longus tendon, and four tendons each of the flexor digitorum profundus and superficialis muscles. (Modified from Moeller JL, Hutchinson MR. Carpal tunnel. In: Rakel RE, ed. *Textbook of Family Medicine*. 7th ed. Saunders; 2007.)

proximal to the medial epicondyle, that is spanned by a ligament called the *ligament of Struthers*, which crosses over the nerve, and may be a rare cause of compression.¹⁶³ The median nerve then courses between the superficial and deep heads of the pronator teres and then between the humeroulnar and radial portions of the flexor digitorum superficialis, supplying these muscles along with the flexor carpi radialis. Distal to the medial epicondyle of the humerus, the median nerve gives off the anterior interosseous nerve (AIN), which supplies the lateral portions of the flexor digitorum profundus, flexor pollicis longus, and pronator quadratus muscles. Proximal to the wrist, the median nerve gives off the palmar cutaneous branch, supplying sensation to the skin over the thenar eminence. At the wrist, the median nerve passes through the carpal canal, which is bordered by the carpal bones and the transverse carpal ligament. The carpal canal contains nine flexor tendons (flexor digitorum superficialis [four], flexor digitorum profundus [four], and flexor pollicis longus) and the median nerve (Fig. 42.6).¹⁰⁷ Distal to the carpal tunnel in the hand, the median nerve supplies the first and second lumbricals, opponens pollicis, abductor pollicis brevis, superficial portion of flexor pollicis brevis, and digital cutaneous branches, which provide sensation to the palmar surface of the thumb and second, third, and lateral portion of the fourth fingers (Fig. 42.7). The median nerve is most vulnerable to injury at the level of the carpal tunnel.

Median Mononeuropathy at the WRIST (Carpal Tunnel Syndrome)

Carpal tunnel syndrome (CTS) is the most common compressive neuropathy. In CTS, the median nerve is compressed within the carpal canal, approximately 1 to 2 cm beyond the distal wrist crease. It is more common in females and is typically more severe in the dominant hand, but is frequently bilateral.¹⁵² Most cases are idiopathic, but some risk factors have emerged. Repetitive hand and wrist movements, occupations requiring high grip force, and the use of vibratory tools have been associated with CTS.⁶⁵ Medical conditions associated with CTS include diabetes, hypothyroidism, and rheumatoid arthritis (RA),¹⁵⁹ as well as obesity¹² and pregnancy. Infrequently, CTS can be caused by a mass

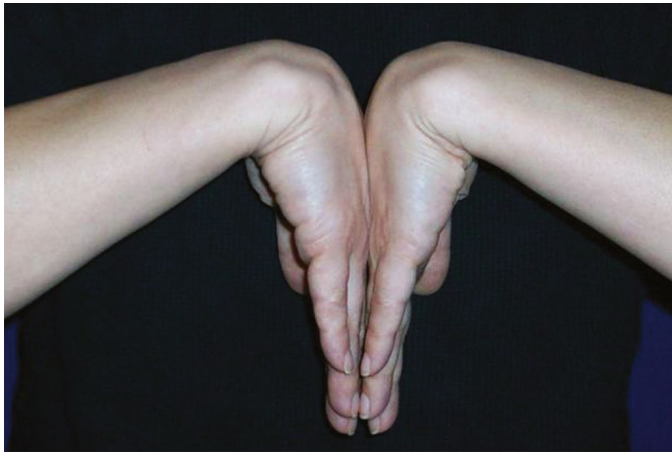


• **Fig. 42.7** Median nerve.

lesion at the wrist, such as a ganglion cyst or neurofibroma, or associated with acute trauma to the wrist.

CTS typically presents with paresthesias and numbness of the second and third digits and variably the thumb and lateral fourth digits, although often the patient will have more diffuse complaints of the entire hand being numb. Sensory supply to the thenar eminence is spared, as the palmar cutaneous branch of the median nerve arises proximal to the carpal tunnel and courses outside it. Classically, in the early phases, symptoms are intermittent, often worse at night, causing nocturnal awakening, and improved by flicking the hands and placing them in a dependent position. Other aggravating activities include driving or grasping objects, such as a telephone, book, or steering wheel, with a flexed or extended wrist and with activation of the finger flexors. Patients may describe pain or discomfort that radiates proximally through the forearm or even to the shoulder. Provocative maneuvers, such as a Tinel or Phalen sign (Fig. 42.8), may reproduce symptoms in the median nerve distribution, but should be interpreted in the appropriate clinical context because they may exacerbate other etiologies of hand and wrist pain. Muscle stretch reflexes will be unaffected by CTS.

More advanced cases with axonal loss may manifest a fixed sensory deficit of the median innervated digits and hand weakness (lateral lumbricals, abductor pollicis brevis, opponens pollicis, and a portion of flexor pollicis brevis), manifesting as difficulty with opening jars and manipulating buttons. [Video 42.1](#) demonstrates weakness of left thumb abduction. Atrophy of the thenar eminence can be seen when the injury is severe and chronic. Long finger flexor strength should be normal, as should ulnar-innervated finger abduction. The differential diagnosis of CTS includes proximal median nerve injury (pronator or anterior interosseous



• **Fig. 42.8** The Phalen test for median nerve compression.

nerve syndromes), C6 or C7 radiculopathy, and brachial plexopathy. Clinical findings suggesting an alternative diagnosis include weakness of non–median innervated muscles or proximal median innervated muscles, asymmetry of muscle stretch reflexes, or prominent neck pain. Electrodiagnostic studies can confirm the diagnosis of CTS and assess the severity, as well as exclude other neurologic diagnoses. EDX has a sensitivity of 80% to 90% for CTS. Median sensory latencies are prolonged in mild disease and are compared with the ulnar sensory latency in the same hand. Midpalmar stimulation might be more sensitive because it records over a shorter segment across the carpal tunnel and contains the larger diameter sensory fascicles that supply the muscle spindles, which are more vulnerable to demyelination.³⁸ Median motor responses are affected in moderate disease and will show decreased compound muscle action potential amplitude and prolonged distal latency, with normal CV across the forearm. Severe CTS will show signs of axonal loss, such as denervation or reinnervation of abductor pollicis brevis or opponens pollicis muscles.

Ultrasound complements EDX testing in evaluating CTS and may show abnormalities in patients with clinical symptoms of CTS but normal EDX studies. Common abnormalities seen on ultrasound include enlargement of the median nerve, flattening of the nerve in the carpal canal, bowing of the flexor retinaculum, increased intraneural vascularity, and decreased median nerve mobility with wrist flexion. A cross-sectional area of greater than 9 mm² is the best criteria for CTS with a sensitivity and specificity of 71% and 86%, respectively.¹⁶⁴ Ultrasound can show structural abnormalities that may impact treatment and surgical management and may sometimes elucidate structural etiologies for persistent symptoms after carpal tunnel release.⁸¹

Mild to moderate CTS usually responds to conservative measures. Exacerbating activities, such as repetitive or excessive wrist flexion and extension and gripping, should be avoided. Wrist splints are a safe and effective treatment. The wrist should be splinted in 0 to 5 degrees of extension at night and as much as tolerated during the day. Nonsteroidal antiinflammatory drugs (NSAIDs) are often used, but evidence is limited; a short course of oral steroids may be helpful.²² Local corticosteroid injections into the carpal tunnel can improve symptoms as well as nerve function (based on electrodiagnostic studies) and are effective for at least 3 months.^{8,62} Two injections may be required if nocturnal paresthesia persists after the initial treatment.⁹⁷ In addition to the antiinflammatory and analgesic properties, steroids may also act

to thin tendons, decreasing carpal canal pressures, although this mechanism in CTS is not confirmed. Surgical decompression is indicated in patients with severe or rapidly progressive weakness and atrophy, for those who fail to respond to conservative measures, or if there is a mass lesion causing the median nerve compression.

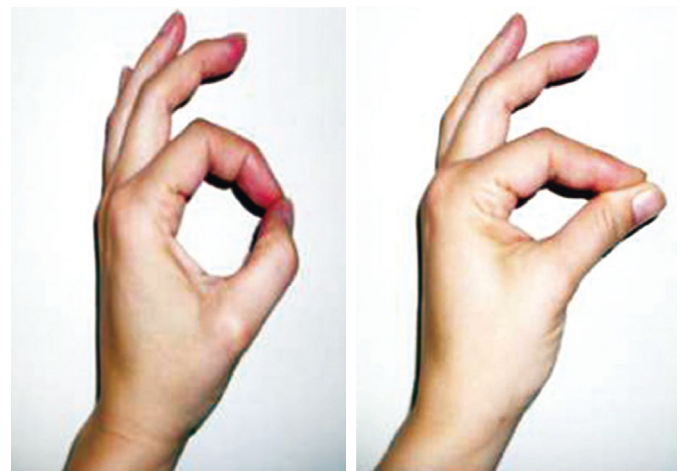
Proximal Median Nerve Injuries

Compression of the median nerve in the forearm is much less common than compression distally at the carpal canal (CTS). The primary levels of compression in the forearm are at the AIN and at the pronator muscle.

The AIN can be compressed by the fibrous arch formed by the flexor digitorum superficialis and the pronator teres muscles. The nerve can also be injured from repetitive forearm flexion or pronation, elbow or forearm fractures, or associated with neuralgic amyotrophy. As the AIN is a purely motor branch, this injury is a purely motor syndrome with weakness of flexion of the interphalangeal joint of the thumb and distal interphalangeal joints of the index and middle fingers. Accordingly, the patient will be unable to form the “OK” sign and will substitute with a key pinch (Fig. 42.9). Pronation may weaken due to involvement of the pronator quadratus, although it is less apparent because of intact strength of the pronator teres, which is spared. Testing pronation with the elbow in flexion will better isolate the action of the pronator quadratus. Although there is no sensory loss, patients may experience an aching pain in the forearm. On electrodiagnostic studies, standard nerve conduction studies are normal, and needle examination will demonstrate denervation in the muscles supplied by the AIN.

Reports of improvement within 3 to 24 months from symptom onset⁵⁴ suggests the etiology is often inflammatory (e.g., neuralgic amyotrophy) rather than mechanical compression. It is therefore recommended that unless there is an identified compressive lesion, surgical exploration and decompression should be delayed unless there is no recovery after 12 months.¹⁴⁰

The median nerve may also be compressed as it passes between the two heads of the pronator teres (pronator syndrome [PS]), a fascial band from the flexor digitorum superficialis, or the biceps aponeurosis. In cases of PS, all median innervated muscles, aside from the pronator teres, will be affected, as well as the entire



• **Fig. 42.9** Anterior interosseous nerve syndrome causes weakness of flexors to distal joints of the fingers and inability to form the “OK” sign (*left*), substituting a “key pinch” (*right*).

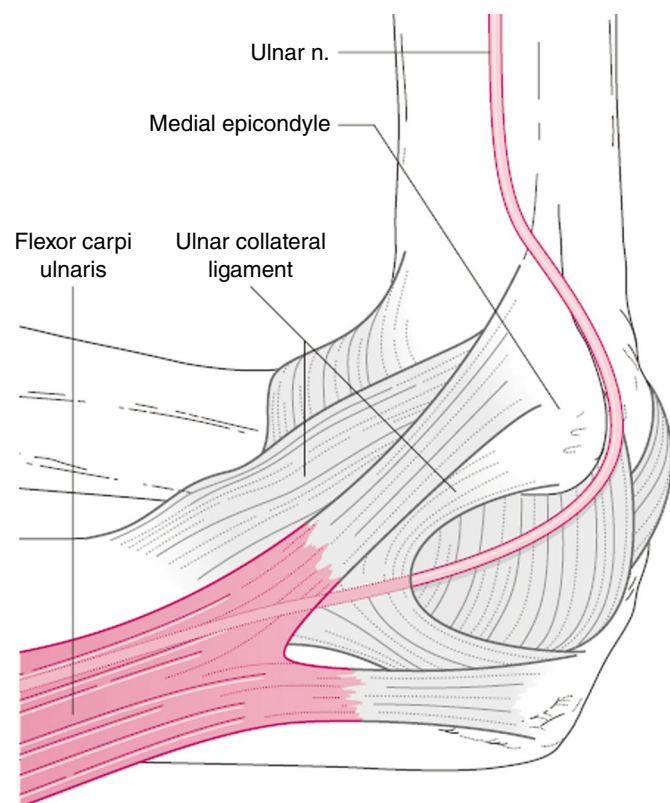
sensory territory of the median nerve. Patients present with pain and paresthesias in the first three fingers of the hand, mimicking CTS. However, there will be involvement of the skin over the thenar eminence, which is usually spared in CTS. Nocturnal awakening is not typically seen in PS. Pain might be provoked by resisted elbow flexion and pronation, and resisted finger flexion. Tenderness over the nerve at the area of compression at the pronator teres or a Tinel sign can often be elicited.

Electrodiagnostic studies may show median sensory amplitudes that are diminished or absent, with normal latency. The amplitude of the median motor response to the abductor pollicis brevis muscle is diminished, and CV over the forearm may be slowed. Needle examination may demonstrate abnormalities in all median-innervated muscles except pronator teres.

Conservative treatment of PS is usually initiated with a trial of NSAIDs, physical therapy, and avoidance of repetitive elbow flexion, pronation, or forced gripping. A posterior elbow splint can be applied. Surgical decompression can be indicated if there is no motor recovery.²⁰

Ulnar Mononeuropathies

The ulnar nerve derives from the medial cord, lower trunk, and C8 and T1 roots. It courses through the medial arm in close proximity to the median nerve. In the elbow region, there are two main sites of vulnerability to injury (Fig. 42.10). The first is the ulnar (retroepicondylar) groove, a depression between the medial epicondyle and the olecranon in which the nerve courses superficially. The second is at the humeroulnar aponeurosis, where the nerve courses deep to the Osborne ligament and enters the space between the two heads of the flexor carpi ulnaris. At each of these

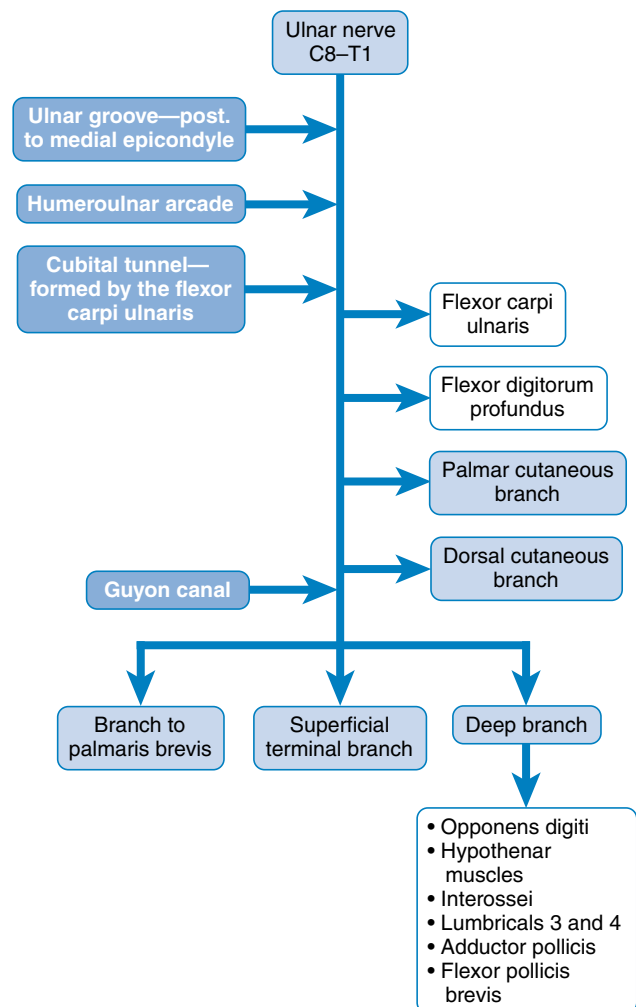


• Fig. 42.10 Ulnar nerve.

sites, the nerve is vulnerable to focal compression and stretch from sustained elbow flexion. Distally, the ulnar nerve innervates the flexor carpi ulnaris and the medial portions of the flexor digitorum profundus. It branches in the distal third of the forearm to supply cutaneous innervation to the palmar and dorsal aspects of the medial hand. This arrangement also has clinical relevance, in that elbow lesions can cause numbness of the entire medial aspect of the hand, whereas wrist injuries primarily affect the digits. The nerve enters Guyon canal in the wrist between the hook of the hamate and the pisiform bone. Upon entering the hand, the ulnar nerve branches to provide sensory and motor innervation, traversing as far laterally as to supply the adductor pollicis and flexor pollicis brevis. Ulnar nerve innervation is shown in Fig. 42.11.

Ulnar Neuropathy at the Elbow

Ulnar neuropathy at the elbow (UNE) is the second most common focal neuropathy after CTS. UNE preferentially affects the left side regardless of hand dominance; in addition to mechanically demanding events, such as repetitive elbow flexion, gripping, or external pressure over the ulnar groove,³⁴ older age, male sex, and smoking cigarettes increase the risk of UNE.^{10,48,133,134} In about 15% of patients, the ulnar nerve will be entrapped 2 to



• Fig. 42.11 The ulnar nerve is vulnerable to compression and stretch injury at the elbow. (From McGann SA, Flores RH, Nashel DJ. Entrapment neuropathies and compartment syndromes. In: Hochberg MC, et al., eds. *Rheumatology*. Elsevier; 2010:785.)

3 cm distal to the medial epicondyle where it courses beneath the humeroulnar aponeurosis or cubital tunnel. The majority of patients experience compression at the medial epicondyle or in the ulnar groove.¹¹⁴

Patients with UNE will typically present with dysesthesias and sensory changes in the ulnar distribution of the hand. Patients report very well-circumscribed numbness in the ulnar half of the fourth digit and the entire fifth digit, as opposed to the more variable and sometimes whole hand numbness seen with CTS. Patients may also report pain or discomfort over the medial aspect of the forearm along the course of the ulnar nerve; however, sensation in this region is spared because it is innervated by the medial antebrachial cutaneous nerve. In more advanced disease, patients report loss of dexterity and grip strength, affecting functional activities.

Physical examination can reveal altered sensation in the ulnar distribution of the hand, especially if there has been axonal loss. A Tinel sign is often elicited over the nerve in the ulnar groove and/or cubital tunnel. Strength of the interossei may be diminished, and with advanced cases, atrophy of the region over the first dorsal interosseous muscle is noted (often in association with some “clawing” of the hand caused by intrinsic muscle weakness).

Electrodiagnostic studies may be useful in localizing UNE to the elbow and assessing severity, as well as ruling out other neurologic causes. Reported sensitivity of EDX testing ranges from 37% to 86% depending on the technique and gold standard used.⁶⁸ Nerve conduction studies should assess for focal slowing or conduction block of ulnar motor fascicles across the elbow. Secondary axonal loss may result in decreased amplitude of SNAP and CMAP responses. In severe disease, axon loss may predominate, causing a decrease in CMAP amplitude across all sites of stimulation, making lesion localization challenging. The first dorsal interosseous is the muscle most likely affected during EMG needle examination, with the abductor digiti minimi and flexor carpi ulnaris, respectively, less involved. C8-T1 radiculopathy and lower trunk plexopathy can be differentiated from UNE through the finding of EMG abnormalities in nonulnar innervated muscles

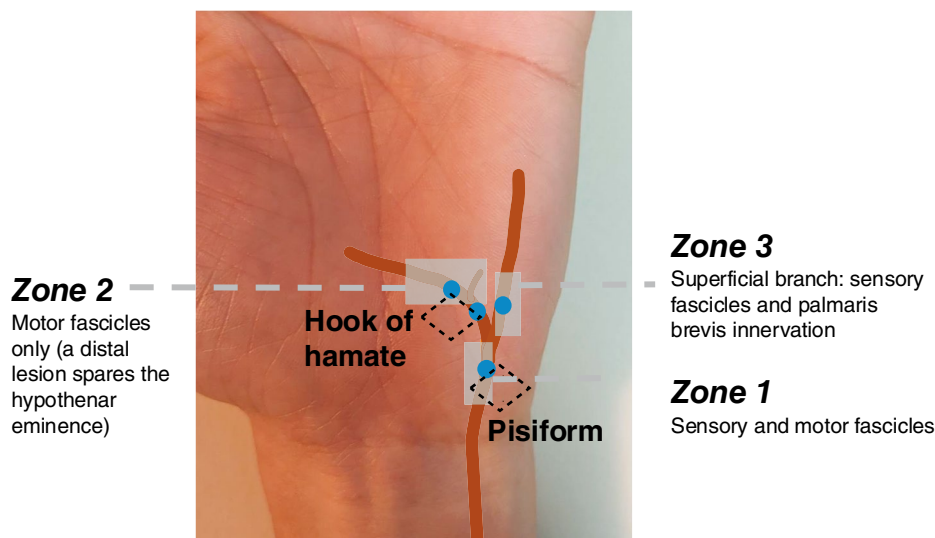
(e.g., opponens pollicis, flexor pollicis longus, and extensor indicis). Lower brachial plexopathy typically shows a low medial antebrachial cutaneous SNAP amplitude; radiculopathy may also show abnormalities in the paraspinal muscles.

In patients with clinical symptoms of UNE and normal EDX studies, ultrasound can be diagnostic. Ultrasound testing demonstrated an 80.5% sensitivity, with UNE present with cross-sectional area of greater than 9.8 mm² at the retroepicondylar groove.⁴² Ultrasound may be particularly valuable in localizing the site of nerve compression and be complementary to short segment nerve conduction studies. Ultrasound and EDX localization concurred in almost 50% of patients and was able to solely identify the site of compression in a quarter of patients in one study.¹¹⁵ Ultrasound also allows for dynamic testing, showing ulnar nerve subluxation or dynamic impingement of the triceps brachii muscle with elbow motion. In those with demonstrated subluxation, nerve transposition may be preferred during surgical decompression.

The results of treatment for UNE, both surgical and nonsurgical, are variable. Strategies include avoidance of strong, repetitive gripping, preventing repetitive or prolonged elbow flexion (using a splint as necessary), cushioning of the elbow to prevent compression over the ulnar nerve, smoking cessation if warranted, and patient education. As smoking appears to be an independent risk factor for UNE and reduces patient satisfaction after surgery, avoidance of tobacco is a reasonable recommendation.¹³⁵ Absent clinical or electrodiagnostic improvement within 2 to 3 months, surgical evaluation for possible decompression or transposition of the nerve may be considered.

Ulnar Neuropathy at the Wrist

Ulnar neuropathy at the wrist (UNW) is much less common than UNE and CTS and is typically caused by direct compression. Repetitive and forceful gripping of a tool that applies pressure over the medial palm is a classic mechanism. Four potential sites of injury within Guyon canal give rise to four distinct clinical presentations with varying degrees of sensory and motor involvement (Fig. 42.12).



• **Fig. 42.12** Anatomy of Guyon canal. Ulnar neuropathy at the wrist can produce four typical patterns of clinical deficit, depending on the site of compression (blue circles). Relevant ulnar nerve branches are shown in brown; osseous landmarks in black; zones of injury in gray. A lesion in zone 2 can cause broad hand weakness (including of abductor digiti minimi) if the lesion is proximal, or it can involve only the deep branch to the lateral hand (if distal). (From Hearn SL, Jorgensen SP, Gabet JM, et al. Occupational nerve injuries. *Muscle Nerve*. 2024.)

When UNW is suspected, the electromyographer should stimulate the ulnar nerve and record from the first dorsal interosseous; assessment for focal slowing and conduction block of these specific fascicles increases sensitivity for this lesion compared with recording abductor digiti minimi alone, given the high proportion of zone 2 lesions.^{29,41,147} The dorsal ulnar cutaneous sensory study, whose fascicles are spared with UNW, is sometimes used; however, it is sometimes spared with UNE^{108,180} and, conversely, can be unobtainable in some individuals without pathology due to anatomic variation.¹⁰⁸ One should exercise similar caution with localizing a lesion to the wrist based solely on EMG abnormalities, as relative sparing of the ulnar forearm muscles is often seen with UNE. Ultrasound can identify nerve enlargement, hypoechogenicity, and a loss of normal fascicular pattern. Space-occupying lesions within the canal may also be identified easily under ultrasound.²⁰ Initial conservative management is appropriate, especially where an aggravating activity can be identified and eliminated. Cases with a severe and progressive motor deficit or with an anatomic cause unlikely to improve with activity modification (mass or ganglion cyst) may benefit from surgical decompression.

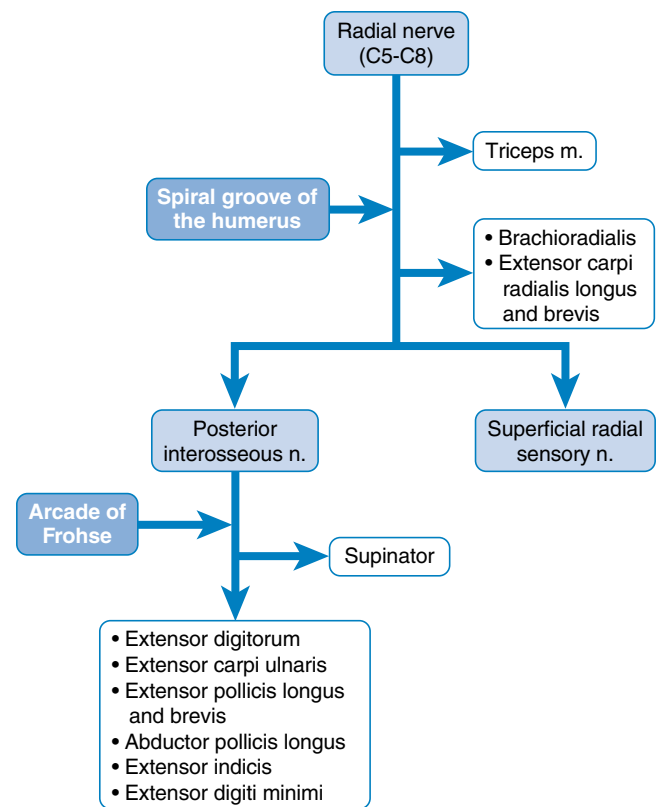
Radial Mononeuropathies

The radial nerve is the largest terminal branch off the brachial plexus, with contributions from C5 to C8 nerve roots. Arising from the posterior cord of the plexus, it runs along the lateral wall of the axilla, where it is first susceptible to compression. It then supplies the triceps brachii, and then winds around the radial groove of the humerus, where it is again susceptible to injury. It pierces the intermuscular septum below the deltoid and supplies the brachioradialis muscle and extensor carpi radialis longus above the elbow. The nerve enters the forearm between the biceps and brachioradialis muscles and gives off the superficial radial nerve at the level of the elbow joint. It then travels around the neck of the radius, passing through the supinator (arcade of Frohse), as the deep radial motor branch. After innervating the supinator and extensor carpi radialis brevis, the nerve emerges as the posterior interosseous nerve (PIN) and enters the extensor compartment of the forearm. Here it supplies the extensor digitorum, extensor carpi ulnaris, extensor pollicis longus and brevis, abductor pollicis longus, extensor indicis, and extensor digiti minimi. The superficial radial nerve traverses the forearm to supply sensation to the dorsolateral portion of the hand and first three digits (Fig. 42.13).

The radial nerve is less susceptible to compression than the median and ulnar nerves. The most common lesion sites are the radial groove and the arcade of Frohse (PIN). Occasionally, the superficial radial branch will be injured at the wrist, causing purely sensory abnormalities on the dorsolateral aspect of the hand. This is encountered with use of handcuffs, tight wristwatches, or even from using wrist braces for other conditions, such as CTS.

Radial Neuropathy at the Radial Groove

The radial nerve is most commonly injured at the radial groove of the humerus as a result of insults, including humeral fractures, tourniquets, or compression during prolonged sleep or unconsciousness. In contrast to direct compression, a different mechanism of injury—intense muscular contraction of the triceps—can lesion the nerve slightly more distally at the tendinous insertion of the lateral head of the triceps brachii. Regardless of mechanism, radial mononeuropathy in the arm presents with



• Fig. 42.13 Radial nerve.

wrist drop and inability to extend the fingers. Elbow extension is usually spared as the triceps is innervated proximal to the radial groove. Slight weakness of elbow flexion may result from involvement of the brachioradialis. Median and ulnar innervated muscle strength will be unaffected; however, finger abduction can falsely appear weak unless the metacarpophalangeal joints of the hand are held in passive extension. Sensation is abnormal over the dorsolateral hand and the dorsum of digits 1 through 4. The brachioradialis reflex is diminished or absent, whereas the triceps and biceps reflexes are spared.

EDX can aid in localization and in exclusion of differential diagnoses, which may include radiculopathy or posterior cord lesions.

Management of proximal radial nerve injuries depends on the etiology of the injury. External compression typically causes neurapraxic injury that usually recovers spontaneously within 2 months.¹⁷⁰ Poorer outcomes are seen with humeral fracture. If any response can be recorded from the extensor indicis on nerve conduction studies, there is a 90% chance of recovering at least antigravity wrist strength.⁹⁸ If there is no recovery within 8 to 10 weeks, surgical exploration is indicated, and if there is no return of function after 1 year, tendon transfer may be considered to improve upper limb functioning. During the recovery period, a cock-up splint is used to maintain the wrist and fingers in an extended position to prevent contracture.

Posterior Interosseous Neuropathy

The PIN is most commonly entrapped at the arcade of Frohse (also known as supinator syndrome).³¹ The PIN can also be injured from elbow fracture, laceration, or compression from soft

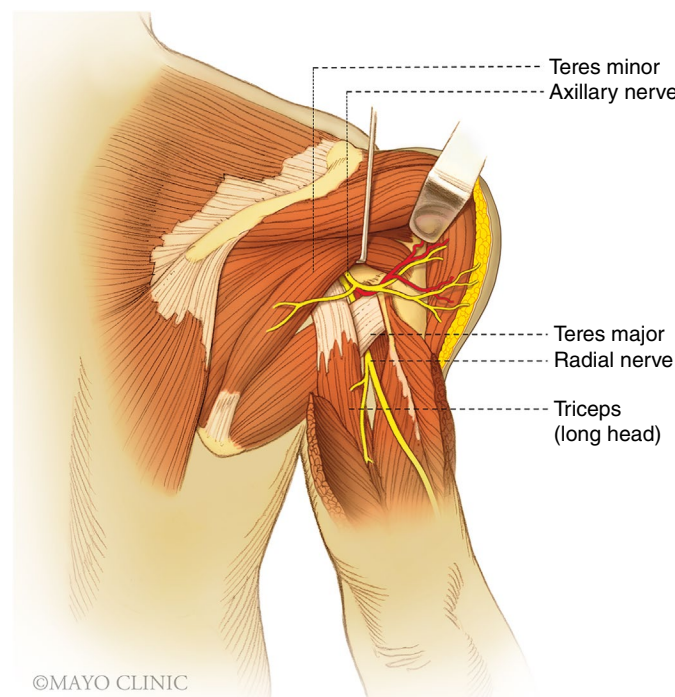
tissue masses such as lipomas,¹⁸⁸ neurofibromas, schwannomas, hematomas, and elbow synovitis from RA.¹⁰⁴ It can also develop insidiously, in association with strenuous use of the forearm, particularly pronation and supination. Finger extensors and extensor carpi ulnaris are weak, while brachioradialis and radial wrist extensors are spared. Accordingly, radial deviation is seen when the wrist is actively extended. The supinator muscle is variably affected. The PIN has no sensory fibers; therefore sensation will be unaffected. Nerve conduction studies to the extensor indicis are abnormal, whereas radial sensory conduction is preserved. The EMG needle examination typically localizes the lesion. Treatment for idiopathic PIN involves avoidance of provocative activities, use of oral NSAIDs, and splinting. If no improvement is seen after 4 to 12 weeks, or if there is progressive or severe weakness and atrophy, surgical treatment brings good to excellent recovery of function in up to 90% of patients.⁶⁴ If a mass lesion is present, surgical exploration is usually required.

Shoulder Region Mononeuropathies

Disorders of Shoulder Abduction

Consideration of nerve disorders affecting shoulder function should include assessment of the axillary and suprascapular nerves.

Axillary nerve. The axillary nerve branches from the posterior cord. It traverses through the quadrilateral space to innervate the deltoid and teres minor muscles and provides cutaneous innervation to a region overlying the lateral proximal arm. It can be injured in the quadrilateral space (Fig. 42.14),¹³ typically in the context of trauma, such as glenohumeral dislocation or direct impact. Axillary mononeuropathy presents with weakness of



• **Fig. 42.14** Course of the axillary nerve through the quadrilateral space. (From Beecher G, Dyck PJB, Zochodne DW. Axillary and musculocutaneous neuropathies. *Handb Clin Neurol*. 2024;201:135-148.)

shoulder abduction, especially beyond the first approximately 15 degrees, and numbness of the aforementioned skin territory.

Suprascapular nerve. The suprascapular nerve is most susceptible to injury at or between the suprascapular and spinoglenoid notches (suprascapular canal).⁴ At the suprascapular notch, a small foramen in the superior scapula bordered superiorly by the transverse scapular ligament,¹³⁰ the nerve angles sharply within its long course.⁴ Scapular depression, retraction, and hyperabduction may lead to a “sling effect” with increased superior compression and angulation of the nerve against the ligament.^{82,130} This movement can result in dynamic kinking of the nerve. Examples of activities with this mechanism include performing an overhead spike during volleyball or heaving loads over the shoulder.⁶⁵ Suprascapular mononeuropathy can present as shoulder pain and/or weakness of external rotation and abduction.

Suprascapular nerve injury localizing to the spinoglenoid notch can yield isolated weakness of infraspinatus and may result from shoulder joint pathology, such as a paralabral cyst compressing the nerve.

Initial management is typically conservative, involving exercises to strengthen the shoulder girdle, utilizing other tools to minimize shoulder loading during occupational tasks, and suspension of repetitive strain activities.

Disorders of Scapular Movement

Nerve lesions affecting innervation to the scapular stabilizers can lead to scapular winging. Assessment of scapular winging requires a careful physical exam, with sufficient clothing removed to observe the scapulae at rest and dynamically. The scapulae should be examined through the ranges of motion (ROM) of shoulder flexion and abduction, as the specific nerve lesion drives the dynamic pattern of weakness. Evaluation of scapular mobility functionally through a modified pushup or wall pushup can often differentiate voluntary scapular winging from a neurologic deficit. Specifically, some individuals can voluntarily produce scapular winging, absent neurologic damage, which can be a point of confusion for patients and practitioners alike during the physical examination; assessment for natural dyskinesia through coordinated, composite movement such as a pushup can help draw the distinction.¹⁵⁰

Table 42.3 depicts three mononeuropathies that create scapular dyskinesia.¹⁰⁰

Other Diagnostic and Management Considerations

In assessing nerve lesions affecting the shoulder region, neuralgic amyotrophy often presents as mononeuropathy or multiple mononeuropathies of the above nerves. This entity is discussed in Regional Nerve Disorders, later. Early recognition guides management.

Treatment of these mononeuropathies largely depends on etiology and extent of axonal loss. Potentially progressive lesions (e.g., nerve compression by mass) require medical or surgical treatment of underlying cause. For static lesions with significant axonal loss (substantial weakness or >70% axonal loss), surgical consultation may be indicated, and serial EDX may be considered to assess for early recovery. Mild lesions can be managed conservatively.

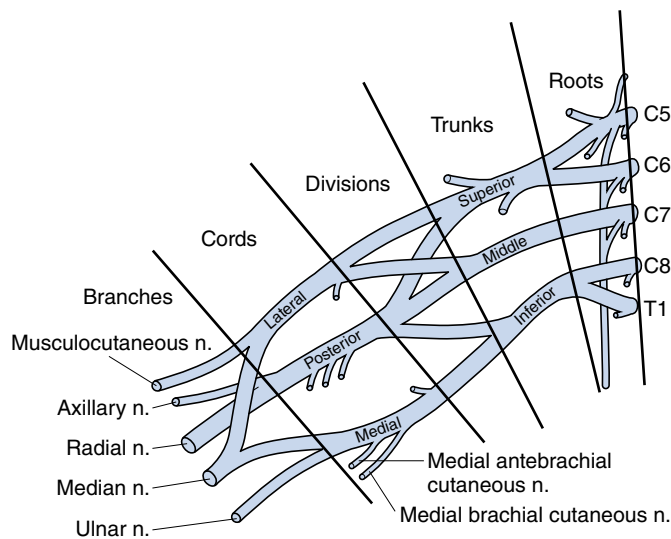
Brachial Plexopathy

The brachial plexus is formed from C5 to C8 and T1 ventral roots. The roots join to form the trunks (upper, middle, and lower), which

TABLE 42.3**Neurogenic Causes of Scapular Winging and the Physical Exam**

	Medial Winging		Lateral Winging
Injured nerve	Long thoracic	Spinal accessory	Dorsal scapular
Muscle palsy	Serratus anterior	Trapezius	Rhomboids
Physical exam	Arm flexion; pushup motion against a wall	Arm abduction; external rotation against resistance	Arm extension from full flexion
Position of the scapula compared with normal	Entire scapula displaced more medial and superior	Superior angle more laterally displaced	Inferior angle more laterally displaced

From Martin RM, Fish DE. Scapular winging: anatomical review, diagnosis, and treatments. *Curr Rev Musculoskelet Med.* 2008;1(1):1-11.



• **Fig. 42.15** Brachial plexus. (Modified from Wedel DJ, Horlocker TT. Nerve blocks. In: Miller RD, ed. *Miller's Anesthesia*. 7th ed. Elsevier; 2010.)

then divide to form the cords (lateral, posterior, and medial). Nerve branches from these structures are shown in Fig. 42.15.¹⁸⁶

The brachial plexus is susceptible to direct trauma, including but not limited to birth injury. Trauma may affect multiple regions of the plexus, and assessment for plexopathy should be considered in the setting of clavicle fracture.

Electrodiagnostic studies of the affected limb show sensory and motor abnormalities in the distribution of the affected portions of the plexus. Effort should be made to obtain a reliable sensory response originating from each of the major components of the plexus, as SNAP abnormalities distinguish plexopathy from radiculopathy. In the context of regional trauma, a combination of normal SNAPs despite dense numbness and poor or absent motor function indicates preganglionic injury and raises suspicion for nerve root avulsion, which carries a poor prognosis for spontaneous recovery. Needle EMG should sample muscles representing each portion of the plexus (trunks, cords, and peripheral nerves) to localize the level of the lesion and to determine axonal continuity (Table 42.4). Imaging studies may be indicated to evaluate for underlying tumor, mass, or disruption of the plexus. Ultrasound can sometimes identify root avulsion and allow for an earlier decision on surgical intervention.

Rehabilitation of brachial plexopathy includes proper positioning with orthoses to stabilize affected joints and improve function. ROM exercises and physiotherapy are important to reduce contractures and strengthen recovering motor function. Medication management is appropriate for neuropathic pain. Surgical treatment can consist of neurolysis or primary nerve repair. Nerve grafting, nerve transfers, or tendon transfers can be used if there is no improvement or if there is a plateau in recovery.

Upper Brachial Plexopathies

Lesions of the upper brachial plexus (C5 and C6 anterior primary rami and upper trunk) manifest clinically as weakness of shoulder abduction, elbow flexion, and forearm supination, also known as Erb palsy.

The upper trunk is particularly vulnerable when the head and neck are forcibly bent away from the shoulder, causing traction of the upper brachial plexus. Milder plexus injuries have been associated with sports, particularly American football. So-called stingers or burners are caused by a blow to the head, neck, or shoulder and cause a sharp burning pain in the shoulder with transient weakness.

Lower Brachial Plexopathies

Lesions of the lower brachial plexus (C8 and T1 anterior primary rami and lower trunk) produce weakness and atrophy of hand intrinsic muscles and sensory deficits along the medial upper limb. This can be a result of a space-occupying lesion, such as a mass, injury during operative procedures, such as a median sternotomy, or they can occur as the trunk passes through the thoracic outlet, especially in the context of anatomic variations in the supracostoclavicular space (e.g., cervical rib). When associated with trauma, the mechanism is typically upward traction of the arm overhead. Cancer can directly invade or compress the lower brachial plexus, as can be observed with a superior pulmonary sulcus (Pancoast) tumor, which can present with an associated Horner syndrome. Lung and breast cancers and lymphomas are the most common metastatic tumors that affect the plexus.

Thoracic outlet syndrome. Neurogenic thoracic outlet syndrome (TOS) arises from compression of the brachial plexus as it passes through, and exits from, the thoracic outlet. This condition is rare, with a prevalence of 1 per 1 million, and most commonly affects the lower trunk of the plexus or the anterior primary rami of T1 or C8.¹⁸⁹ The plexus can be compressed at three major points. The first is in the interscalene triangle, as it travels between

TABLE 42.4 Localization of Focal Brachial Plexus Lesions

Plexus Lesion	Nerves Affected	Muscles Affected	Sensory Nerve Conduction Studies Affected
Trunks			
Upper	Suprascapular Lateral pectoral Musculocutaneous Lateral portion median Portion of radial Axillary	Supraspinatus, infraspinatus Upper portion pectoralis major Biceps Pronator teres, flexor carpi radialis Brachioradialis deltoid	Lateral antebrachial cutaneous (LAC) Median Radial
Middle	Thoracodorsal Subscapular Radial Lateral portion median	Latissimus dorsi Teres major All radial muscles (except brachioradialis) Pronator teres, flexor carpi radialis	Median Radial
Lower	Medial pectoral Ulnar Medial portion median	Lower pectoralis major All ulnar muscles All median muscles except pronator teres and flexor carpi radialis	Ulnar to fifth digit Medial
Cords			
Lateral	Musculocutaneous Lateral portion median	Biceps Pronator teres Flexor carpi radialis	LAC Median to first digit
Posterior	Thoracodorsal Subscapular Axillary Radial	Latissimus dorsi Teres major Deltoid All radial muscles	Median to third digit Radial to first and third digits
Medial	Ulnar Medial half of median	All ulnar muscles All median muscles except pronator teres and flexor carpi radialis	Ulnar to fifth digit Medial antebrachial cutaneous

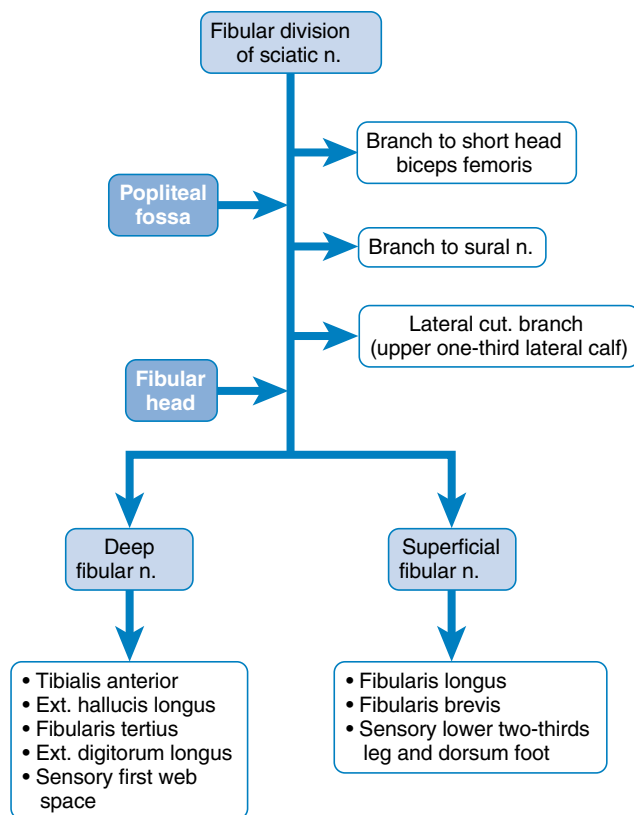
the medial and anterior scalene muscles, over the first rib. In its midcourse, the plexus can be compressed as it runs between the first rib and the clavicle. At its distal point, it passes beneath the pectoralis muscle close to its insertion on the coracoid process. Common causes of neurogenic TOS include the presence of a cervical rib, fibrous bands, accessory and hypertrophied scalene muscles, anatomic variations in the course of the muscle, and postural factors.

Radiographs, computed tomography (CT), and magnetic resonance imaging (MRI) scans may reveal a cervical rib or an elongated C7 transverse process, often with a fibrous band connecting it to a rib. Electrodiagnostic studies reveal a lesion of the lower trunk or C8/T1 anterior primary rami, with the T1 fascicles most affected as they are lifted superiorly or anteriorly by the fibrous band or rib. Accordingly, a typical electrodiagnostic pattern is abnormally low median antebrachial cutaneous sensory, median and ulnar motor, and ulnar sensory responses, with relatively normal median sensory response. Needle EMG will classically demonstrate abnormalities in lower trunk muscles, with the T1-innervated thenar eminence muscles more affected than the C8-innervated ulnar hand intrinsic muscles, although electrophysiologic abnormalities may be broader in some cases and may evidence fascicular selectivity.¹⁰⁹ In cases with neurologic and electrophysiologic abnormalities and a clear structural lesion, surgical management, which may include removal of a fibrous band and/or rib resection, should be considered.

Focal Neuropathies of the Lower Extremities

Fibular (Peroneal) Mononeuropathy

The fibular nerve arises from the L4 to S1 nerve roots and runs with the tibial nerve in the posterior thigh as part of the sciatic nerve. While still a part of the sciatic nerve at the popliteal fossa, it gives off a muscular branch to the short head of the biceps femoris, the only muscle of the thigh that is innervated by the fibular division of the sciatic nerve. Within the popliteal fossa, after dividing from the tibial nerve, it contributes a cutaneous branch which, along with a branch from the tibial nerve, forms the sural sensory nerve. Another sensory branch, the lateral cutaneous nerve of calf, also arises in the popliteal fossa and supplies the proximal lateral aspect of the calf. The common fibular nerve then courses around the fibular head where it is firmly tethered by a fibroosseous band; it then forms its terminal branches, the superficial and deep fibular nerves. The superficial fibular nerve innervates the fibularis longus and brevis muscles and supplies sensory innervation to the lower two-thirds of the lateral leg and the dorsum of the foot, sparing the first dorsal web space. In 12% to 35% of the population, the superficial fibular nerve gives off an accessory fibular nerve, which provides some motor innervation to the extensor digitorum brevis (EDB).¹⁷³ The deep fibular nerve supplies the anterior tibialis, extensor hallucis longus, fibularis tertius, EDB, and sensation to the first web space (Fig. 42.16).



• Fig. 42.16 Fibular (peroneal) nerve.

Fibular neuropathy is the most common nerve injury in the lower limb.⁸³ Lesions to the common fibular nerve at the fibular head are common because of its vulnerable superficial location and because it is relatively tethered by a fibrous band, which makes it susceptible to traction with knee and hip flexion. Causes of direct compression at the fibular head include prolonged passive positioning during surgery or bedrest, repeated leg crossing (especially with a history of recent weight loss), and improperly placed splints, braces, and casts. Prolonged squatting can place excessive traction on the nerve. Acute knee trauma, particularly with fibular head fracture, can also cause injury. Importantly, an intraneural or extraneural ganglion cyst can be a cause of direction compression or neural injury. These typically arise when a defect in the tibiofibular joint capsule enables joint fluid to accumulate within a cyst through a one-way valve mechanism.¹⁵⁸ An extra-neural ganglion cyst from this joint can directly compress the fibular nerve. Alternatively, the cyst may communicate with an intraarticular branch of the fibular nerve, leading to an intraneural ganglion cyst, with fluid tracking proximally within the fibular nerve and compressing its fascicles.

The prominent clinical finding is weakness of ankle dorsiflexion, often with minimal pain. In severe injury, complete foot drop can result in a steppage gait, a compensatory mechanism that allows the patient to clear the toe during the swing phase of gait. When weakness is more subtle, premature foot flat or foot slap can be seen and heard after heel strike on the affected side. Often the deep fibular branch is more affected than the superficial branch, with foot eversion less impaired than dorsiflexion, and with a motor deficit outweighing the sensory deficit. Sensation is diminished in the lower two-thirds of the lateral leg and dorsum of the foot. Tapping over the

common fibular nerve as it travels around the fibular neck may produce a positive Tinel sign.

Electrodiagnosis should include a fibular motor study, with stimulation both above and below the fibular head to evaluate for focal slowing or conduction block. The fibular motor response can be recorded over either the EDB or the anterior tibialis. Although the EDB is routinely used, it can be atrophic as a result of local trauma and may be less reliable. Focal conduction block, with a greater than 20% drop in motor amplitude when stimulating above compared to below the fibular head, is the best indicator for localizing a fibular nerve injury at the knee and is present in 60% to 70% of lesions.¹²² The presence of any response obtained at the anterior tibialis or EDB is associated with good recovery of grade 4 to 5 strength compared with absence of response.³³

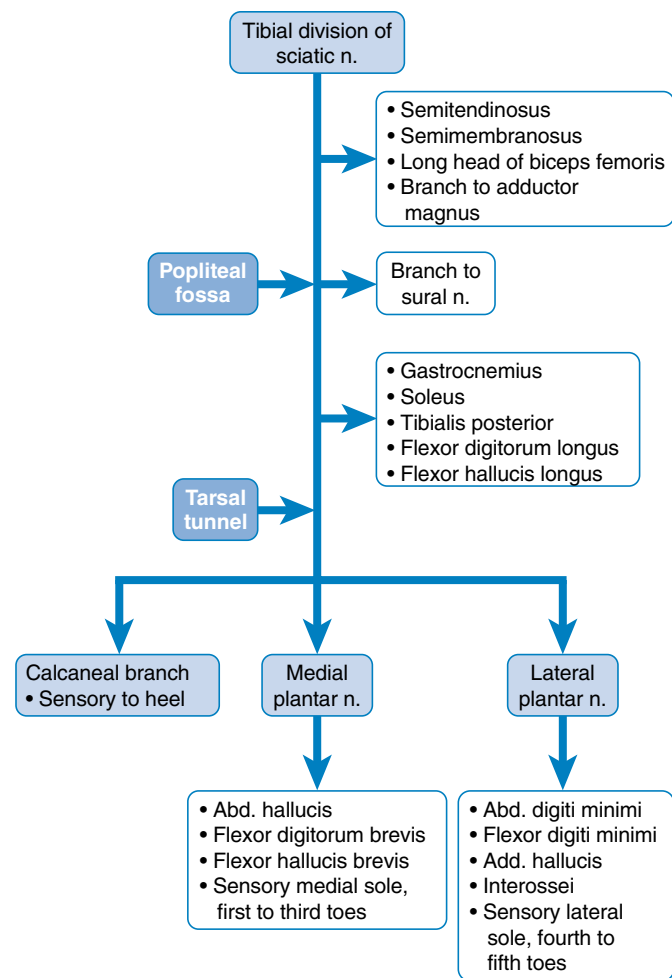
Several other neurologic problems can present as foot drop. In addition to fibular neuropathy, the differential diagnosis also includes L5 radiculopathy, lumbosacral plexopathy, and sciatic neuropathy. The presence of back pain, prominent leg pain, a depressed hamstring reflex, or weakness of the hamstrings or plantarflexion suggest an alternate diagnosis.

If fibular nerve injury is associated with trauma, radiographs may demonstrate a fracture around the fibular head. MRI is useful for visualizing a space-occupying lesion, directly visualizing the fibular nerve in the popliteal fossa and identifying denervated muscles. Ultrasound can also visualize the nerve as it passes near the fibular head and can assist in identification of space-occupying lesions, such as intraneural and/or extraneural ganglion cysts or parameniscal cysts.

Awareness of fibular neuropathy enables prevention, especially in the rehabilitation setting. As many cases of fibular neuropathy are the result of improper positioning during prolonged hospitalization or surgery, care should be taken to offload the lateral aspect of the knee, such as padding the lateral knee with pillows and avoiding casts or orthoses that apply direct pressure distal to the fibular head. Casts and braces should be carefully evaluated and periodically reevaluated for pressure points at the fibular head. Atraumatic patients presenting with fibular neuropathy should be questioned regarding prolonged knee flexion activities, repetitive squatting, severe weight loss, and habitual leg crossing. Ankle-foot orthoses are often needed for more severe foot drop. Surgical management includes neurolysis and decompression, nerve repair, and nerve or tendon transfers.

Tibial and Plantar Mononeuropathy

The tibial nerve arises from the L5 to S2 nerve roots to form the tibial division of the sciatic nerve. While part of the sciatic nerve, it innervates all the hamstring muscles with the exception of the short head of the biceps femoris. It also partially innervates the adductor magnus muscle, with the obturator nerve contributing remaining innervation. Within the popliteal fossa, it contributes a sensory branch (together with a fibular nerve contribution) to form the sural sensory nerve, which innervates the lateral lower leg and foot. The tibial nerve continues on to supply the muscles of the posterior compartment of the lower leg. At the ankle, it passes just posterior to the medial malleolus through the tarsal tunnel, which is formed by the flexor retinaculum that spans the space between the medial malleolus and the calcaneus. Accompanying the tibial nerve is the posterior tibial artery and the tendons of the posterior tibialis, flexor digitorum longus, and flexor hallucis longus. The three terminal branches of the tibial nerve in the



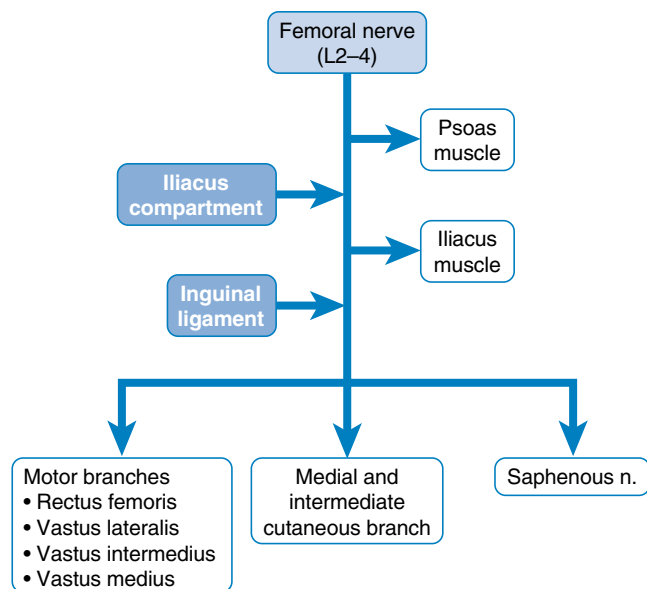
• Fig. 42.17 Tibial nerve.

foot are the calcaneal branch and the medial and lateral plantar nerves (Fig. 42.17).

Tibial Neuropathy at the Ankle

The tibial nerve can be lesioned at the medial ankle where it courses under the flexor retinaculum, yielding foot numbness or dysesthesias in the distribution of the affected fascicles, exacerbated by standing and walking. When arising insidiously, it is often called *tarsal tunnel syndrome*. Compression can affect any of the three terminal branches.¹¹³ Etiologies can include ankle trauma, arthritis, deformity of the heel, talocalcaneal coalition,¹⁶⁵ vascular compression, and masses such as ganglion cysts or lipomas. Physical examination can reveal sensory loss in the lateral and/or medial plantar distribution, often sparing the heel. A Tinel sign might be provoked by percussion over the flexor retinaculum at the medial malleolus.

Electrodiagnostic studies should evaluate for dysfunction of tibial innervation distal to the ankle. Medial and lateral plantar conduction studies can be performed as sensory, motor, or mixed nerve studies and can be selected based on the patient's symptoms. Given the wide range of healthy nerve conduction response amplitudes and the high prevalence of mild neuropathy or age-related distal neuromuscular decline, both needle EMG of foot intrinsic muscles and comparison to contralateral nerve conduction studies where symptoms are unilateral are important study planning considerations.³² MRI can assess the tarsal tunnel and evaluate for tenosynovitis,



• Fig. 42.18 Femoral nerve.

structural abnormalities, hypertrophied or accessory muscles, or other mass lesions.

Treatment consists of NSAIDs, local corticosteroid injections, and orthoses to correct biomechanical abnormalities such as overpronation or excessive supination to reduce biomechanical factors at the tarsal tunnel. Surgical release can be indicated for those who do not respond to conservative management, but it may be more effective in those with space-occupying lesions as the source of compression and when decompression is done within 10 months of symptom onset.¹⁷⁴

Femoral Neuropathy

The femoral nerve is formed from the L2 to L4 spinal roots. The nerve then travels between the psoas and iliacus muscles, innervating the iliacus just before it passes under the inguinal ligament. After passing under the inguinal ligament, it divides into its terminal branches, which supply the four heads of the quadriceps muscle, the sartorius and pectineus muscles, the medial and intermediate cutaneous branches to the anterior thigh, and the saphenous nerve, which innervates the medial leg (Fig. 42.18). The femoral nerve is usually injured within the retroperitoneal space in the pelvis (e.g., compression by retroperitoneal hematoma) or under the inguinal ligament. Injury can be iatrogenic, as with intraabdominal and intrapelvic surgeries, often caused by self-retaining retractors, gynecologic and urologic procedures, ilioinguinal nerve blocks, and catheterization procedures.³⁶ The lithotomy position, with extreme hip flexion and external rotation, can result in compression of the femoral nerve at the inguinal ligament.¹ Thin body habitus predisposes to injury from retractors and the lithotomy position,¹⁸³ while obese patients are at greater risk of femoral nerve injury in high-energy pelvic trauma.⁹³

Femoral neuropathy usually presents with unilateral thigh weakness and numbness of the anterior thigh and medial leg. A sensation of instability around the knee is usually noted, with particular difficulty with stairs and inclines. The patellar reflex can be depressed or absent. If the nerve is compressed within the pelvis, hip flexion strength can also be affected. Adduction and abduction of the thigh should be unaffected, as should knee

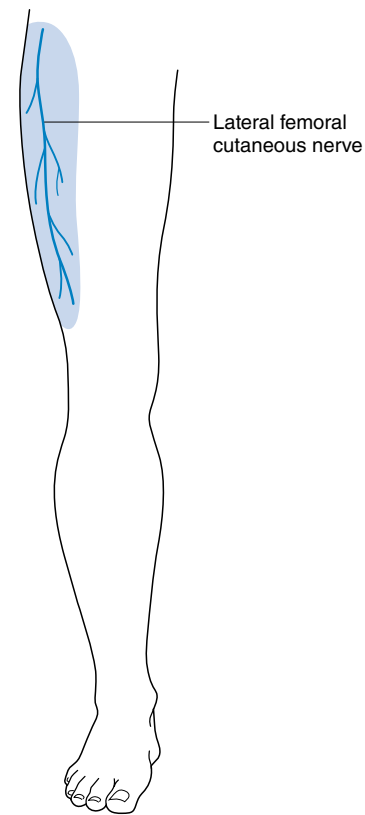
flexion and strength around the ankle. Pain with extension of the hip can be seen with femoral nerve injury.

Electrodiagnostic studies can include a femoral motor conduction study, stimulating at the groin. A CMAP of at least 50% of the opposite side favors good prognosis for improvement within 1 year, whereas fewer than half of those with a CMAP of less than 50% will improve.⁸⁹ Needle examination may help in localizing the site of the lesion to the inguinal ligament or intrapelvic based on involvement of the iliopsoas muscle. Study of the hip adductor muscles and paraspinal muscles will help exclude high lumbar radiculopathy or plexopathy. MRI and CT imaging are indicated if there is suspicion for a mass lesion.

Many causes of femoral neuropathy are iatrogenic and may be avoidable. In operative procedures, extremes of flexion, external rotation, and abduction of the hip should be avoided to prevent compression as the nerve passes under the inguinal ligament. Self-retaining retractors are also a potential source of injury during surgery, and care should be taken with placement to avoid compressing the nerve as it passes the psoas muscle.¹⁶ When a femoral neuropathy is suspected, imaging studies can assess for compression caused by hematoma, masses, or pseudoaneurysm. If a correctable cause is found, immediate surgical intervention might be indicated. Early physical therapy for strengthening and ROM should be initiated. Training to activate the gluteal muscles and/or plantar flexors in stance phase can help to augment weak knee extension. With mild to moderate knee extension weakness, an ankle-foot orthosis with dorsiflexion stop can create an extension moment at the knee to compensate. If the weakness is severe, a knee-ankle-foot orthosis may be indicated; however, these are cumbersome and often poorly tolerated. Therefore a walker to keep the ground reaction force anterior to the knee axis of rotation during stance is often more functional. Serial clinical and electrodiagnostic examinations should be performed. If there is no improvement after 3 to 6 months, either clinically or electrodiagnostically, then surgical intervention might be indicated.³⁶ Because the quadriceps is such a large muscle, it is important to extensively sample the muscle for signs of reinnervation, as early recovery in the more proximal portions may be missed if only sampled in the more typically studied middistal portions.

Lateral Femoral Cutaneous Neuropathy

Mononeuropathy of the lateral femoral cutaneous nerve, also known as *meralgia paresthetica*, is a common and benign cause of anterolateral thigh numbness. The lateral femoral cutaneous nerve is a purely sensory nerve arising from the L2 and L3 nerve roots. There are several anatomic variations as the nerve exits the pelvis, most commonly in close relationship with the anterior superior iliac spine (ASIS) and the inguinal ligament, making the nerve vulnerable to compression in this area.⁷ The nerve then divides into anterior and posterior divisions and supplies the sensory territory of the anterolateral thigh (Fig. 42.19).³⁰ Meralgia paresthetica is typically idiopathic and is associated with obesity, pregnancy, and diabetes.⁷⁴ The nerve may be externally compressed by heavy tool belts, habitually leaning against a table, or wearing tight waist bands. Rarely, injury can occur during surgery, particularly during hernia repair, renal transplantation, harvesting of iliac bone graft, hip surgery, and aortic valve and coronary bypass surgery.¹⁵² A tumor involving the iliac crest can also compress the nerve at the ASIS, or more proximally it can be affected by an abdominal aneurysm or other pelvic mass.



• **Fig. 42.19** The sensory territory of the lateral femoral cutaneous nerve. (Modified from Craig EJ, Clinchot DM. Lateral femoral cutaneous neuropathy. In: Frontera WR, Silver JK, Rizzo TD, eds. *Essentials of Physical Medicine and Rehabilitation: Musculoskeletal Disorders, Pain, and Rehabilitation*. Saunders; 2008.)

Patients with lateral femoral cutaneous nerve dysfunction present with numbness and pain in a generally well-circumscribed distribution over the anterolateral thigh. Symptoms can be exacerbated by walking or extension of the hip and by wearing tight clothing or undergarments. No weakness is seen. On physical examination, in addition to numbness or paresthesia in the territory of the lateral femoral cutaneous nerve, a Tinel sign may be elicited when percussing just medial and inferior to the ASIS. Manual motor testing and muscle stretch reflexes are not affected. Meralgia paresthetica must be differentiated from femoral neuropathy, lumbar plexopathy, and high lumbar radiculopathy. This can often be done clinically, based on the distribution of sensory symptoms. Femoral and lumbar radiculopathies, unlike meralgia paresthetica, can also cause weakness of thigh muscles and a depressed patellar reflex.

Nerve conduction studies of the lateral femoral cutaneous nerve may reveal asymmetry when comparing the symptomatic with the unaffected side. However, this is a technically difficult sensory response to obtain even in healthy, nonobese individuals.^{1,90} Needle EMG is normal in those with meralgia paresthetica. Although not routinely performed, ultrasound can improve the ability to obtain a nerve conduction response by visualizing the nerve.¹⁴ Anesthetic nerve block of the lateral femoral cutaneous nerve may help confirm the diagnosis, although the anatomic variability of the nerve may lead to a negative block.

Meralgia paresthetica is a benign and self-limited condition. It often improves with conservative treatment over a few months. Eliminating exacerbating factors, such as tight clothing or tool belts, and weight loss will often resolve the symptoms. When painful

paresthesia is present, topical agents such as capsaicin cream and lidocaine patches can be helpful. Oral agents used for neuropathic pain can also control symptoms. Local corticosteroid injections might provide temporary relief.³⁹ Surgery might be indicated in refractory cases, with high success rates reported for both decompression and neurectomy,⁸⁵ although patients will be left with a persistent area of numbness with the latter procedure.^{18,86,143}

Sciatic Mononeuropathy

The sciatic nerve is made up of fibers from the L4 to S2 nerve roots, forming the tibial and fibular divisions of the sciatic nerve. These divisions run as distinctly separate nerves that do not interchange fascicles. The fibular division is much more susceptible to injury than the tibial division, for two main reasons. First, the fibular nerve is tethered at both the sciatic notch and the fibular head, making it more vulnerable to traction than the tibial nerve, which is not similarly restricted distally. Second, the fascicular pattern of the fibular nerve contains less supporting tissues and has larger and fewer fascicles than the tibial nerve, which is more invested with elastic epineurial tissue and is therefore more tolerant of external pressure.¹⁶¹ This propensity for greater injury to the fibular division of the sciatic nerve can make it challenging to differentiate a sciatic lesion from a common fibular lesion. The tibial innervated hamstring and plantar flexor muscles are powerful, making subtle weakness less obvious than with muscles innervated by the fibular nerve. Causes of sciatic nerve injury include hip fracture, posterior hip dislocation, and hip surgery, particularly total hip arthroplasty. Poorly placed intramuscular gluteal injections, knife or gunshot wounds, compression resulting from positioning during hospitalization in patients with thin body habitus, and the lithotomy position¹⁸³ can also injure the nerve. Space-occupying lesions

such as tumors, hematoma, aneurysm, and heterotopic ossification can also compress the nerve.

As discussed previously with fibular neuropathy, electrodiagnostic studies are designed to differentiate between sciatic neuropathy, common fibular or tibial mononeuropathy, L5 or S1 radiculopathy, and lumbosacral plexopathy. In severe sciatic neuropathy, all the fibular and tibial motor conduction responses are absent or reduced, as are sensory responses of the sural and superficial fibular nerves. In mild cases, side-to-side asymmetry of sensory and motor nerve conduction amplitudes should be evaluated. Needle examination of both fibular and tibial innervated muscles show denervation, usually with fibular muscles more affected. Occasionally, the fibular division can be exclusively injured in a sciatic lesion, in which case sampling of the short head of the biceps femoris is useful to differentiate it from a more distal common fibular lesion.⁸⁴ Gluteal and paraspinal muscles will be spared in sciatic lesions. MRI can demonstrate compressive lesions and, along with a CT scan, characterize traumatic injury. The sciatic nerve itself can be well visualized with MRI, showing focal enlargement and abnormal signal intensity. Ultrasound is not useful in evaluating the nerve within the pelvis; however, it may be useful in evaluating the nerve below the piriformis muscle.¹⁰¹

Treatment of sciatic nerve injury is largely supportive, with more specific management directed at etiology of the injury. If a compressive lesion is identified, such as tumor or hematoma, decompression may be indicated. Nerve grafting outcomes are guarded because of the size and length of the sciatic nerve; key functional muscles are often at great distance from the nerve lesion, precluding functional reinnervation.

Other Lower Limb Mononeuropathies

Several additional mononeuropathies of the lower extremity are listed in Table 42.5. Initial management includes identification

TABLE 42.5 Lower Limb Mononeuropathies

Nerve	Muscular Innervation	Sensory Innervation	Points of Vulnerability	Hallmarks of Presentation
Fibular nerve at the ankle	Extensor digitorum brevis, extensor hallucis brevis	First dorsal web space	Deep to the inferior extensor retinaculum Deep to the extensor hallucis longus (EHL) overlying talonavicular joint Deep to the EHL overlying the tarsometatarsal joints	Acute or repetitive microtrauma (inversion sprain) at the ankle Sensory changes in the first dorsal web space
Tibial nerve in the leg	Dependent on localization of lesion	Plantar surface of the foot and toes, posterolateral leg and lateral foot via sural nerve	Popliteal fossa	Acute traumatic injury (i.e., knee dislocation) or space-occupying lesions (i.e. Baker cyst) Weakness and sensory changes distal to the lesion
Obturator nerve	Adductor longus, adductor brevis, gracilis, pectineus, obturator externus, adductor magnus (partial innervation)	Distal two-thirds of the medial thigh, articular branches to the hip and knee joints	Within the pelvis, along anterior aspect of the sacroiliac joint Fascia overlying the adductor brevis	Weakness of adduction and internal rotation Sensory changes of the medial thigh
Sural nerve	None	Distal third of the posterolateral leg and lateral foot	Deep to the gastrocnemius aponeurosis With injury to the Achilles tendon, lateral ankle, or fifth metatarsal Iatrogenic injury via biopsy or harvesting	Sensory changes in nerve distribution, distal to the lesion
Lateral sural cutaneous nerve	None	Proximal lateral leg, with occasional extension to proximal lateral knee	Within popliteal fossa in the setting of abnormal anatomy	Sensory changes in nerve distribution, distal to the lesion

of an underlying etiology, if present. Conservative therapy may involve attention to strengthening and ROM, bracing, and diagnostic or therapeutic injection. Surgical decompression can be considered after a trial of conservative management or earlier if a structural lesion is identified or if there is progressive neurologic deficit.

Lumbosacral Plexopathies

The lumbosacral plexus can be divided into the lumbar and sacral plexuses. The lumbar portion is formed within the psoas muscle from the L1 to L3 nerve roots with contributions from T12 and L4. The lumbar plexus provides direct branches to the psoas muscle. The terminal branches are the ilioinguinal, iliohypogastric, genitofemoral, femoral, lateral femoral cutaneous, and obturator nerves. The sacral plexus forms within the pelvis anterior to the piriformis muscle. Terminal branches include the nerves to the superior/inferior gemelli, obturator internus, piriformis, superior and inferior gluteal nerves, posterior femoral cutaneous nerve, and sciatic nerve (Fig. 42.20).

Lumbar Plexus

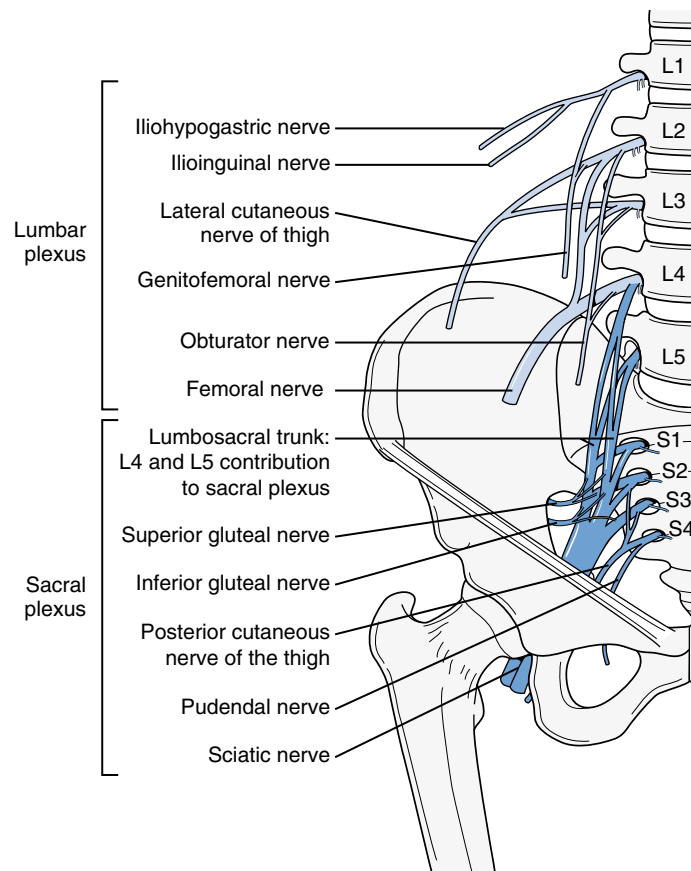
Retroperitoneal hemorrhage preferentially affects the lumbar portion of the plexus, usually in association with anticoagulation, blood dyscrasias (e.g., hemophilia), or leaking aortic aneurysm. This lesion affects the femoral innervation disproportionately, relative to obturator nerve innervation. Vascular aneurysms can also lesion the lumbar plexus directly, even without overt bleeding.

The lumbar and lumbosacral plexuses can also be affected an inflammatory or vasculitic disease processes. In particular, a diabetic lumbosacral radiculoplexus neuropathy (DLRPN; formerly known as diabetic amyotrophy) can involve the lumbar plexus. The pathophysiology is felt to reflect microscopic vasculitis, with ischemia leading to axonal nerve injury. A typical presentation is one of asymmetric pain in an upper lumbar distribution, followed by weakness and atrophy, often disproportionate to the sensory loss. It can be associated with weight loss and autonomic derangements.

Lumbosacral Trunk

The lumbosacral trunk carries the L4 and L5 contributions to the sacral plexus, coursing close to the sacrum as it traverses the pelvic outlet. The structure can be injured during labor and delivery, with risk factors including cephalopelvic disproportion, abnormal fetal presentation, and use of forceps during delivery. The lesion presents as a foot drop, with a broader span of weakness, including knee flexors, hip abductors, and internal rotators. Unlike an L4 radiculopathy, muscles innervated through the lumbar plexus (quadriceps, adductors) are spared with this lesion localization, while L4 contribution to the sciatic nerve (e.g., innervation to anterior tibialis) is affected.

In the pregnancy or delivery context, the lesion is typically caused by compression or stretch. There is often a two-stage recovery, with recovery of the demyelinating component within days to weeks, and a slower recovery for any axonal loss (months to years).



• Fig. 42.20 The lumbosacral plexus.

Sacral Plexus

The sacral plexus is well protected within the bony pelvis. When lesioned traumatically, the mechanism is typically one of high force, often with concomitant pelvic, hip, or sacroiliac joint trauma. Concurrent root avulsion can occur, usually in association with separation of the sacroiliac joint. Incomplete lesions are often initially overlooked because of pain and immobility from the orthopedic injury or from altered mental status from concurrent brain injury or intubation. It is not uncommon to detect these injuries during consultation in the trauma unit or after the patient has begun the rehabilitation course; therefore it is important to have a high index of suspicion for plexus and other peripheral nerve injury in the setting of trauma.

The sacral plexus can also be injured by compressive lesions. Tumor can injure the plexus through compression by mass effect from primary disease or nodal metastases or by direct infiltration of the neural structure as by tumor cells. Commonly involved malignancies include those of the prostate, bladder, uterus, cervix, colon, and rectum. Lymphoma and chordoma can also be implicated. Plexopathy caused by malignancy usually presents with pain as the predominant symptom, with weakness and sensory loss occurring later.

EDX is helpful in confirming a plexopathy and documenting the severity and extent of the lesion. It may also be useful in differentiating between plexopathy, radiculopathy, and individual mononeuropathy. MRI and CT imaging are important to evaluate for injuries in the setting of trauma or compressive lesions in nontraumatic cases.

Regional Nerve Disorders

Neuralgic Amyotrophy

Neuralgic amyotrophy (also known as Parsonage-Turner syndrome and acute brachial neuritis) classically presents with abrupt onset of severe pain often affecting the scapula and upper arm or neck. Within days to about 2 weeks, this pain is followed by weakness of the muscles of the shoulder girdle and upper limb, and paresthesias in a patchy, nondermatomal distribution. An antecedent immunologic event, such as immunization, infection, surgery, or pregnancy in the days or weeks prior, is commonly identified. Mechanical factors may also play a role, with focal strain causing disturbances in the perineural blood barrier. This, in the context of immune system activation, leads to autoimmune damage to nerve segments.⁷²

Neuralgic amyotrophy has a predilection for certain neural structures, likely driven by mechanical and anatomic factors; these include the upper brachial plexus and the suprascapular, long thoracic, axillary, radial, posterior interosseous, anterior interosseous, and musculocutaneous nerves.¹⁷¹ Though typically unilateral, some involvement of the contralateral side is seen in up to one-third of cases. Recent literature estimates the incidence of classic neuralgic amyotrophy to be 1 per 1000 persons per year in the general population.¹⁷⁵ The average age of onset is 40 years, although an autosomal dominant form, or hereditary neuralgic amyotrophy, has an average age of onset around 20 years.²⁰ Neuralgic amyotrophy should be especially considered when patients report severe pain that resolves and is followed by weakness, when sensory abnormalities are vague and patchy, and when the degree of weakness is significant. EDX can assist with localization and exclude differential diagnoses. Sensory nerve conduction studies are often normal, although they can show axonal loss in cases

where patients have distinct sensory fascicular involvement.^{148,149} Needle EMG may demonstrate denervation potentials in the distribution of nerves affected. Ultrasound can aid in the diagnosis when it reveals focal nerve or fascicular swelling, nerve constriction, and fascicular entwinement, although not all cases will show abnormality. These abnormalities may represent a continuum of disease, with nerve swelling occurring first, followed by local adhesion formation and constriction, and finally fascicular torsion.²⁴

Initial treatment of neuralgic amyotrophy focuses on pain management with analgesics. A role for corticosteroids, administered early, in decreasing the immune-mediated damage has been suggested.^{176,179} Shoulder ROM should be maintained, and orthoses should be used as appropriate to assist function. Prognosis is generally good; however, recovery might occur over several years, with 85% recovering over 3 years.¹⁷¹ Residual weakness can lead to musculoskeletal disorders, as commonly occurs when scapular stabilizer weakness predisposes to rotator cuff pain. Accordingly, exercises to build endurance of scapular muscles and activity modification that includes frequent rest breaks in a supine position can be helpful.¹⁷⁸ A potential role for surgical management, especially where recovery is poor and imaging reveals hourglass-like constrictions or fascicular torsion, is emerging and warrants further study.⁸⁸

Diabetic and Nondiabetic Lumbosacral Radiculoplexus Neuropathy

DLRPN is also known by the terms *diabetic amyotrophy* and *Bruns-Garland syndrome*. It classically presents acutely or subacutely in older patients with type 2 diabetes with severe pain that is unilateral or bilateral, and it is often accompanied by weight loss. The pain tends to attenuate over weeks to months, accompanied by marked atrophy of the thigh muscles; it affects particularly the quadriceps, adductors, and iliopsoas, with relative sparing of gluteal and hamstring muscles. Sensory abnormalities are usually seen in the femoral and occasionally saphenous distribution. A proximal neuropathy affecting the thoracic dermatomes can less commonly occur, presenting with severe abdominal or chest pain, associated with weight loss, which may prompt concern for malignancy. The mechanism of injury is thought to be an immune-mediated microvasculitis affecting the nerve roots and plexus. It is usually seen in diabetic patients with preexisting distal neuropathy. A variant can present with acute and debilitating pain in a younger patient with early, mild diabetes (or previously undiagnosed diabetes), with no evidence of distal neuropathy. The presentation can also arise absent diabetes mellitus, in which case it is termed lumbosacral radiculoplexus neuropathy.

Electrodiagnostic studies show axonal, sensorimotor lesions of the nerve supply to affected muscles, often with paraspinal muscle abnormalities, reflective of an underlying radiculoplexus neuropathy in an upper lumbar distribution. Distal nerve conduction studies often show evidence of a concomitant distal neuropathy even in patients without clinical complaints of distal neuropathic symptoms.

Management of DLRPN consists of tight glycemic control and aggressive management of neuropathic pain. Corticosteroids, plasmapheresis, and intravenous immunoglobulin (IVIG) have not shown efficacy.²¹ Even though the condition is extremely debilitating, the prognosis for improvement is good, with recovery progressing for up to 3 years, although it may be incomplete.²⁵

Radiation-Induced Nerve Injury

Radiation-induced nerve injury is a late effect of treatment, presenting 0.5 to 20 years after treatment,⁷⁵ and is dose related. Radiation injury is caused by both direct toxic effects on the axon and damage to the vasa nervorum, causing ischemia. Radiation-induced nerve injury tends to present with numbness, sensory changes, and weakness, without much pain and with gradual progression over many months or years. Electrodiagnostic findings include denervation and myokymic discharges, although these findings are not specific to this disorder.

Management of radiation-induced nerve injury is primarily supportive, with therapy to maintain ROM and to prevent contractures, bracing, adaptive measures, management of lymphedema, and prevention of skin breakdown. Occasionally, nerve transfer or reconstruction has been utilized in select patients with radiation-induced nerve injury to improve function.¹⁷²

Other Pain Syndromes

Several syndromes of limb pain may represent peripheral nerve entrapment syndromes, which can sometimes be confirmed structurally (via imaging or surgical exploration) or physiologically (via EDX). Where diagnostic testing is unrevealing, diagnosis of these entities as mononeuropathies is controversial. Clinical presentations within a given category likely comprise a heterogeneous array of pathologies, ranging from pain syndromes unrelated to peripheral nerves, other musculoskeletal or neurologic diagnoses, nerve irritation without frank damage, to physiologically confirmable

mononeuropathies with focal neurologic deficits. The variability of diagnostic criteria across studies limits systematic study of these disorders. Accordingly, where these pain disorders are suspected, treatment should be tailored based on the presenting symptoms and the presence or absence of objective diagnostic findings. Table 42.6 lists common pain syndromes that may, in some cases, represent underlying peripheral nerve entrapments.

Lower Abdominal Neuropathies

Pain affecting the lower abdomen and genital region can be caused by injury to the ilioinguinal, iliohypogastric, and genitofemoral nerves. These can be diagnostically challenging, and patients often present after an unfruitful gastroenterologic, urologic, or gynecologic workup. Localization is challenging because these nerves have overlapping and variable innervation patterns (Table 42.7). Typically, these neuropathies will present with pain or abnormal sensation in their cutaneous distributions. Weakness of the anterior abdominal muscles can contribute to hernia formation. A Tinel sign can sometimes be elicited near the ASIS. The cremasteric reflex may be lost. Extension of the spine puts tension on these nerves, leading patients to adopt a forward flexed posture.

Frequent areas of entrapment or injury include the psoas muscle, the transverse abdominus muscle, and the inguinal ligament and canal. These injuries are frequently iatrogenic and associated with Pfannenstiel incisions, the placement of lateral trocars in inguinal hernia repairs, and surgeries with low transverse incisions such as abdominoplasty and cesarean section. The nerves may be compressed by space-occupying lesions of the psoas muscle, entrapment

TABLE 42.6 Pain Syndromes That Can Reflect Nerve Entrapment

Syndrome	Possible Nerve and Site of Involvement	Clinical Presentation
Radial tunnel syndrome	Radial nerve or posterior interosseous nerve in the forearm	Dorsal forearm pain and/or paresthesia
Pronator teres syndrome	Median nerve deep to pronator teres	Volar forearm pain and hand paresthesia
Neurogenic thoracic outlet syndrome	Brachial plexus	Upper limb pain and paresthesia
Tarsal tunnel syndrome	Tibial nerve at the medial ankle	Medial ankle pain and plantar foot paresthesia
Piriformis syndrome	Sciatic nerve within or deep to the piriformis muscle	Radiating lower limb pain from buttocks

TABLE 42.7 Anatomic Pattern of the Innervation to the Lower Abdominal Muscles

Nerve	Root Level	Course	Typical Innervation Pattern
Ilioinguinal	L1 (variable T12)	Lateral border psoas, along iliac crest, pierces transversus abdominus near ASIS, accompanies spermatic cord/round ligament in inguinal canal	Upper portion inner thigh, base of penis and upper scrotum, mons pubis and lateral labia
Iliohypogastric	L1 (variable T12)	Passes over lateral border of psoas, along inner surface of ilium, between transversus and external oblique	Posterolateral gluteal and abdomen above pubis
Genitofemoral	L1 and L2	Travels through psoas, femoral branch passes under inguinal ligament, genital branch passé through the inguinal ligament	Round ligament, labia majora, base of scrotum, and cremasteric muscle

ASIS, Anterior superior iliac spine.

in the transversus abdominus near the ASIS, or blunt trauma in this area. The nerves may be entrapped in scar tissue after surgery in the inguinal region. These nerves cannot be directly visualized on imaging studies; however, a compressive space-occupying lesion may be seen if clinically suspected. EMG cannot directly assess these nerves but can rule out a high lumbar radiculopathy or plexopathy. If neuropathic pain cannot be managed conservatively, nerve blocks may be useful both therapeutically and diagnostically. The use of ultrasound or nerve stimulation for localization can increase accuracy.¹⁶⁸ Neurolysis or resection can be successful in recalcitrant cases.

Generalized Peripheral Neuropathies

Generalized peripheral polyneuropathies are characterized by their pattern of involvement and by their underlying pathophysiology. The majority of insidious and gradually progressive peripheral polyneuropathies involve both sensory and motor fascicles in a length-dependent manner, with distal segments most affected. The regional pattern and spread are aptly described as a “stocking” and, eventually, “stocking-glove” distribution. The primary pathophysiology is typically axonal rather than demyelinating. An understanding of these typical features (Fig. 42.21) can aid in recognizing this pattern and can serve as a reference point for understanding the atypical patterns of neuropathy.

Typical Neuropathy: Clinical Presentation

Patients typically describe a gradual onset and distal to proximal progression of numbness, tingling, or sensory loss over months or years. Onset, beginning in the toes or forefeet, is gradual enough that there is usually not a hallmark date or event. Positive symptoms include paresthesias and dysesthesias. Patients may experience negative symptoms (diminished ability to feel), describing their feet as feeling like “wood” or that the plantar surface of their feet feels indistinct when walking. Pain, if present, may be a dull, cramping ache. Functionally, patients may describe difficulty maintaining balance with higher level tasks that rely on proprioceptive input, such as walking on uneven terrain or at night in the dark. Impaired balance (and even falls) can be the initial presenting complaint and has significant and sometimes dangerous functional consequences. Any patient who presents with repeated falls

or an injurious fall in a routine context warrants a careful clinical examination for peripheral neuropathy.

Causes

Common causes of typical axonal neuropathy include diabetes mellitus, alcohol toxicity, hypothyroidism renal failure, and metabolic syndrome. Impaired glucose tolerance and, specifically, the severity and duration of hyperglycemia are major risk factors.^{156,52} Other risk factors include hypertension, elevated triglycerides, tobacco use, obesity, microvascular disease, and cardiovascular disease.¹⁶⁶ Where impaired glucose tolerance is implicated, counseling on the importance and role of lifestyle modifications is important. Strict glycemic control has been shown to reduce the prevalence of polyneuropathy by almost 70% and autonomic dysfunction by over 50%.³⁵ It has also been shown to reverse polyneuropathy.^{37,121}

As neuropathy progresses, more body regions become involved in a length-dependent pattern. When lower limb symptoms have progressed as far proximally as the knees, distal finger involvement may begin. In advanced stages, the intercostal and abdominal nerves can be affected, leading to numbness over the more distal cutaneous regions, near the sternum and central abdomen.

Physical Examination

Exam for suspected neuropathy should incorporate components of the sensory and motor systems and an assessment for function and balance.

Sensory Examination

On physical exam, assessment of vibratory sensation can quantify sensory function. The use of a tuning fork for detection of vibratory loss can detect neuropathy even when monofilament testing is normal.¹¹⁷ The Romberg test can also detect impaired balance related to decreased sensory function. Skin lesions or ulceration may be seen in an insensate foot.

Motor Examination

In early typical neuropathy, the motor examination is normal. Longstanding neuropathy may evidence atrophy of the foot—and eventually hand—intrinsic muscles. Chronic atrophy and weakness lead to foot deformities, such as claw toes, hammertoes, and even deformity and collapse of the midfoot architecture. Functionally, motor deficits are often characterized by easy fatigability. Testing the muscle multiple consecutive times can improve detection of subtle weakness. Occasionally, fasciculations can be seen.

Reflexes

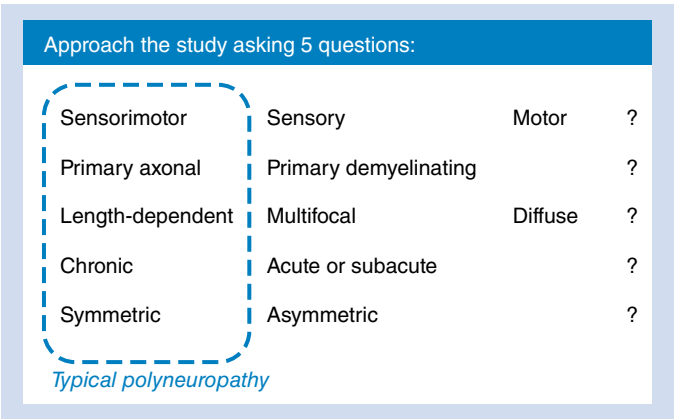
Reflexes are typically impaired in a length-dependent fashion, with the Achilles reflex first involved.

Autonomic Assessment

Autonomic dysfunction can result in decreased sweating and dry, cold feet. Small fiber neuropathy can result in increased blood flow and “hot foot.”

Assessment of Gait and Balance

Unipedal stance time is a sensitive test of balance impairment. Decreased unipedal stance time is associated with a diffuse peripheral neuropathy in young and middle-aged males⁷¹ and is strongly correlated with frontal plane sensory and motor function



• Fig. 42.21 Five key questions in the characterization of peripheral polyneuropathies.

at the ankle among older persons with neuropathy.^{58,157} The patient should be observed walking, looking carefully for increased variability of lateral foot placement and frank crossover steps. These suggest instability during single-limb stance and increase the risk of foot collisions during ambulation. When motor involvement is severe and has progressed proximally to involve the leg muscles, patients may develop a steppage gait.

Challenges in the Elderly Population

Distinguishing the normal distal decline in neuromuscular function associated with aging from that resulting from disease can be difficult. Physical findings predictive of electrodiagnostic confirmation of peripheral neuropathy in this population are loss of the Achilles reflex, inability to perceive at least 8 of 10 movements of 0.5 to 1 cm, or being unable to detect a vibrating 128-Hz tuning fork for at least 8 seconds at the great toe (Box 42.1).¹³¹

Electrophysiologic Assessment

Electrophysiologic examination should be designed to characterize the neuropathy: whether the typical axonal pattern or a less typical subtype. Confirmation and characterization of an atypical subtype is a critical step toward accurate diagnosis and appropriate management.

The electrophysiologic examination is tailored to the patient and designed to assess the following features.

Fiber Type Involvement: Sensory, Motor, or Both?

Sensory and motor nerve fascicles should be studied distally to assess this parameter. In its earliest phase, a typical axonal polyneuropathy may evidence sensory abnormalities alone (subtly decreased SNAP amplitudes). However, in more advanced neuropathy, these should be accompanied by motor abnormalities, possibly reinnervation changes observed in distal muscles on EMG (MUAPs with increased amplitude, duration and/or polyphasia). A pattern of very low amplitude or absent SNAPs combined with robustly normal CMAPs and no EMG abnormalities raises questions of a sensory neuropathy or neuronopathy. These pathophysiologic patterns are important to identify because they may reflect a treatable disorder (such as a nutritional neuropathy), or they may herald a broader disease process important to identify (such as Sjögren syndrome or a paraneoplastic phenomenon). Conversely, the finding of motor-predominant or motor-only abnormalities on EDX study is also notable and may indicate a hereditary disorder or atypical motor predominant process.

Notably, standard EDX testing does not assess small fibers. Pure small fiber neuropathy is likely rare; many patients with small fiber predominant symptoms have been shown to also have large fiber involvement. However, a pure small fiber neuropathy, manifesting primarily with burning pain or changes in

temperature perception, cannot be adequately assessed by standard EDX testing.

Regional Distribution: Is There Length Dependence or a Different Pattern?

Length dependence can be assessed electrophysiologically, comparing distal lower limbs (longest nerves) with distal upper limbs, and, sometimes, comparing relatively more proximal responses with distal ones (e.g., a fibular motor study recording anterior tibialis assesses a more proximal segment than the motor study recording EDB). Needle EMG should include a selection of proximal and distal muscles to assess for a length-dependent gradient of abnormality. Peripheral polyneuropathies in their mildest phases may manifest abnormalities only in distal muscles, such as foot intrinsics.

Notably, muscles affected very chronically by advanced neuropathy may not show fibrillation potentials. In advanced neuropathy, the most distal muscles may reveal decreased insertional activity (reflective of muscle fiber atrophy) in lieu of fibrillation potentials. Fibrillation potentials and reinnervation changes may be observed only in more proximal muscles that still have viable fibers (e.g., the leg segment).

Findings of nonlength dependence, including prominent proximal muscle involvement early in disease course, or marked asymmetry with any given level of length dependence, should raise question of a different pathophysiologic pattern (e.g., nerve by nerve, multifocal, or myotomal).

Axonal or Demyelinating?

Typical peripheral polyneuropathy is predominantly axonal, manifesting low-amplitude SNAPs and CMAPs. Latencies may be moderately prolonged, and conduction velocities may be moderately slowed, commensurate with loss of the fastest fibers. Slowing of CV beyond about 75% of the lower limit of reference values, or prolongation of distal latency beyond about 130% of the upper limit of reference values should raise the question of demyelinating pathophysiology, especially where amplitudes remain moderately well preserved. Demyelination can also be assessed using late responses (e.g., F-waves), which offer increased sensitivity by assessing a longer conduction distance and by inclusion of proximal segments (such as nerve roots, which can be involved early in inflammatory demyelinating neuropathies).

In contrast to axonal neuropathies, acquired demyelinating disorders, in addition to showing slowing of CV and prolongation of distal latencies, often show conduction block, abnormal temporal dispersion, and distal CMAP dispersion. Abnormal temporal dispersion and increased distal CMAP dispersion both reflect desynchrony of conduction within a nerve segment.

On the other hand, hereditary demyelinating neuropathies typically show uniform slowing of conduction velocities, without significantly increased temporal dispersion or conduction block. In these disorders, the myelin conducts diffusely slowly, but without the conduction blocks or desynchrony observed in patchy, acquired demyelination.

Finally, it is notable that some conduction block reflects nodo-paranodopathy rather than segmental demyelination of the internodes. An understanding of this pathophysiology has recently emerged, and it is tied to several atypical neuropathies that involve immune-mediated destruction at the nodal or paranodal regions of nerves. It is possible for a nodo-paranodopathy to manifest with conduction block, absent the temporal dispersion, distal CMAP dispersion, and marked conduction slowing more

• BOX 42.1 Exam Finding Predictive of Electrodiagnostically Confirmed Peripheral Neuropathies

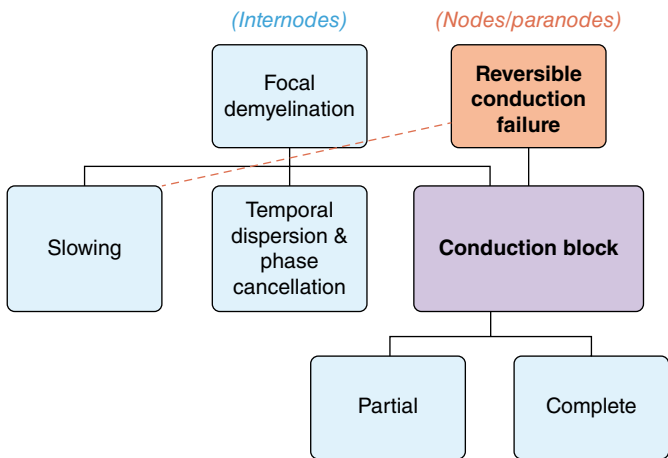
- Loss of Achilles reflex
- Perceive <8 of 10 movements at great toe
- <8-sec detection of 128-Hz tuning fork vibration at great toe

characteristic of intranodal demyelination. Fig. 42.22 presents the findings of focal demyelination and reversible conduction failure, including how both can give rise to conduction block.

Time Course? Is It Chronic, Subacute, or Acute?

The chronicity of a typical axonal peripheral polyneuropathy can be reflected in the MUAP morphologies, which typically show reinnervation reflective of chronicity. Very slowly progressive

neuropathies show MUAPs with high amplitudes and moderately prolonged durations, and sometimes a paucity of fibrillation potentials, reflecting reinnervation keeping pace with denervation. Faster progressing chronic axonal polyneuropathies show MUAPs with varying morphologies, often with more polyphasia and motor unit instability, reflecting reinnervation in its early phases. In the context of neuropathy, absence of reinnervation changes (i.e., decreased recruitment of MUAPs with normal morphologies) could support an acute or subacute underlying process, which is atypical and usually worthy of further evaluation.



• Fig. 42.22 Origin of demyelinating phenomena and conduction block.

Atypical Neuropathy Patterns

Table 42.8 depicts common patterns of atypical neuropathies and neuropathy mimics.

Inflammatory Neuropathies

Guillain-Barré Syndrome

GBS is the most common cause of acute neuromuscular paralysis in the Western world, with an annual incidence of 1 to 2 per 100,000.¹²⁶ GBS is recognized to be a group of related disorders with various subtypes. The most common subtype is the acute inflammatory demyelinating polyradiculoneuropathy (AIDP) accounting for 95% of all cases of GBS in Europe and North America, and it affects both motor and sensory nerves.⁶⁰ Purely axonal forms are rare (<5% overall) but are more common in

TABLE 42.8 Common Patterns of Atypical Neuropathies and Neuropathy Mimics

Atypical Pattern	Hallmark Features	Examples
Symmetric, demyelinating, with some proximal features	Weakness without significant muscle atrophy	Chronic inflammatory demyelinating polyradiculoneuropathy
Acute, symmetric, demyelinating, with some proximal features		Acute inflammatory demyelinating polyradiculoneuropathy
Acute, axonal		Axonal variants of Guillain-Barré syndrome acute motor axonal neuropathy, acute motor and sensory axonal neuropathy, many toxic neuropathies
Multifocal, demyelinating		Multifocal motor neuropathy, multifocal acquired demyelinating sensory and motor neuropathy
Multifocal, axonal, rapid		Vasculitic mononeuropathy multiplex
Motor predominant		Some toxins, chemotherapeutics (vincristine)
Sensory predominant, length dependent		Many nutritional neuropathies (vitamin deficiencies)
Sensory predominant, nonlength dependent	Can have pseudoathetosis	Sensory neuronopathy (toxic, immune mediated [associated with Sjögren syndrome], paraneoplastic, cobalamin deficiency)
Hereditary	Less pain, fewer positive symptoms. Children with neuropathy often present when they are not meeting normal developmental milestones. Older children may be noted to be clumsier than their peers or have difficulty with coordination and participation in normal childhood activities.	Hereditary motor and sensory neuropathy, various types
Small fiber	Often with burning pain or dysesthesia; alterations of pain and temperature sensation	Diabetes mellitus, amyloidosis, toxins, and inherited conditions

Asia and South America, making up 30% of the cases in those populations.¹¹⁹ Axonal forms include acute motor axonal neuropathy (AMAN) and acute motor and sensory axonal neuropathy (AMSAN). The Miller Fisher variant is characterized by a triad of ataxia, areflexia, and ophthalmoplegia, although bulbar abnormalities, ptosis, facial weakness, and pupillary dysfunction can be seen (Table 42.9).⁴⁷

GBS presents with progressive onset of limb weakness, both proximally and distally, that is typically symmetric, with strength nadir in 2 to 4 weeks. Reflexes are lost early in the disease, although reflexes can be retained or even brisk with axonal forms. Sensory loss is variable. The cranial nerves can be affected, with facial palsy and bulbar weakness. Muscles of respiration are frequently affected, leading to decline of vital capacity and requiring ventilatory support in 25% of cases. Autonomic dysfunction can result in tachycardia, hypertension, and cardiac arrhythmias. Pain can be significant and can precede the onset of clinical weakness. In the acute stages, the pain is described as deep and aching, affecting the back, buttocks, and posterior thighs, caused by nociceptive pain from inflammation. Later, neuropathic pain can be described, with degeneration and regeneration of sensory nerves. Fatigue is common and persists even after good neurologic recovery. Amantadine and other pharmacologic agents have not helped, but bicycle exercise training has been shown to reduce fatigue and improve functional outcome and quality of life.⁵¹

Two-thirds of patients with GBS have an antecedent infection within 3 weeks of onset of weakness. The most commonly identified infectious agent is *Campylobacter jejuni*. Other causative agents include cytomegalovirus (CMV), Epstein-Barr virus, *Mycoplasma pneumoniae*, and *Haemophilus influenzae*.⁶¹ Although studies of the swine flu vaccine during the 1976 epidemic demonstrated an increased risk of one excess case of GBS for every 100,000 people vaccinated,¹⁴⁴ there is little evidence to link GBS with other vaccines, including newer influenza vaccines.⁵⁹ The strong association with antecedent infection suggests that at least in some cases there is an immune response to these infectious agents that triggers damage to peripheral nerves. The preceding

infectious agent is also associated with the subtype of GBS that develops. Nerves are initially affected at the root level, followed by distal segments and the intervening areas later in the course of the disease.⁹

Required clinical features for the diagnosis of GBS are progressive weakness in both arms and legs, and areflexia. Supportive features include progression of weakness over days to 4 weeks, symmetry of symptoms, mild sensory symptoms, cranial nerve involvement, autonomic involvement, pain (which can be prominent), high protein concentration in the cerebrospinal fluid (CSF), and typical electrodiagnostic findings. A number of features raise doubts about the diagnosis of GBS (Box 42.2).

Lumbar puncture is routinely performed in the workup of patients presenting with rapidly progressive weakness. The characteristic finding in GBS is an elevated CSF protein with normal white blood cell count ($<5\text{--}10 \times 10^6$ cells/L), referred to as *albuminocytologic dissociation*. EDX is the most useful confirmatory test for GBS and differentiates between the more common demyelinating form (AIDP) and the axonal forms (AMAN, AMSAN). Nerve conduction studies can be normal in early AIDP, and electrodiagnostic findings usually lag behind clinical severity. As the nerve roots are typically affected first, prolongation of the F-wave latencies is the earliest abnormality seen, as they measure conduction in the proximal nerve segments.² Nerve conduction studies demonstrate a demyelinating pattern, with motor involvement occurring earlier and more extensively than sensory abnormalities. Sensory changes tend to be patchier in distribution, and median and ulnar sensory responses tend to be affected more than the sural response. Temporal dispersion and partial conduction block can be seen between proximal and distal stimulation sites. Needle examination shows decreased firing of motor units on maximal contraction. An axonal variant of GBS is suggested if there is significant reduction in CMAP amplitudes and early and significant findings of abnormal insertional activity on needle examination. CMAP amplitudes that fall below 20% of the lower limit of normal predict poor functional outcome.¹⁰⁶ Nerve ultrasound may demonstrate diffuse or segmental nerve enlargement and may be useful in the diagnosis of atypical presentations.¹⁵¹

Management of GBS requires a multidisciplinary approach consisting of supportive care, disease-modifying treatment, and rehabilitation. About one-third of those affected require intensive care unit admission.⁷⁷ Mortality for GBS is 10%, with infections, pulmonary embolus, and cardiac arrhythmia the most common causes of death. Deep venous thrombosis prophylaxis should be initiated, particularly in nonambulatory patients. Vital capacity should be sequentially monitored every 2 to 4 hours initially. Mechanical ventilation should be initiated if vital capacity falls below 20 mL/kg.⁹² Autonomic involvement can result in labile

TABLE 42.9 Variants of Acute Inflammatory Demyelinating Polyradiculoneuropathy

	Clinical Features	Antibodies
Acute inflammatory demyelinating polyradiculoneuropathy	Symmetric weakness of all limbs Hyporeflexia, areflexia Mild sensory symptoms Respiratory involvement Facial nerve palsy	Anti-GM1
Axonal (AMAN, ASMAN)	Symmetric weakness Reflexes may be normal or increased Variable sensory symptoms Respiratory involvement	Anti-GM1, GD1a, Ga1Nac-GD1a
Miller Fisher	Ophthalmoplegia Ataxia Areflexia Minimal weakness	Anti-GQ1b

AMAN, Acute motor axonal neuropathy; AMSAN, acute motor and sensory axonal neuropathy.

• BOX 42.2 Features Not Consistent With Guillain-Barré Syndrome

Severe pulmonary involvement early, without significant weakness
Bowel or bladder involvement at onset
Severe sensory symptoms with minimal weakness
Fever
Well-delineated sensory level
Slow progression, limited weakness, no respiratory involvement
Marked asymmetry
Increased mononuclear cell in cerebrospinal fluid (CSF) ($>50 \times 10^6/L$)
Polymorphonuclear cells in CSF

blood pressure and cardiac arrhythmias. Serious bradyarrhythmias can develop, requiring atropine or transcutaneous pacemaker.¹⁹¹ Gastrointestinal involvement is the most common autonomic abnormality with ileus occurring in 15% of those with GBS.⁵ In some patients, pain can be difficult to treat, requiring opioids, gabapentin, and other agents for neuropathic pain. Prevention of pressure ulcers with proper positioning and turning, as well as ROM and splinting to prevent contractures, should be instituted. Swallowing function should be monitored, particularly in those with signs of facial or bulbar involvement.

Both plasma exchange (PE) and IVIG have been proven to be of benefit in altering the course of GBS.⁵⁶ IVIG has been shown to be as effective as PE⁴⁵ and is usually the preferred treatment because of greater convenience and availability, and fewer complications. Combining PE and IVIG has not been shown to be superior to monotherapy.¹²³ Forty percent to 50% of patients do not improve with either treatment; however, there is no evidence of benefit to repeating treatment or changing to an alternate treatment.¹⁸¹ Eight percent to 16% of patients develop neurologic deterioration after initial improvement after treatment.⁸⁷ It is common practice to administer a second course of IVIG in these patients. Oral or intravenous steroids have not been shown to be effective in altering the course or functional outcome of GBS.⁶⁹

Rehabilitation of patients begins in the acute phase with prevention of contractures with proper ROM, positioning, and splinting. Mobilization as soon as the patient's medical condition permits is essential in minimizing the deleterious effects of bedrest. Intensive pulmonary hygiene reduces the risk of atelectasis and pneumonia. Formal inpatient rehabilitation is initiated when immunotherapy has been completed, and the neurologic deterioration has reached its nadir. Autonomic and pulmonary status should be stabilized, and the patient should either be weaned from the ventilator or transferred to a facility that has ventilator capabilities. Orthostatic hypotension is a common problem, and use of a tilt table may be necessary before initiating standing and gait training. Lower extremity orthoses are often needed when initiating ambulation. Functional and weight-training exercises are used for upper extremity strengthening, and assistive devices are employed for self-care activities. Exercise in GBS patients should be nonfatiguing because patients can be prone to declines in strength if exercise is advanced too rapidly (particularly in muscles with less than antigravity strength).

Prognosis of GBS is generally good, especially given the often-marked initial degree of weakness. Twenty percent of patients will have persistent disability, remaining nonambulatory or requiring an assistive device to walk at 6 months, and many have residual fatigue and pain.⁷⁰ Poor outcome is associated with advanced age, male sex, axonal involvement, antecedent diarrhea, and CMV infection.¹²⁶ Relapse is rare at a rate of 3% to 5%.⁶¹

Chronic Inflammatory Demyelinating Polyradiculoneuropathy

CIDP is an immune-mediated disorder of the peripheral nervous system. Classically, CIDP is a symmetric neuropathy. Unlike typical polyneuropathies, which are length dependent, CIDP presents with both proximal and distal involvement. Variants include Lewis-Sumner syndrome, which has a multifocal presentation variably affecting motor and sensory nerve, and purely sensory forms, as well as forms associated with immunoglobulin G (IgG) and IgA paraproteins, and central nervous system (CNS) demyelination. A number of conditions also are important to

differentiate from CIDP variants, especially because they respond differently to treatment. These include multifocal motor neuropathy and IgM-related neuropathies, such as POEMS (*p*olyneuropathy, *o*rganomegaly, *e*ndocrinopathy, *M* protein, and *s*kin changes).

CIDP typically progresses over at least 2 months, which differentiates it from GBS. Motor symptoms generally predominate, and both proximal and distal muscles are affected. Because CIDP is a primarily demyelinating disorder, muscle atrophy is minimal. Muscle stretch reflexes are reduced or absent. Cranial involvement occurs in 10% to 20% of cases, and a small number of patients have painful dysesthesias. Sensory involvement preferentially affects large fiber nerves, consequently preferentially affecting vibration and proprioception. Sensory symptoms progress distally to proximally, although hand involvement is often perceived as early as that in the feet. Respiratory and cranial nerve involvement is rare. CIDP typically shows a pattern of recurrence and remittance (polyphasic) in younger patients.¹¹² Older adults are more likely to have a progressive form of CIDP (monophasic). A small number (20%) may have a subacute onset (4 to 8 weeks) or an acute GBS-like symptom onset before it eventually evolves into a relapsing form. Unlike AIDP, an antecedent infection is typically not identified.⁴⁰

Electrodiagnostic studies are critical for confirmation of CIDP. Nerve conduction studies show a pattern of primary demyelination affecting multiple segments. Findings consistent with demyelination include conduction block, conduction slowing that is greater than can be explained by axon loss, prolonged F-wave and H-reflex latencies, and temporal dispersion of CMAP between proximal and distal stimulation points. A nonuniform pattern is also suggestive of CIDP. Upper limb sensory responses are often more affected than the sural responses. Because CIDP is multifocal and affects motor nerves and proximal segments predominantly, sural biopsy can be normal and therefore is not routinely done. However, sural nerve biopsy, when affected, shows demyelination or findings pointing to other disorders that can mimic CIDP, such as vasculitis or amyloidosis. As in AIDP, a lumbar puncture will show albuminocytologic dissociation in 90% of patients. MRI with gadolinium of the brachial and lumbar plexuses and cauda equina can reveal enhancing or enlarged nerves.

The European Academy of Neurology/Peripheral Nerve Society proposed diagnostic criteria for "typical" CIDP and "CIDP variants" based on review of available evidence and expert consensus.¹⁷⁷ Diagnostic criteria include both clinical (Box 42.3) and electrodiagnostic components, which can be used to identify probable or possible disease. CIDP and CIDP variants are diagnosed based on the presence of clinical features, motor and sensory nerve conduction criteria, and supportive features.

The 2021 consensus criteria¹⁷⁷ should be referenced to guide electrophysiologic workup and diagnostic impression. Broadly, the threshold of abnormality to establish systemic demyelinating disease exceeds that of identifying a single demyelinating feature that could be explained by mononeuropathy alone. In particular, diagnosis of CIDP requires demyelinating range findings across multiple nerves, not exclusively at common entrapment sites (e.g., median nerve at the wrist) and not inclusive of conduction blocks in segments that often exhibit such findings (e.g., tibial nerve in the leg). Motor electrodiagnostic criteria for CIDP include the following in at least two nerves:

- Motor distal latency prolongation $\geq 50\%$ of the upper limit of normal
- CV $< 30\%$ of the lower limit of normal

• BOX 42.3 Clinical Criteria for CIDP

Typical CIDP

All the following:

- Progressive or relapsing, symmetric, proximal and distal muscle weakness of upper and lower limbs, and sensory involvement of at least two limbs
- Developing over at least 8 weeks
- Absent or reduced tendon reflexes in all limbs

CIDP Variants

One of the following, but otherwise as in typical CIDP (tendon reflexes may be normal in unaffected limbs):

- Distal CIDP: distal sensory loss and muscle weakness predominantly in lower limbs
- Multifocal CIDP: sensory loss and muscle weakness in a multifocal pattern, usually asymmetric, upper limb predominant, in more than one limb
- Focal CIDP: sensory loss and muscle weakness in only one limb
- Motor CIDP: motor symptoms and signs without sensory involvement
- Sensory CIDP: sensory symptoms and signs without motor involvement

CIDP, Chronic inflammatory demyelinating polyradiculoneuropathy.

From Van den Bergh PYK, van Doorn PA, Hadden RDM, et al. European Academy of Neurology/Peripheral Nerve Society guideline on diagnosis and treatment of chronic inflammatory demyelinating polyradiculoneuropathy: report of a joint task force-second revision [published corrections appear in *J Peripher Nerv Syst.* 2022;27(1):94 and *Eur J Neurol.* 2022;29(4):1288]. *J Peripher Nerv Syst.* 2021;26(3):242-268.

- F-wave latency $\geq 20\%$ of the upper limit of normal if CMAP amplitude is well preserved; $\geq 50\%$ of the upper limit of normal if CMAP amplitude is $< 80\%$ of the lower limit of normal
- Motor conduction block of $\geq 30\%$
- Abnormal temporal dispersion with $\geq 30\%$ duration increase for proximal waveform

The following features are also diagnostic in one nerve, in the presence of another demyelinating feature in a different nerve:

- Distal CMAP dispersion
- Absent F-wave despite CMAP amplitude $\geq 20\%$ of the lower limit of normal

In addition, sensory nerve conduction abnormalities should be present.

Similar to GBS, CIDP is responsive to IVIG and PE. CIDP also responds to corticosteroids, but GBS does not. IVIG and PE can be used to bring rapid improvement in symptoms in severe attacks; however, they rarely result in complete remission, and most patients require repeated treatment at intervals of 2 to 6 weeks to prevent relapse.⁹⁴ IVIG is usually favored over PE. Corticosteroids are more likely than IVIG or PE to produce remission of disease in the more common demyelinating form; however, corticosteroids are not effective and can even be detrimental to pure motor variants of CIDP. Metanalysis showed that the children have an 80% to 100% response rate to standard treatments of IVIG and corticosteroids, and complete functional recovery is seen in most children.¹⁰³ After stabilization of initial onset or recurrence of CIDP, appropriate physical and occupational therapy, orthotic management, and treatment of neuropathic pain should be addressed. A course of inpatient rehabilitation might be appropriate after severe episodes.

Infectious Neuropathies

Infections are the most common cause of neuropathy in the world. Historically, leprosy has been the most common worldwide cause of neuropathy. Although the incidence of leprosy had

declined by 50% between 2003 and 2009,¹⁹² it remains a significant problem in several countries. HIV is responsible for a number of neuropathic syndromes, and the agents used to treat HIV can also be responsible for polyneuropathy. Chronic hepatitis C viral infection is a growing global health issue that is associated with polyneuropathy. Lyme disease, a tickborne infection common in the United States and Europe, is also responsible for polyneuropathy. Infection can cause neuropathy either through the inflammatory reactions induced by the infectious agent or through the body's immune reaction to the organism.

Leprosy (Hansen Disease)

Leprosy is caused by *Mycobacterium leprae*, which is found primarily in tropical and subtropical developing countries. There are three clinical presentations, which are determined by the host immune response to the bacillus: tuberculoid, lepromatous, and borderline leprosy. Tuberculoid leprosy causes single or multiple well-circumscribed cutaneous lesions. High levels of cell-mediated immunity kill the bacilli and cause focal destruction of the nerves that travel through the granulomatous tuberculoid lesions. The most commonly involved nerves are the ulnar, median, common fibular, facial, superficial radial, digital, posterior auricular, and sural nerves.¹⁴¹ The bacillus tends to proliferate in parts of the body with areas of lower temperature. The lepromatous form occurs when there is deficient cell-mediated immune response. No granulomas are formed, and nerve destruction occurs from direct nerve invasion of the bacillus, particularly affecting the Schwann cell. Nerve destruction is typically diffuse and symmetric. In both forms, the predominant finding is sensory loss affecting nerves in areas of lower temperature such as extensor surfaces, earlobes, and over the maxilla. The soles of the feet and the palms are typically spared. Motor weakness develops late and only after there is significant sensory loss. Leprosy is treated with dapsone, which itself can cause neuropathy. Dapsone primarily causes a progressive motor neuropathy that can preferentially affect the hip girdle muscles or both hand and foot muscles in a symmetric fashion.

Lyme Disease

Lyme disease is a tickborne illness that is caused by the spirochete *Borrelia burgdorferi*. There are three stages of disease: early infection that characterized by a local erythematous lesion (erythema migrans), disseminated infection, and late-stage infection. Meningitis is the most common neurologic manifestation in Lyme disease. Peripheral nerve abnormalities vary with the stage of the disease. Cranial mononeuropathies can be seen in the second stage, affecting the facial nerve in up to 80%.¹⁶⁰ Radiculoneuritis also is reported that progresses to a mononeuritis multiplex pattern (i.e., plexopathy). The lower limbs are affected more than the upper limbs. Although much less common now because of better detection, in the late stage of untreated Lyme disease about half of the patients develop a distal symmetric neuropathy, presenting with a stocking glove distribution of paresthesia and sensory loss. Other presentations can also occur, including mononeuropathies, acute polyradiculoneuropathies, and asymmetric distal neuropathies. Although the clinical presentation of peripheral nerve dysfunction is broad, the underlying injury is an asymmetric axonal neuropathy. Treatment of Lyme disease with neurologic involvement consists of intravenous ceftriaxone for 2 to 4 weeks. Improvement in acute neurologic symptoms is fairly good with antibiotic treatment; however, chronic neuropathies respond more slowly and can result in residual deficits.¹⁶⁷

HIV-Associated Neuropathies

HIV infection is associated with a range of neurologic disorders that can occur at any stage of the disease. Early in HIV infection, these neurologic insults are caused by immune dysregulation. AIDP can occur at the time of seroconversion, whereas CIDP manifests later, usually when CD4 counts are less than 50 cells/mm.⁵⁰ The clinical features are identical to those in HIV-negative patients and are treated in the same manner and respond to treatment similarly to patients without HIV. Immune-mediated vasculitic neuropathies, cranial mononeuropathies, brachial plexopathies, and multiple mononeuropathies are also seen early in HIV infection. In mid- to late-stage infection, neuropathies are usually related to the HIV replication, most commonly a sensory polyneuropathy or an autonomic neuropathy. Later-stage disease brings neuropathies associated with opportunistic infection, such as CMV, syphilis, and zoster. Neuropathies associated with malignancy and nutritional deficiency are also seen. Drugs used to treat HIV infection are also known to cause neuropathy. As an example, nucleoside reverse transcriptase inhibitors cause a distal sensory neuropathy that is clinically and electrodiagnostically identical to that associated with primary HIV infection. The use of antiretroviral therapy (ART) can also lead to exaggerated immune response to an already present opportunistic infection such as *M. leprae*. This is known as the immune reconstitution inflammatory syndrome (IRIS) and is associated with a CD4 count lower than 50 cells/ μ L at the initiation of ART. IRIS is associated with greater morbidity and mortality and is usually managed by the discontinuation of ART and the use of corticosteroids or thalidomide.⁹⁹ Diffuse infiltrative lymphocytosis syndrome (DILS) can be seen in HIV-positive patients with a persistent CD8 hyperlymphocytosis and Sjögren-like syndrome. In these individuals, nerve biopsy will show CD8+ infiltration causing a painful sensorimotor axonal neuropathy that responds to ART and steroids. It is believed to be an exaggerated immune response to HIV (Box 42.4).

A symmetric distal sensory polyneuropathy can be seen either because of HIV distal sensory polyneuropathy (HIV-DSP) or antiretroviral drug toxicity distal sensory polyneuropathy (ARV-DSP). Since the introduction of highly active ARV treatment, the incidence of clinical DSP is estimated to be 30%, although on autopsy close to 100% of patients demonstrated histologic evidence of DSP.⁶⁸ Risk factors for HIV-DSP are older age, a CD4 count less than 50 cells/mm³, coexisting diabetes mellitus or nutritional deficiency, and use of neurotoxic drugs and alcohol.²⁷ DSP causes distal axonal degeneration of small unmyelinated fibers. Pain and paresthesias progress in a symmetric “stocking glove” distribution. Proprioception and strength are relatively preserved. Achilles reflexes are absent or decreased. Electrodiagnostic studies, which are relatively insensitive to small fiber dysfunction, might be normal or demonstrate a length-dependent sensory axonal neuropathy. Needle EMG examination is typically normal, although it occasionally can show chronic denervation and reinnervation in foot intrinsic muscles. Skin biopsy with epidermal nerve fiber density will detect a small fiber neuropathy and correlates with severity of painful symptoms.¹⁵⁴ It is important to differentiate between HIV-DSP and ARV-DSP because discontinuation of the drug can reverse ARV-DSP. Differentiation is usually made on historical grounds based on a close temporal relationship between drug initiation and the onset of symptoms. Discontinuation or reduction of dose leads to improvement within 8 weeks, although there can be an initial period of worsening after stopping the drug.⁵³

• BOX 42.4 Neuropathies Associated With HIV-1

Immune Mediated

- Acute inflammatory demyelinating polyradiculoneuropathy
- Chronic inflammatory demyelinating polyradiculoneuropathy
- Cranial mononeuropathies
- Vasculitic neuropathy
- Brachial plexopathy
- Multiple mononeuropathies
- Immune reconstitution inflammatory response
- Diffuse infiltrative lymphocytosis syndrome

HIV-1 Replication Associated

- Distal sensory polyneuropathy
- Autonomic neuropathy

Drug Related

- Antiretroviral-nucleoside reverse transcriptase inhibitors (ddI, ddC, d4T)
- Other drugs (ethambutol, vincristine, isoniazid)

Opportunistic Infection/Malignancy

- Cytomegalovirus (CMV) progressive polyradiculopathy
- CMV mononeuritis multiplex
- Syphilitic polyradiculopathy
- Tuberculous polyradiculomyelitis
- Zoster
- Lymphomatous polyradiculopathy

Nutritional

- AIDS cachexia neuropathy
- Nutritional deficiency (vitamins B6, B12)

Typical distal symmetric polyneuropathy is more common in adult patients with HIV-1 infection than in children. Polyneuropathy in children with HIV has not been extensively studied but occurs often in the form of AIDP at the time of seroconversion as an immune reconstitution phenomenon and in relation to secondary infection with CMV.¹⁹⁰

CMV infection in HIV can cause a severe, rapidly progressive polyradiculomyelopathy, which is a true neurologic emergency. It presents with symptoms of cauda equina syndrome developing over days to weeks. It initially presents with low back pain and unilateral lower limb radiation with urinary incontinence, followed quickly by progressive leg weakness and saddle anesthesia. If untreated, flaccid paraplegia develops with death within weeks. CSF reveals CMV infection by culture, polymerase chain reaction assay, elevated protein, low glucose, and polymorphonuclear pleocytosis. Electrodiagnostic studies are consistent with an axonal loss involving the lumbosacral roots. MRI may show enhancement of the lumbosacral roots. CMV-associated polyradiculoneuropathy is treated with ganciclovir or other antiviral agents. A similar clinical picture can be seen in syphilitic, tubercular, or cryptococcal infection, lymphoma, and concurrent HTLV1 infection as well as in IRIS and DILS.⁵⁰

Hepatitis C Virus

Hepatitis C virus (HCV) affects 2% to 3% of the world's population,⁹¹ and polyneuropathy is the most common neurologic complication. Approximately 15% of those with HCV will have polyneuropathy detected on standard nerve conduction studies, although one-third have subclinical disease.¹⁴² Small fiber neuropathy has a prevalence of over 40% in the HCV-infected

population.¹⁹³ Almost 80% of HCV patients have mixed cryoglobulinemia.^{110,111} An acute sensorimotor mononeuritis multiplex occurs with the finding of ischemic changes of nerve caused by vasculitis. In those without cryoglobulins, autoimmune mechanisms may be the pathologic mechanism of injury. Sensory neuropathy is the most common form of polyneuropathy associated with HCV. Both large- and small-fiber variants can be seen. Rare variants include demyelinating neuropathies, pure motor, and autonomic forms. A polyarteritis nodosa-like systemic vasculitis can present with sudden onset of multifocal mononeuropathy with severe motor deficit and painful sensory symptoms.⁸⁰ CIDP is also seen in association with HCV. CIDP can also be seen associated with HCV and responds to treatment with immunoglobulins.²³

Toxic Neuropathies

Although toxins are a less common cause of polyneuropathy, prompt diagnosis and identification of toxic insult can lead to improvement in withdrawal of the offending agent. Recognition of a source of chronic exposure in the household or workplace can also protect others at risk. Toxic exposure can be (1) occupational, as occurs with industrial solvents and glues; (2) environmental, as in lead exposure; (3) secondary to abuse, as seen in alcohol abuse or the “huffing” of household solvents; (4) the

result of intentional poisoning (suicide or homicide attempts); or (5) iatrogenic from medications used to treat a variety of diseases (Table 42.10).

As with any neuropathy evaluation, a careful exposure history should be obtained, which includes understanding the patient's workplace and any chemicals there that may lead to exposure. Recreational activities are another source of exposure, particularly ones that involve glues or solvents, or exposure to pesticides. In the case of children, especially those with pica, exposure to toys containing lead-based paint may be a particular hazard. A recreational drug history may reveal heavy alcohol use or intentional inhalation of glues and aerosols (huffing). Toxic neuropathies frequently present with other systemic effects, ranging from subtle to severe and life threatening. Gastrointestinal complaints are seen in arsenic, lead, and thallium poisoning. Cutaneous findings can be seen, such as Mees lines in the fingernails in lead and arsenic poisoning; gingival abnormalities in phenytoin, lead, and mercury exposure; and irritant dermatitis with acrylamide, thallium, and toluene exposure.⁹⁵ Cardiovascular, hepatic, and renal abnormalities are associated with many toxic exposures. Exposure to organophosphates causes symptoms of cholinergic excess, including bradycardia, sweating, excess salivation, miosis, bronchospasm, and diarrhea. The CNS and the peripheral nervous system can be affected. Chronic alcohol abuse can cause midline cerebellar degeneration and a triad of ataxia, dementia, and ophthalmoplegia

TABLE 42.10 Features of Toxic Neuropathies

Agent	Clinical Symptoms	Systemic Effects	Diagnostic Findings
Sensory and Sensory > Motor, Axonal			
Nitrofurantoin	Pain, paresthesias, gait dysfunction, can progress to profound quadriparesis		Electromyography (EMG): absent or severely decreased sensory and motor amplitudes, relatively preserved conduction velocity (CV; when obtained), significant neuropathic changes on needle electrodiagnostic examination (NEE)
Ethyl alcohol	Distal symmetric sensory loss, affects hands late	Dementia/ophthalmoplegia/ataxia (Wernicke syndrome), cirrhosis, cerebellar degeneration	
Colchicine	Mild decrease in touch, vibration, and proprioception, pinprick and temperature sensation less affected, proximal weakness (caused by severe myopathy)		EMG: ↑ sensory and motor amplitudes, normal distal latency; myopathic changes proximally on needle examination
Amitriptyline	Stocking glove distribution (rare)		Significantly decreased sensory nerve action potential (SNAP), motor CV 80% or lower limit of normal (LLN), neuropathic changes on NEE
Lithium	Distal sensory loss, weakness	Tremor, confusion, seizure	Absent SNAP and compound muscle action potential (CMAP) in lower extremity, markedly reduced amplitude in upper extremity with normal CV
Phenytoin	Mild sensory symptoms in stocking glove distribution, with toxicity numbness, tingling, gait difficulties with mild weakness	Gingival hyperplasia, cerebellar ataxia	Mildly reduced SNAP, mildly reduced motor CV, NEE normal or mildly neuropathic distally
Cisplatin	Affect dorsal root ganglion and large myelinated sensory nerves, numbness, painful paresthesias, ↓ proprioception, ↓ deep tendon reflexes (DTR)	Nephrotoxicity, ototoxicity, nausea, vomiting, myelosuppression	Reduced or absent SNAP in upper and lower extremities, mildly prolonged latency, motor conduction and NEE normal

Continued

TABLE 42.10 Features of Toxic Neuropathies—cont'd

Agent	Clinical Symptoms	Systemic Effects	Diagnostic Findings
Pyridoxine	Gait dysfunction, Lhermitte sign, decreased/absent DTR, normal strength, affects dorsal root ganglion		Significantly decreased to absent SNAP with normal CV, CMAP normal to borderline low, NEE-decreased recruitment
Thallium	Small fiber predominant, painful dysesthesias, preserved reflexes	Gastrointestinal symptoms, cardiac and respiratory failure, renal insufficiency, encephalopathy, alopecia, and Mees lines (late findings)	EMG: can be normal initially as a result of small fiber loss, sensory axonal findings late
Motor and Motor > Sensory, Axonal			
Lead	Early upper extremity weakness, preferential involvement of radial nerve, atrophy of hand and foot intrinsic, sparing all sensory modalities	Abdominal pain, constipation, gingival discoloration	Nerve conduction study: mildly decreased amplitude and CV, NEE with neuropathic motor unit changes and fibrillation potentials and positive sharp waves
Organophosphates	Length-dependent weakness followed by proximal weakness, spasticity and upper motor neuron signs resulting from corticospinal tract involvement	Symptoms of cholinergic excess, myasthenia-like syndrome, pulmonary edema, dermatitis, corticospinal tract dysfunction	EMG: sensorimotor axonal neuropathy with conduction slowing
Vincristine	Early pain and small fiber sensory loss, distal symmetric weakness	Autonomic dysfunction	EMG: ↓ motor and sensory amplitude
Motor and Motor > Sensory With Conduction Slowing			
Amiodarone	Symmetric, distal sensorimotor neuropathy Rapidly progressive motor predominant (mimic acute inflammatory demyelinating polyneuropathy [AIDP])	Action tremor, myopathy, optic neuropathy, encephalopathy, basal ganglia dysfunction, pseudotumor cerebri	EMG: varies from predominant conduction slowing to predominant sensory amplitude reduction Sural nerve biopsy: either axonal or demyelinating findings
Arsenic	High dose: mimics AIDP, flaccid quadriplegia, facial weakness, respiratory failure Chronic: dermatologic symptoms first, length-dependent painful neuropathy, fewer motor symptoms	Gastrointestinal symptoms, hyperpigmentation, Mees lines, hyperkeratosis, desquamation of hands and feet, hepatomegaly, cardiomyopathy, renal failure, anemia, basophilic stippling of red blood cells	EMG: ↓ amplitude, motor > sensory, borderline low CV, prolonged F-waves, partial conduction block Laboratory: urine arsenic >25 µg/24 h, hair and nail levels (chronic)
Hexacarbon “huffing”	Length-dependent distal sensory loss, weakness, atrophy, absent Achilles reflex, cranial nerve involvement Massive exposure: more cranial and motor involvement Chronic: degeneration of distal corticospinal tract and dorsal columns	Autonomic dysfunction, impairment of color vision (chronic)	Marked decreased SNAP, mildly prolonged latency, significantly decreased CMAP with CV 70–80% of LLN, NEE with fibrillation potential and positive waves, fasciculations
Mononeuritis Monoplex			
Dapsone	Motor predominant, asymmetric, occurring with chronic (years of) exposure		Asymmetric, motor without conduction slowing or increased latencies
Trichloroethylene	Cranial mononeuropathies, with ptosis, facial and bulbar weakness, extraocular muscle dysfunction, trigeminal involvement	Dermatitis, cirrhosis, cardiac failure	Normal standard limb peripheral nerve conduction studies

known as Wernicke syndrome. Some toxins produce different syndromes depending on whether the exposure is high dose and acute or low dose and chronic. Arsenic presents with a syndrome of flaccid quadriplegia that resembles AIDP with acute high-dose exposure.⁵⁵ Chronic arsenic poisoning presents with more dermatologic abnormalities and a painful length-dependent neuropathy with less motor involvement.⁴⁴ Typical electrodiagnostic classifications of toxic polyneuropathies are summarized in Table 42.11.

Lead is an important cause of environmental poisoning. Lead neuropathy will have different presentation depending on the length of exposure. Lead-induced neuropathy after about 5 years of sustained exposure usually presents as a subacute motor neuropathy (with minimal sensory involvement) with segmental demyelination and axonal degeneration, with predominantly upper limb involvement at onset (wrist drop) but also involving the lower extremities in due course (foot drop). However, a more

TABLE 42.11 Typical Electrodiagnostic Classifications of Toxic Neuropathies

Pattern	Causes
Distal sensorimotor, axonal	Metronidazole, lithium, nitrofurantoin, phenytoin, vincristine, hydralazine, isoniazid Alcohol abuse Chronic arsenic poisoning Carbon monoxide Mercury
Motor, axonal	Organophosphates Vincristine
Sensory, axonal	Cisplatin, nitrofurantoin, thalidomide, pyridoxine (vitamin B6) Thallium Alcohol
Motor, with conduction block	Acute arsenic poisoning Amiodarone hexacarbons
Sensory greater than motor, with conduction block	Saxitoxin (shellfish), tetrodotoxin (puffer fish)
Multiple mononeuropathies	Dapsone Trichloroethylene

indolent and slowly progressive sensorimotor neuropathy may be seen after 10 to 20 years of low-level exposure. Monophasic high-intensity exposure may also result in an acute neuropathy without other systemic features.¹⁶⁹ The developing brain is particularly vulnerable to the CNS effects of lead, and cognitive, learning, and behavioral impairments may be seen in children even with blood lead levels around 10 µg/dL. Polyneuropathy occurs more frequently in adults with lead toxicity, but it has been reported in children—especially in those with sickle cell disease. Lead levels between 20 and 30 µg/dL have been implicated in peripheral nerve dysfunction. Early diagnosis and removal from continuing lead exposure is important, with the initiation of chelation therapy for levels greater than or equal to 45 µg/dL to enhance the chances of clinical and neurophysiologic improvement in lead poisoning and acute neuropathy, although the effects on subacute or chronic polyneuropathy are unclear.

The patient's medication list should always be carefully reviewed because a number of pharmaceutical agents can cause nerve toxicity. Many factors can increase the risk of medication-induced polyneuropathy, including impaired renal or hepatic function causing decreased drug metabolism or clearance and the presence of an unrelated neuropathy such as Charcot-Marie-Tooth (CMT) disease.

Chemotherapy-induced polyneuropathy is a very important source of morbidity in cancer patients. Not only does it impact quality of life, but also it can limit treatment. Clinical features will vary with the specific chemotherapeutic agent. Taxanes typically cause a bilateral symmetric axonal neuropathy that is sensory predominant.¹⁴⁵ Incidence and severity depend on both dose per cycle and the total cumulative dose received. Other risk factors include rate of infusion, use of other neurotoxic drugs, and comorbidities such as diabetes or CMT. Although mild symptoms improve with dose reduction, mild sensory symptoms may persist in 14% of patients after cessation of treatment taxels.¹¹⁶ Vinca alkaloids, particularly vincristine, cause some decrease of a mixed sensorimotor neuropathy in almost all treated

patients.¹²⁴ Autonomic neuropathies are common, and mononeuropathies may occur, most commonly affecting the oculomotor nerve.⁸ Platinum-containing agents primarily affect the dorsal root ganglion. Cisplatin-induced polyneuropathy predominantly affects large sensory fibers. Oxaliplatin causes an acute, reversible neuropathy and a chronic polyneuropathy characterized by distal symmetric axonal sensory involvement.¹⁸⁷ Symptoms of neuropathy can appear or progress for up to 2 months after cessation of the chemotherapeutic agent, known as the “coasting phenomenon.”⁵⁷ Dose reduction, longer infusion times, or longer periods between doses may improve symptoms, but sometimes treatment must be completely stopped, and recovery can sometimes be incomplete.

Vasculitic and Connective Tissue Disease–Associated Neuropathies

Vasculitis and connective tissue diseases produce neuropathies with varied presentations. Distal symmetric neuropathy and single mononeuropathies do occur, but the prototypical neuropathy caused by vasculitic and connective tissue disorders is an asymmetric mixed motor and sensory pattern referred to as a *mononeuritis multiplex*. The classic presentation begins with vasculitic ischemia and acute onset of pain and sensory abnormality in the distribution of the affected nerve. Occasionally the symptoms progress in a more gradual fashion, although it is typically asymmetric. Lower limb nerves are more commonly affected. The cranial nerves are rarely affected. The nerves affected are more likely to have injury at more proximal sites than those seen with typical compression neuropathies.

Vasculitic disorders that commonly cause polyneuropathy include polyarteritis nodosa, Churg-Strauss syndrome, microscopic polyangiitis, and Wegener granulomatosis (Box 42.5). Nerves can also be affected by nonsystemic vasculitis of the peripheral nerves (i.e., a vasculitis without other organ or system involvement). The vasculitic injury to the peripheral nervous system usually results in the asymmetric mononeuritis multiplex pattern. Connective tissue disorders, such as RA, systemic lupus erythematosus (SLE), and Sjögren syndrome, more often cause a diffuse symmetric polyneuropathy.

RA and SLE tend to cause a polyneuropathy that is mild and a late presentation in the disease. RA can progress to rheumatoid vasculitis, causing multiple mononeuropathies. Compression neuropathies, especially CTS, are more common in both SLE and RA. Sjögren syndrome can also present as a pure sensory neuropathy.

The hallmark of mononeuritis multiplex on physical examination is weakness or sensory loss that represents a side-to-side asymmetry of the same peripheral nerve or proximal greater than distal involvement of different peripheral nerves in sites that are not typical locations of anatomic compression. EDX in mononeuritis multiplex shows decreased SNAP amplitudes in affected nerves, with normal to borderline low CV. The findings are similar for the motor conduction. Marked asymmetry between the same nerves on the right and left sides may be seen. Nerves in the lower extremities are usually more affected. Conduction block can be seen in the affected nerves, in segments not typically involved in compression. F-waves can be prolonged. Needle EMG shows positive waves and fibrillation potentials with reduced recruitment and neuropathic motor unit changes in the territory of the affected nerves. As the disease progresses and involves more nerves, the findings can coalesce, making differentiation from a diffuse, symmetric pattern difficult. In patients with known vasculitic or connective tissue disease, the findings of new

• BOX 42.5 Neuropathies Associated With Vasculitic and Connective Tissue Disease

Polyarteritis Nodosa (PAN)

- 50% with peripheral neuropathies
- Mononeuritis multiplex 57%
- Distal symmetric peripheral neuropathies 25%
- Single mononeuropathy 17%

Eosinophilic Granulomatosis With Polyangiitis

- Peripheral neuropathies more common than in PAN
- Similar distribution as PAN

Microscopic Polyangiitis

- Peripheral neuropathies less likely (14–36%)

Granulomatosis With Polyangiitis

- Peripheral neuropathies less common (<15%)

Mixed Cryoglobulinemia

- 30–60% with peripheral neuropathies
- Distal symmetric sensory or sensorimotor neuropathy

Giant Cell Arteritis

- Carpal tunnel syndrome

Rheumatoid Arthritis

- Compression neuropathies (carpal tunnel syndrome)
- Mild distal symmetric sensory neuropathy (late complication)
- Mononeuritis multiplex in rheumatoid vasculitis

Sjögren Syndrome

- 10–50% with peripheral neuropathies
- Sensorimotor neuropathy
- Pure distal sensory
- Trigeminal sensory neuropathy
- Autonomic neuropathy
- Carpal tunnel syndrome

Systemic Lupus Erythematosus

- 25–50% with peripheral neuropathies
- Symmetric diffuse sensory or sensorimotor axonal peripheral neuropathies (late)
- Carpal tunnel syndrome

Systemic Sclerosis

- Peripheral neuropathies uncommon

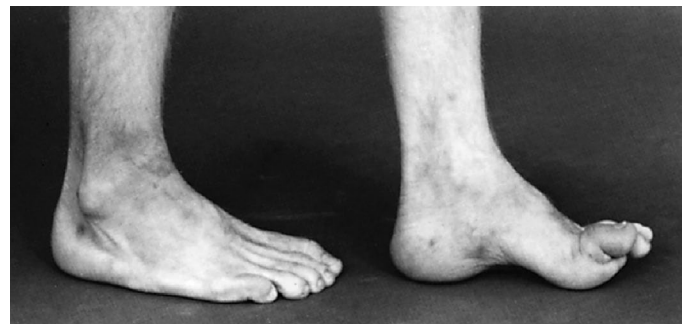
PN can indicate more active disease and the need for more aggressive management. PN can also be the presenting sign of these disorders. In those without known systemic disease, the finding of mononeuritis multiplex requires a search for systemic involvement and might require nerve or muscle biopsy. Prompt identification of a vasculitic-associated neuropathy is important because management with immunomodulators, such as corticosteroids and cyclophosphamide, can halt progression. Specific rehabilitation recommendations vary based on the specific nerves affected.

Hereditary Neuropathies

Charcot-Marie-Tooth Disease

CMT disease is the most common inherited polyneuropathy with a prevalence of 17 to 40 per 100,000.¹⁰² CMT is a genetically heterogeneous disorder characterized by wasting of the distal limb muscles, particularly the fibular innervated muscles, with distal sensory loss, decreased or absent muscle stretch reflexes, and skeletal deformity. CMT is often referred to as hereditary motor and sensory neuropathy (HMSN) and has been subclassified as HMSN types I through VI. HMSN-I, now known as CMT-1, is characterized by slow CVs and diffuse demyelination. HMSN-II/CMT-2 is the axonal form with axonal degeneration and preserved or only mildly slowed CVs. HMSN-III is more commonly known as Déjérine-Sottas neuropathy and is characterized by early onset with severe hypertrophic demyelination. HMSN-IV, HMSN-V, and HMSN-VI describe various motor-only forms now referred to as *distal hereditary motor neuropathies*. At least 40 different genetic forms of CMT have been identified, and any one genotype can have variable penetrance and phenotypical expression.⁷⁶

CMT typically shows a pattern of slow progression of onset, starting in the first 2 decades of life. Children and teenagers can first present with steppage gait caused by weakness of the fibular innervated muscles and the development of pes cavus (Fig. 42.23).¹⁸⁴ Weakness of the ankle plantar flexors is less severe,



• **Fig. 42.23** Pes cavus deformity and hammer toes associated with Charcot-Marie-Tooth disease. (From Warner WC. Neuromuscular disorders. In: Canale ST, Beaty JH, eds. *Campbell's Operative Orthopaedics*. 11th ed. Mosby; 2009.)

and atrophy of the lower limbs is often described as resembling an inverted champagne bottle. Upper limb involvement with claw hand deformity occurs later in the course of the disease. Although gait is affected, loss of independent ambulation is rare.⁶³ Sensory symptoms are less prominent. Loss of sensation for pain, touch, and vibration is seen in the feet, with clinical proprioception relatively preserved. Pain or paresthesias are less common and less severe, but complaints of cold feet and muscle cramps may be noted. Muscle stretch reflexes are decreased to absent in the demyelinating CMT-1 but are less affected in CMT-2. Skeletal deformities are seen in 66% of all CMT patients, with pes cavus and hammer toes the most common abnormality. Scoliosis is less common. Foot deformities occur as a result of the imbalance in strength of the foot intrinsic and leg muscles. Late-onset CMT tends to result in subtle foot abnormalities if at all.¹²⁰

Nerve conduction tests are the primary diagnostic tests for CMT. CMT-1, the demyelinating form of CMT, is characterized by marked slowing of nerve conduction studies, defined as median nerve conduction values less than 38 m/sec.⁶³ Discriminating hereditary from acquired neuropathies can be challenging

when the onset of noticeable signs and symptoms does not appear until the adult years. Nerve conduction slowing in hereditary neuropathies tends to be uniform and diffuse, whereas in acquired demyelinating neuropathies (such as CIDP), the demyelination is patchy and asymmetric with temporal dispersion and partial conduction blocks in locations atypical for entrapment.⁹⁴ There is an exception in that X-linked CMT-1 might present with nonuniform demyelination and temporal dispersion, and can be misdiagnosed as CIDP.¹⁹ CMT-2 demonstrates normal or mildly slowed conduction velocities (>38 m/sec) with reduced motor and sensory amplitudes. Needle examination shows signs of chronic denervation distally greater than proximally in all types of CMT, and fibrillation potentials and positive waves are present in severe or rapidly progressing forms.

There is a growing body of evidence demonstrating the merits of exercise for people with CMT and showing that not only will it not hurt, but it actually has the potential to improve functional ability, independence, and quality of life. Because function is likely highest in children with CMT as compared with adults, regular exercise and activity may help preserve physical ability into adulthood. Progressive strengthening of the hip flexors improves hip flexor fatigue, resulting in improved walking duration, and strengthening of the ankle musculature results in improved muscle strength, walking speed, cadence, step time, and stride length. Children, like adults with CMT, have shown responsiveness to progressive strength training with improvement in muscle strength and walking speed, cadence, step time, and stride length.^{17,129}

Autonomic Neuropathy

Autonomic neuropathy, often associated with diabetes, is a significant cause of morbidity and even mortality, affecting the cardiovascular, gastrointestinal, genitourinary, and thermoregulatory systems. Involvement of the cardiovascular system limits activity tolerance and possibly life-threatening consequences. Patients with cardiac autonomic neuropathy have 40% greater mortality compared with those without,⁷⁸ which is caused by silent ischemia, lethal arrhythmias, or prolonged QT interval. Those with cardiac involvement have resting tachycardia greater than 100 beats/min and orthostasis, a decrease of systolic blood pressure greater than 20 mmHg on standing, which occurs without appropriate heart rate response. This leads to exercise tolerance and increases the risk of sudden death and silent myocardial ischemia. Both hypotension and hypertension can develop after exercise. Orthostasis is the most common clinical complaint. Management includes education regarding the need for slow postural changes, elevation of the head of the bed, and use of compression garments, such as compressive stockings and abdominal binders. If these measures are inadequate, mineralocorticoids, such as fludrocortisone or the alpha agonist midodrine, can be used.

Autonomic dysfunction of the gastrointestinal system can lead to esophageal dysmotility, gastroparesis, constipation or diarrhea, and bowel incontinence. Gastroparesis causes a sensation of bloating, nausea, heartburn, early satiety, and erratic glycemic control. These symptoms are typically managed with metoclopramide.

Autonomic neuropathy can lead to irreversible erectile dysfunction.⁴³ Treatment consists of assessing for secondary causes of erectile dysfunction, counseling, medications, suction erectile devices, and penile implants. Neurogenic bladder leads to overflow incontinence, recurrent urinary tract infection, and pyelonephritis. Urinary retention can be diagnosed with postvoid residuals and might need more formal urodynamic studies to determine the

severity. In mild cases, the patient can be instructed to void every 2 to 3 hours while awake. In more severe cases, intermittent catheterization may be necessary.

In those with abnormalities in sweating and thermoregulation, there should be counseling about adequate hydration and caution with exertion in hot or cold environments. Lack of sweating can lead to dryness and cracking and fissuring of the feet, increasing susceptibility to foot ulceration and infection. Patients should be taught to keep their feet well moisturized.

Complications of Neuropathies

Foot Complications

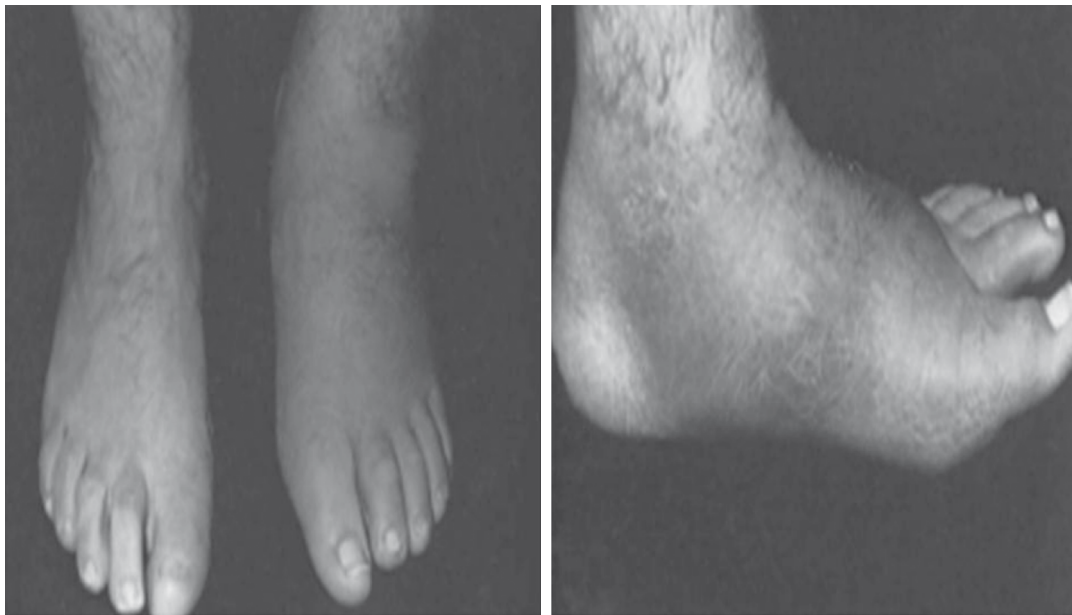
Foot ulceration is a major and serious complication in patients with polyneuropathy, particularly in those with diabetes mellitus, who manifest a lifetime risk of 15%.⁴⁹ The presence of neuropathy is associated with an 8- to 18-fold increased risk of foot ulcer and a 2- to 15-fold greater risk of amputation.¹¹⁸ Loss of protective sensation in polyneuropathy may lead to unrecognized foot trauma and skin breakdown. Motor neuropathy causes muscle atrophy and weakness, distorting normal foot architecture and leading to abnormal pressure distribution and overloaded plantar areas, further increasing risk of ulceration.

Autonomic dysfunction further compromises skin integrity by causing decreased sweating, resulting in skin dryness and cracking. The detrimental impact of autonomic dysfunction is accentuated by arteriovenous shunting, leading to alteration of perfusion with secondary osteopenia caused by bone resorption and impaired healing. Neuropathy can also lead to Charcot changes. Charcot neuroarthropathy is characterized by destruction of the joints of the foot, with pathologic fracture and joint dislocation usually in the midfoot region, which results in severe foot deformity (Fig. 42.24).⁷³

Patients with neuropathy should be educated on the importance of daily inspection of their feet. Protective socks and appropriate shoe wear, including extra-depth toe boxes and custom orthoses, should be used to offload high-pressure areas. Feet should be kept dry and the skin well moisturized, and extreme temperatures should be avoided, particularly hot water. Nail care and callus management is important and may require regular care by a podiatrist or diabetic foot clinic. Prompt management of any skin breakdown and aggressive management of infection are important to avoid progression to amputation. Acute Charcot foot can be difficult to differentiate from cellulitis, presenting with a warm, erythematous, and swollen foot. Radiographs may be normal in the acute phase. Fractures and joint dislocation will become apparent later in the course of the disease; however, a bone scan is often positive early on. Strict immobilization with a total contact cast is used to manage acute Charcot arthropathy and maintained until clinical signs of warmth and tenderness have significantly decreased and the bone scan becomes negative or less active, indicating that bone remodeling is near complete. At this point, careful weaning from the orthosis can begin. This is often accompanied by physical therapy using a gradual resumption of weight bearing and strengthening foot and ankle musculature. Prevention and treatment programs can significantly reduce lower extremity breakdowns and amputations.

Pain

Neuropathic pain is common and can be quite incapacitating. Large fiber neuropathies often manifest with dull, deep, toothache-like, or cramping pain. Small fiber neuropathies affect small



• **Fig. 42.24** Charcot neuropathy causing pathologic fracture, dislocation, and deformity of the midfoot. (From Ishikawa SN. Diabetic foot. In: Canale ST, Beatty JH, eds. *Campbell's Operative Orthopaedics*. 11th ed. Mosby; 2009.)

unmyelinated C fibers that control thermal perception, pain, and autonomic function and initially manifest with significant pain in the feet. Small fiber neuropathy is typically perceived as superficial, burning, and hypersensitive. Other sources of foot pain should be excluded. Other causes of painful feet are lumbar radiculopathy, vascular claudication, tarsal tunnel syndrome, plantar fasciitis, osteoarthritis, and Morton neuroma. It should also be kept in mind that each of these diagnoses can coexist with symptomatic painful neuropathy.

Treatment of neuropathic pain can be challenging, and the side effects of the medications can be particularly problematic in elderly patients. Topical agents, such as capsaicin cream, have the advantage of having no systemic side effects, but the application can be cumbersome (requiring application three to four times per day) and can cause initial irritation. Transdermal lidocaine patches can be useful in those with a relatively discrete distribution of pain and are easier to apply. Transcutaneous electrical stimulation may also be beneficial. Oral neuropathic pain medications, including gabapentinoids, tricyclic antidepressants, and duloxetine, have been shown to be effective in treatment of neuropathic pain. It is important to consider the side effect profile of each class when starting a medication for treatment of neuropathic pain. Tricyclic antidepressant dosing is often limited by anticholinergic effects, which include sedation, dry mouth, urinary retention, orthostatic hypotension, and potentially dangerous arrhythmias. Low-dose nortriptyline, dosed at 10 to 50 mg at bedtime, may be better tolerated than amitriptyline. It has been shown to have equivalent effect on some types of neuropathic pain,¹⁸⁵ with fewer side effects. Gabapentin and pregabalin can cause somnolence and dizziness, particularly on initiation of treatment, and they can exacerbate gait and cognitive problems in older patients. Use of duloxetine can be limited by gastrointestinal side effects. Tramadol, a weak mu-opioid receptor agonist and a weak norepinephrine and serotonin reuptake inhibitor, has been shown to have beneficial effects on the allodynia associated with diabetic polyneuropathy.¹⁵⁵

Tramadol can cause cognitive impairment, and serotonin syndrome may occur with concomitant use of other serotonergic medications. Opioid analgesics have been shown to improve pain and sleep but not function or mood.¹²⁸ Because opioids can cause cognitive side effects, physiologic dependency, and addiction, they should be used with extreme caution. Second-line medications for the treatment of neuropathic pain include lamotrigine, carbamazepine, other selective serotonin reuptake inhibitors, and clonidine.

Functional Impairment

Polyneuropathy can significantly affect functional mobility and quality of life. Patients with polyneuropathy are about 20 times more likely to fall than those without neuropathy.^{132,136} Neuropathy impairs proprioceptive sensation at the ankle, and those with neuropathy are less able to rapidly develop the ankle torque needed to correct for lateral lean.⁵⁸ Patients with neuropathy are additionally penalized when their center of mass is displaced as a relatively greater displacement occurs before loss of balance is perceived, and then there is a delay in corrective ankle torque generation. As a result, the patient with neuropathy experiences dynamic postural instability, particularly on uneven surfaces and in dim lighting, where visual cues cannot assist in compensating for proprioceptive loss. Both the perceived and real risk of instability can lead to increasingly limited physical activity, physical deconditioning, isolation, and depression.¹⁸²

Although neuropathy-related impairments in ankle sensory and motor function have been recognized for some time, an understanding of the relationship between poor distal neuromuscular function and proximal strength has only recently developed. It has been recognized that the ratio of proximal strength at the hip to ankle proprioceptive precision predicts unipedal stance time,³ falls, and fall-related injury in older patients with diabetic neuropathy.¹³³ The clinical relevance of the ratio suggests that increased hip strength can compensate for imprecise ankle proprioceptive precision. Of note, patients with neuropathy caused by type 2 diabetes

mellitus respond well to resistance training.¹²⁵ There are a number of other interventions that can improve gait and balance. Vision should be regularly tested, and corrective eyewear should be used as needed. Patients should consider using glasses for walking that correct distance vision only because the use of bifocals has been independently associated with falls.⁹⁶ Proper lighting of the patient's environment is important, particularly when getting up at night. Proper shoe wear that has a wide base of support and thin soles can improve balance. A cane can stabilize gait, but to achieve maximal benefit, patients should be able to support up to 25% of their body weight on the cane to prevent a fall.⁶ The cane should be brought down with each contralateral footstep. Acceptance of the use of a cane can be improved if presented as a substitute for the loss of a special sense, much like eyeglasses, and used only as needed on uneven surfaces and in unfamiliar areas. However, a disadvantage of canes is that they are associated with decreased walking speed. Use of ankle orthoses with medial and lateral support has been shown to improve gait and stability parameters.¹³⁸ They also have the advantage of freeing up both hands and better walking speed than a cane, but the skin in contact with the orthoses must be monitored for breakdown. Exercise trials involving neuropathic older subjects demonstrated that tai chi and functional balance training have shown improvement in clinical measurements of balance in patients with neuropathy.^{127,137} Specific exercises aimed at strengthening hip abductors and adductors, as well as abdominals, can improve frontal plane hip and trunk stability. Strengthening of grip, shoulder depression, and elbow extension can improve support when using a cane or other means of upper limb stabilization.

Key References

8. Balayssac D, Ferrier J, Descoeur J, et al. Chemotherapy-induced peripheral neuropathies: from clinical relevance to preclinical evidence. *Expert Opin Drug Saf.* 2011;10(3):407-417.
9. Barohn RF, Saperstein DS. Guillain-Barré syndrome and chronic inflammatory demyelinating polyneuropathy. *Semin Neurol.* 1998; 18(1):49-61.
11. Baute Penry V, Cartwright MS. Neuromuscular ultrasound for peripheral neuropathies. *Semin Neurol.* 2019;39(5):542-548.
12. Becker J, Nora DB, Gomes I, et al. An evaluation of gender, obesity, age, and diabetes mellitus as risk factors for carpal tunnel syndrome. *Clin Neurophysiol.* 2002;113(9):1429-1434.
15. Brown MC, Holland RL, Hopkins WG. Motor nerve sprouting. *Annu Rev Neurosci.* 1981;4:17-42.
17. Burns J, Raymond J, Ouvrier R. Feasibility of foot and ankle strength training in childhood Charcot-Marie-Tooth disease. *Neuromuscul Disord.* 2009;19(12):818-821.
21. Chan YC, Lo YL, Chan ES. Immunotherapy for diabetic amyotrophy. *Cochrane Database Syst Rev.* 2017;7.
24. Cignetti NE, Cox RS, Baute V, et al. A standardized ultrasound approach in neuralgic amyotrophy. *Muscle Nerve.* 2023;67(1):3-11.
29. Cowdery SR, Preston DC, Herrmann DN, et al. Electrodiagnosis of ulnar neuropathy at the wrist: conduction block versus traditional tests. *Neurology.* 2002;59(3):420-427.
31. Cravens G, Kline DG. Posterior interosseous nerve palsies. *Neurosurgery.* 1990;27:397-402.
33. Derr JJ, Mickelsen PJ, Robinson LR. Predicting recovery after fibular nerve injury: which electrodiagnostic features are most useful? *Am J Phys Med Rehabil.* 2009;88:547-553.
38. Dumitru D. Focal peripheral neuropathies. In: Dumitru D, ed. *Electrodiagnostic Medicine.* Hanley & Belfus; 1995.
40. Dyck PJB, Tracy JA. History, diagnosis, and management of chronic inflammatory demyelinating polyradiculoneuropathy. *Mayo Clin Proc.* 2018;93(6):777-793.
45. Fergusson D, Hutton B, Sharma M, et al. Use of intravenous immunoglobulin for treatment of neurologic conditions: a systematic review. *Transfusion.* 2005;45:1640-1657.
46. Ferrante M, Ferrante N. The thoracic outlet syndromes: part 1. Overview of the thoracic outlet syndromes and review of true neurogenic thoracic outlet syndrome. *Muscle Nerve.* 2017;55:782-793.
53. Gonzalez-Duarte A, Robinson-Papp J, Simpson DM. Diagnosis and management of HIV-associated neuropathy. *Neurol Clin.* 2008;26:821-832.
57. Guitiérrez-Guitiérrez G, Sereno M, Miralles A, et al. Chemotherapy-induced peripheral neuropathy: clinical features diagnosis, prevention and treatment strategies. *Clin Transl Oncol.* 2009;12:81-91.
63. Harding AE, Thomas PK. The clinical features of hereditary motor and sensory neuropathy (types I and II). *Brain.* 1980;103: 259-280.
65. Hearn SL, Jorgensen SP, Gabet JM, et al. Occupational nerve injuries. *Muscle Nerve.* 2024. <https://doi.org/10.1002/mus.28099>.
66. Hearn SL, Stino AM, Howard IM, et al. Serial electrodiagnostic testing: utility and indications in adult neurological disorders. *Muscle Nerve.* 2024;69(6):670-681.
67. Hobson-Webb LD, Padua L. Ultrasound of focal neuropathies. *J Clin Neurophysiol.* 2016;33:94-102.
76. Jani-Acsadi A, Krajewski K, Shy ME. Charcot-Marie neuropathies: diagnosis and management. *Semin Neurol.* 2008;2:185-194.
77. Jasti AK, Selmi C, Sarmiento-Monroy JC, et al. Guillain-Barré syndrome: causes, immunopathogenic mechanisms and treatment. *Expert Rev Clin Immunol.* 2016;12:1175-1189.
81. Kapritskaya Y, Novak C, Mackinnon S. Prevalence of smoking, obesity, diabetes mellitus and thyroid disease in patients with carpal tunnel syndrome. *Ann Plast Surg.* 2002;48(3):269-279.
83. Katiirji MB, Wilbourn AJ. Common peroneal mononeuropathy: a clinical and electrophysiologic study of 116 lesions. *Neurology.* 1998;38:1723-1728.
88. Krishnan KR, Sneag DB, Feinberg JH, et al. Outcomes of micro-neurolysis of hourglass constrictions in chronic neuralgic amyotrophy. *J Hand Surg Am.* 2021;46(1):43-53.
95. London Z, Albers JW. Toxic neuropathies associated with pharmacologic and industrial agents. *Neurol Clin.* 2007;25:257-276.
98. Malikowski T, Micklesen PJ, Robinson LR. Prognostic values of electrodiagnostic studies in traumatic radial neuropathy. *Muscle Nerve.* 2007;36(3):364-367.
100. Martin RM, Fish DE. Scapular winging: anatomical review, diagnosis, and treatments. *Curr Rev Musculoskelet Med.* 2008;1(1): 1-11.
106. Miller RG, Peterson GW, Daube JR, et al. Prognostic value of electrodiagnosis in Guillain-Barre syndrome. *Muscle Nerve.* 1988; 11:769-774.
109. Mul K, Pesser N, Vervaart K, et al. Variability in electrodiagnostic findings associated with neurogenic thoracic outlet syndrome. *Muscle Nerve.* 2022;65(1):34-42.
115. Omejec G, Zgur T, Podnar S. Diagnostic accuracy of ultrasonographic and nerve conduction studies in ulnar neuropathy at the elbow. *Clin Neurophysiol.* 2015;126:1797-1804.
117. Oyer D, Saxon D, Shah A. Quantitative assessment of diabetic peripheral neuropathy with use of the clanging tuning fork test. *Endocr Pract.* 2007;13:5-10.
120. Pareyson D, Scaiola V, Laurà M. Clinical and electrophysiological aspects of Charcot-Marie-Tooth disease. *Neuromolecular Med.* 2006;8:3-22.
123. Plasma Exchange/Sandoglobulin Guillain-Barré Syndrome Trial Group. Randomised trial of plasma exchange, intravenous immunoglobulin, and combined treatments in Guillain-Barré syndrome. *Lancet.* 1997;349:225-230.
127. Quigley PA, Bulat T, Schulz B, et al. Exercise interventions, gait, and balance in older subjects with distal symmetric polyneuropathy: a three-group randomized trial. *Am J Phys Med Rehabil.* 2014;93:1-16.
131. Richardson JK. The clinical identification of peripheral neuropathy among older persons. *Arch Phys Med Rehabil.* 2002;83:205-209.

137. Richardson JK, Sandman D, Vela S. A focused exercise regimen improves clinical measures of balance in patients with peripheral neuropathy. *Arch Phys Med Rehabil.* 2001;82:205-209.
146. Seddon HJ. Three types of nerve injury. *Brain.* 1943;66(4):237-288.
147. Seror P. Electrophysiological pattern of 53 cases of ulnar nerve lesion at the wrist. *Clin Neurophysiol.* 2013;43(2):95-103.
158. Spinner RJ, Atkinson JL, Tiel RL. Peroneal intraneural ganglia: the importance of the articular branch. A unifying theory. *J Neurosurg.* 2003;99(2):330-343.
161. Sunderland S. A classification of peripheral nerve injuries producing loss of function. *Brain.* 1951;74(4):491-516.
171. Tsairis P, Dyck PJ, Mulder DW. Natural history of brachial plexus neuropathy: report on 99 patients. *Arch Neurol.* 1972;27:109-117.
176. van Alfen N, van Engelen BG. The clinical spectrum of neuralgic amyotrophy in 246 cases. *Brain.* 2006;129(Pt 2):438-450.
177. Van den Bergh PYK, van Doorn PA, Hadden RDM, et al. European Academy of Neurology/Peripheral Nerve Society guideline on diagnosis and treatment of chronic inflammatory demyelinating polyradiculoneuropathy: report of a joint task force-second revision. *J Peripher Nerv Syst.* 2021;26(3):242-268.
178. Van Eijk JJ, Groothuis JT, Van Alfen N. Neuralgic amyotrophy: an update on diagnosis, pathophysiology, and treatment. *Muscle Nerve.* 2016;53(3):337-350.

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References

- Al Hakim M, Katirji MB. Femoral mononeuropathy induced by the lithotomy position: a report of 5 cases and a review of the literature. *Muscle Nerve*. 1997;20:205–211.
- Albers JW, Donofrio PD, McGonagle TK. Sequential electrodiagnostic abnormalities in acute demyelinating polyneuropathy. *Muscle Nerve*. 1978;1:292–296.
- Allet LA, Kim H, Ashton-Miller JA, et al. Frontal plane hip and ankle sensorimotor function, not age, predicts unipedal stance time. *Muscle Nerve*. 2012;45(4):578–585.
- Al-Redouan A, Holding K, Kachlik D. “Suprascapular canal”: anatomical and topographical description and its clinical implication in entrapment syndrome. *Ann Anat*. 2021;233:151593.
- Anandan C, Khuder SA, Koffman BM. Prevalence of autonomic dysfunction in hospitalized patients with Guillain-Barré syndrome. *Muscle Nerve*. 2017;56(2):331–333.
- Ashton-Miller JA, Yeh MWL, Richardson JK, et al. A cane reduces loss of balance in patients with peripheral neuropathy: results from a challenging unipedal balance test. *Arch Phys Med Rehabil*. 1996;77(5):446–452.
- Aszmann OC, Dellon ES, Dellon AL. Anatomical course of the lateral femoral cutaneous nerve and its susceptibility to compression and injury. *Plast Reconstr Surg*. 1997;100:600–604.
- Balayssac D, Ferrier J, Descoeur J, et al. Chemotherapy-induced peripheral neuropathies: from clinical relevance to preclinical evidence. *Expert Opin Drug Saf*. 2011;10(3):407–417.
- Barohn RE, Saperstein DS. Guillain-Barré syndrome and chronic inflammatory demyelinating polyneuropathy. *Semin Neurol*. 1998;18(1):49–61.
- Bartels RHMA, Verbeek ALM. Risk factors for ulnar nerve compression at the elbow: a case control study. *Acta Neurochir*. 2007;149:669–674.
- Baute Penry V, Cartwright MS. Neuromuscular ultrasound for peripheral neuropathies. *Semin Neurol*. 2019;39(5):542–548.
- Becker J, Nora DB, Gomes I, et al. An evaluation of gender, obesity, age, and diabetes mellitus as risk factors for carpal tunnel syndrome. *Clin Neurophysiol*. 2002;113(9):1429–1434.
- Beecher G, Dyck PJB, Zochodne DW. Axillary and musculocutaneous neuropathies. *Handbook of Clinical Neurology*. 2024;201:135–48.
- Boon AJ, Bailey PW, Smith J, et al. Utility of ultrasound-guided surface electrode placement in lateral femoral cutaneous nerve conduction studies. *Muscle Nerve*. 2011;44:525–553.
- Brown MC, Holland RL, Hopkins WG. Motor nerve sprouting. *Annu Rev Neurosci*. 1981;4:17–42.
- Burnett AL, Brendler CB. Femoral neuropathy following major pelvic surgery: etiology and prevention. *J Neurosurg*. 1995;83:783–790.
- Burns J, Raymond J, Ouvrier R. Feasibility of foot and ankle strength training in childhood Charcot-Marie-Tooth disease. *Neuromuscul Disord*. 2009;19(12):818–821.
- Campagnolo DI, Romello MA, Park YI, Foye PM, Delisa JA. Technique for studying conduction in the lateral cutaneous nerve of calf. *Muscle Nerve*. 2000;23(8):1277–9.
- Capasso M, DiMuzzio A, Farrarini M, et al. Internerves and intra-nerve conduction heterogeneity in CMTX with Arg(15)GLN mutation. *Clin Neurophysiol*. 2004;115:64–70.
- Carter GT, Weiss MD. Diagnosis and treatment of work-related proximal median and radial nerve entrapment. *Phys Med Rehabil Clin N Am*. 2015;26(3):539–49.
- Chan YC, Lo YL, Chan ES. Immunotherapy for diabetic amyotrophy. *Cochrane Database Syst Rev*. 2017;7.
- Chang MH, Chiang HT, Lee SS, et al. Oral drug of choice in carpal tunnel syndrome. *Neurology*. 1998;51:390–393.
- Chin RL, Sander HW, Brannagan TH, et al. Demyelinating neuropathy in patients with hepatitis C virus infection. *J Clin Neuromuscul Dis*. 2010;11:209–212.
- Cignetti NE, Cox RS, Baute V, et al. A standardized ultrasound approach in neuralgic amyotrophy. *Muscle Nerve*. 2023;67(1):3–11.
- Coppack SW, Watkins PJ. The natural history of diabetic femoral neuropathy. *Quart J Med*. 1991;79(288):307–313.
- Coraci D, Tsukamoto H, Granata G, et al. Fibular nerve damage in knee dislocation: spectrum of ultrasound patterns. *Muscle Nerve*. 2015;51:859–863.
- Cornblath DR, McArthur JC. Predominantly sensory neuropathy in patients with AIDS and AIDS-related complex. *Neurology*. 1988;38:794–796.
- Cosgrove JL, Vargo M, Reidy ME. A prospective study of peripheral nerve lesions occurring in traumatic brain-injured patients. *Am J Phys Med Rehabil*. 1989;68:15–17.
- Cowdery SR, Preston DC, Herrmann DN, et al. Electrodiagnosis of ulnar neuropathy at the wrist: conduction block versus traditional tests. *Neurology*. 2002;59(3):420–427.
- Craig EJ, Clinchot DM. Lateral femoral cutaneous neuropathy. In: Frontera WR, Silver JK, Rizzo TD, eds. *Essentials of Physical Medicine and Rehabilitation: Musculoskeletal Disorders, Pain, and Rehabilitation*. Philadelphia: Saunders; 2008.
- Cravens G, Kline DG. Posterior interosseous nerve palsies. *Neurosurgery*. 1990;27:397–402.
- DeLisa JA, Saeed MA. The tarsal tunnel syndrome. *Muscle Nerve*. 1983;6:664–670.
- Derr JJ, Mickelsen PJ, Robinson LR. Predicting recovery after fibular nerve injury: which electrodiagnostic features are most useful? *Am J Phys Med Rehabil*. 2009;88:547–553.
- Descatha A, Leclerc A, Chastang JF, et al. Incidence of ulnar nerve entrapment at the elbow in repetitive work. *Scand J Work Environ Health*. 2004;30(3):234–240.
- The Diabetes Control and Complications Trial Research Group. The effect of intensive diabetes therapy on the development and progression of neuropathy. *Ann Intern Med*. 1995;122:561–568.
- Ducic I, Dellon L, Larson EE. Treatment concepts for idiopathic and iatrogenic femoral nerve mononeuropathy. *Ann Plast Surg*. 2005;55(4):397–401.
- Duckworth W, Abaira C, Moritz T, et al. Glucose control and vascular complications in veterans with type 2 diabetes. *N Engl J Med*. 2009;360:129–139.
- Dumitru D. Focal peripheral neuropathies. In: Dumitru D, ed. *Electrodiagnostic Medicine*. Hanley & Belfus; 1995.
- Dureja GP, Galaya V, Jayalakshmi TS, Mandal P. Management of meralgia paresthetica: a multimodality regimen. *Anesth Analg*. 1995;80(5):1060–1061.
- Dyck PJB, Tracy JA. History, diagnosis, and management of chronic inflammatory demyelinating polyradiculoneuropathy. *Mayo Clin Proc*. 2018;93(6):777–793.
- Ebeling P, Gilliat RW, Thomas PK. A clinical and electrical study of ulnar nerve lesions in the hand. *J Neurol Neurosurg Psychiatry*. 1960;23(1):1–9.
- Ellegaard HR, Fuglsang-Frederiksen A, Hess A, et al. High-resolution ultrasound in ulnar neuropathy at the elbow: a prospective study. *Muscle Nerve*. 2015;52:759–766.
- Fairburn CG, Wu FC, McCulloch DK, et al. The clinical features of diabetic impotence: a preliminary study. *Br J Psychiatry*. 1982;140:447–452.
- Feldman RG, Niles CA, Kelly-Hayes M, et al. Peripheral neuropathy in arsenic smelter workers. *Neurology*. 1979;29(7):939–944.
- Fergusson D, Hutton B, Sharma M, et al. Use of intravenous immunoglobulin for treatment of neurologic conditions: a systematic review. *Transfusion*. 2005;45:1640–1657.
- Ferrante M, Ferrante N. The thoracic outlet syndromes. part 1. Overview of the thoracic outlet syndromes and review of true neurogenic thoracic outlet syndrome. *Muscle Nerve*. 2017;55:782–793.
- Fisher C. An unusual variant of acute idiopathic polyneuritis (syndrome of ophthalmoplegia, ataxia, and areflexia). *N Engl J Med*. 1956;225:57–75.

48. Frost P, Johnsen B, Fuglsang-Frederiksen A, Svendsen SW. Lifestyle risk factors for ulnar neuropathy and ulnar neuropathy-like symptoms. *Muscle Nerve*. 2013;48:507–515.
49. Frykberg RG, Lavery LA, Pham H, et al. Role of neuropathy and high foot pressure in diabetic foot ulceration. *Diabetes Care*. 1998;21:1714–1719.
50. Gabbai AA, Castelo A, Bulle Oliveria AS. HIV peripheral neuropathy. In Gerard S, Krarup C, eds. *Handbook of clinical neurology*, vol 115. Elsevier; 2013.
51. Garssen MP, Bussmann JB, Schmitz PI, et al. Physical training and fatigue, fitness, and quality of life in Guillain-Barré syndrome and CIDP. *Neurology*. 2004;3:2393–2395.
52. Genuth S. Insights from the diabetes control and complications trial/epidemiology of diabetes interventions and complications study on the use of intensive glycemic treatment to reduce the risk of complications of type 1 diabetes. *Endocr Pract*. 2006;12(Suppl 1):34–41.
53. Gonzalez-Duarte A, Robinson-Papp J, Simpson DM. Diagnosis and management of HIV-associated neuropathy. *Neurol Clin*. 2008;26:821–832.
54. Goulding PJ, Schady W. Favourable outcome in non-traumatic anterior interosseous nerve lesions. *J Neurol*. 1993;240:83–86.
55. Greenberg SA. Acute demyelinating polyneuropathy with arsenic ingestion. *Muscle Nerve*. 1996;19:1611–1613.
56. Guillain-Barré Syndrome Study Group. Plasmapheresis and acute Guillain-Barré syndrome. *Neurology*. 1985;35:1096–1104.
57. Guitiérrez-Gutiérrez G, Sereno M, Miralles A, et al. Chemotherapy-induced peripheral neuropathy: clinical features diagnosis, prevention and treatment strategies. *Clin Transl Oncol*. 2009;12:81–91.
58. Gutierrez MS, Helber MB, Dealva D, et al. Mild diabetic neuropathy affects ankle motor function. *Clin Biomech*. 2001;16(6):522–528.
59. Haber P, Sejvar F, Mikaeloff Y, et al. Vaccines and Guillain-Barré syndrome. *Drug Safety*. 2009;32(4):309–323.
60. Hadden RD, Cornblath DR, Hughes RA, et al. Electrophysiological classification of Guillain-Barré syndrome: clinical associations and outcome: plasma Exchange/Sandoglobulin Guillain-Barré syndrome Trial Group. *Ann Neurol*. 1998;44:780–788.
61. Hadden RD, Karch H, Hartung HP, et al. Preceding infections, immune factors, and outcome in Guillain-Barré syndrome. *Neurology*. 2001;56:758–756.
62. Hagebeuk EE, DeWeerd AW. Clinical and electrophysiologic follow-up after local steroid injection in carpal tunnel syndrome. *Clin Neurophysiol*. 2004;115(6):1464–1468.
63. Harding AE, Thomas PK. The clinical features of hereditary motor and sensory neuropathy (types I and II). *Brain*. 1980;103:259–280.
64. Hashizume H, Nishida K, Nanba Y, et al. Non-traumatic paralysis of the posterior interosseous nerve. *J Bone Joint Surg Br*. 1996;78:771–776.
65. Hearn SL, Jorgensen SP, Gabet JM, et al. Occupational nerve injuries. *Muscle Nerve*. 2024. <https://doi.org/10.1002/mus.28099>.
66. Hearn SL, Stino AM, Howard IM, et al. Serial electrodiagnostic testing: utility and indications in adult neurological disorders. *Muscle Nerve*. 2024;69(6):670–681.
67. Hobson-Webb LD, Padua L. Ultrasound of focal neuropathies. *J Clin Neurophysiol*. 2016;33:94–102.
68. Hoke A, Cornblath DR. Peripheral neuropathies in human immunodeficiency virus infection. *Suppl Clin Neurophysiol*. 2004;57:195–210.
69. Hughes RA, Brassington R, Gunn AA, et al. Corticosteroids for Guillain-Barré syndrome. *Cochrane Database Syst Rev*. 2016.
70. Hughes RA, Swan AV, Raphael JC, et al. Immunotherapy for Guillain-Barré syndrome: a systematic review. *Brain*. 2007;130:2245–2257.
71. Hurvitz EA, Richardson JK, Werner RA. Unipedal stance testing in the assessment of peripheral neuropathy. *Arch Phys Med Rehabil*. 2001;82(2):199–204.
72. Ijspeert J, Janssen RMJ, van Alfen N. Neuralgic amyotrophy. *Curr Opin Neurol*. 2021;34(5):605–612.
73. Ishikawa SN. Diabetic foot. In Canale ST, Beaty JH, eds. *Campbell's operative orthopaedics*, 11th ed. Philadelphia: Mosby; 2009.
74. Ivins GK. Meralgia paresthetica, the elusive diagnosis: clinical experience with 14 adult patients. *Ann Surg*. 2000;232:281–286.
75. Jaekle KA. Neurologic manifestations of neoplastic and radiation-induced plexopathies. *Semin Neurol*. 2010;30:254–262.
76. Jani-Acsadi A, Krajewski K, Shy ME. Charcot-Marie neuropathies: diagnosis and management. *Semin Neurol*. 2008;2:185–194.
77. Jasti AK, Selmi C, Sarmiento-Monroy JC, et al. Guillain-Barré syndrome: causes, immunopathogenic mechanisms and treatment. *Expert Rev Clin Immunol*. 2016;12:1175–1189.
78. Jermendy G, Toth L, Voros P, et al. Cardiac autonomic neuropathy and QT interval length. A follow-up study in diabetic patients. *Acta Cardiol*. 1991;46:189–200.
79. Jobe MT, Martinez SF. Peripheral nerve injuries. In Canale ST, Beaty JH, eds. *Campbell's operative orthopaedics*. 11th ed. Philadelphia: Mosby; 2009:3062–3123.
80. Kanda T. Chronic hepatitis C infection and peripheral neuropathy: is mixed cryoglobulinemia really important? *Internal Medicine*. 2003;42(5):377–378.
81. Kapritskaya Y, Novak C, Mackinnon S. Prevalence of smoking, obesity, diabetes mellitus and thyroid disease in patients with carpal tunnel syndrome. *Ann Plast Surg*. 2002;48(3):269–279.
82. Karatas GK, Gogus F. Suprascapular nerve entrapment in newsreel cameramen. *Am J Phys Med Rehabil*. 2003;82(3):192–6.
83. Katiirji MB, Wilbourn AJ. Common peroneal mononeuropathy: a clinical and electrophysiologic study of 116 lesions. *Neurology*. 1998;38:1723–1728.
84. Katiirji MB, Wilbourn AJ. High sciatic lesions mimicking peroneal neuropathy at the fibular head. *J Neurol Sci*. 1994;121:172–175.
85. Khalil N, Nicotra A, Rakowicz W. Treatment for meralgia paresthetica. *Cochrane Database Syst Rev*. 2008;3:1–14.
86. Khalil NM, Nicotra A, Kaplan C, O'Neill KS. Entrapment of the lateral cutaneous nerve of the calf. *Case Rep*. 2013;2013:bcr2013008847.
87. Kleyweg RP, van der Meché. Treatment related fluctuations in Guillain-Barré syndrome after high-dose immunoglobulins or plasma-exchange. *J Neurol Neurosurg Psychiatry*. 1991;54:957–960.
88. Krishnan KR, Sneag DB, Feinberg JH, et al. Outcomes of micro-neurolysis of hourglass constrictions in chronic neuralgic amyotrophy. *J Hand Surg Am*. 2021;46(1):43–53.
89. Kuntzer T, van Melle G, Regli F. Clinical and prognostic features in unilateral femoral neuropathies. *Muscle Nerve*. 1997;20:205–211.
90. Laguëny A, Deliac MM, Deliac P, et al. Diagnostic and prognostic value of electrophysiologic tests in meralgia paresthetica. *Muscle Nerve*. 1991;14:51–56.
91. Lavanchy D. The global burden of hepatitis C. *Liver Int*. 2009;29(Suppl 1):74–81.
92. Lawn N, Fletcher D, Henderson R, et al. Anticipating mechanical ventilation in Guillain-Barré syndrome. *Arch Neurol*. 2001;58:8893–8898.
93. Lazar MA, Plocher EK, Egol KA. Obesity and its relationship with pelvic and lower-extremity orthopedic trauma. *Am J Orthop*. 2010;39(4):175–182.
94. Lewis RA, Sumner AJ. The electrodiagnostic distinctions between chronic familial and acquired demyelinating neuropathies. *Neurology*. 1982;32:592–596.
95. London Z, Albers JW. Toxic neuropathies associated with pharmacologic and industrial agents. *Neurol Clin*. 2007;25:257–276.
96. Lord SR, Dayhew J, Howland A. Multifocal glasses impair edge-contrast sensitivity and depth perception and increase the risk of falls in older people. *J Am Geriatr Soc*. 2002;50:1760–1766.
97. Ly-Pen D, Andreu JL, de Blas G, et al. Surgical decompression versus local steroid injection in carpal tunnel syndrome. *Arthritis Rheum*. 2005;52(2):612–619.
98. Malikowski T, Micklesen PJ, Robinson LR. Prognostic values of electrodiagnostic studies in traumatic radial neuropathy. *Muscle Nerve*. 2007;36(3):364–367.
99. Marais S, Wilkinson RJ, Pepper DJ, et al. Management of patients with the immune reconstitution inflammatory syndrome. *Curr HIV/AIDS Rep*. 2009;6:162–171.

100. Martin RM, Fish DE. Scapular winging: anatomical review, diagnosis, and treatments. *Curr Rev Musculoskelet Med*. 2008;1(1):1-11.
101. Martinoli C, Miguel-Perez M, Padua L, et al. Imaging of neuropathies about the hip. *Eur J Radiol*. 2013;82:17-26.
102. Martyn CN, Hughes RAC. Epidemiology of peripheral neuropathy. *J Neurol Neurosurg Psychiatry*. 1997;62:310-318.
103. McMillan HJ, Kang PB, Royden Jones H, Darras BT. Childhood chronic inflammatory demyelinating polyradiculoneuropathy: Combined analysis of a large cohort and eleven published series. *Neuromuscular Disord*. 2013;23(2):103-111.
104. Millender LH, Nalebuff EA, Holdsworth DE. Posterior interosseous-nerve syndrome secondary to rheumatoid synovitis. *J Bone Joint Surg Am*. 1973;55(4):753-757.
105. Miller RD. Anatomy of the peripheral nerve. In: Miller RD, ed. *Miller's anesthesia*. 6th ed. Philadelphia: Elsevier; 2005.
106. Miller RG, Peterson GW, Daube JR, et al. Prognostic value of electrodiagnosis in Guillain-Barre syndrome. *Muscle Nerve*. 1988;11:769-774.
107. Moeller JL, Hutchinson MR. Carpal tunnel. In Rakel RE, ed. *Textbook of family medicine*. 7th ed. Philadelphia: Saunders; 2007.
108. Mondelli M, Ginanneschi F, Aretini A. Diagnostic accuracy of sensory clinical findings of the hand dorsum and of neurography of the dorsal ulnar cutaneous nerve in ulnar neuropathy at the elbow. *Arch Phys Med Rehabil*. 2019;100(5):908-13.
109. Mul K, Pesser N, Vervaaert K, et al. Variability in electrodiagnostic findings associated with neurogenic thoracic outlet syndrome. *Muscle Nerve*. 2022;65(1):34-42.
110. Nemni R, Corbo M, Fazio R, et al. Cryoglobulinaemic neuropathy. A clinical, morphological and immunocytochemical study of 8 cases. *Brain*. 1988;111(Pt 3):541-552.
111. Nemni R, Sanvito L, Quattrini A, et al. Peripheral neuropathy in hepatitis C virus infection with and without cryoglobulinaemia. *J Neurol Neurosurg Psychiatry*. 2003;74:1267-1271.
112. Odaka M, Yuki N, Hirata K. Patients with chronic inflammatory demyelinating polyneuropathy initially diagnosed as Guillain-Barre syndrome. *J Neurol*. 2003;250(8):913-916.
113. Oh SH, Meyer RD. Entrapment neuropathies of the tibial (posterior tibial) nerve. *Neurol Clin*. 1999;17:593-615.
114. Omejec G, Podnar S. Long-term outcomes in patients with ulnar neuropathy at the elbow treated according to the presumed aetiology. *Clin Neurophysiol*. 2018;129(8):1763-9.
115. Omejec G, Zgur T, Podnar S. Diagnostic accuracy of ultrasonographic and nerve conduction studies in ulnar neuropathy at the elbow. *Clin Neurophysiol*. 2015;126:1797-1804.
116. Osmani K, Vignes S, Aissi M, et al. Taxane-induced peripheral neuropathy has good long-term prognosis: a 1- to 13-year evaluation. *J Neurol*. 2012;259(9):1936-1943.
117. Oyer D, Saxon D, Shah A. Quantitative assessment of diabetic peripheral neuropathy with use of the clanging tuning fork test. *Endocr Pract*. 2007;13:5-10.
118. Paola LD, Faglia E. Treatment of diabetic foot ulcer: an overview Strategies for clinical approach. *Curr Diabetes Rev*. 2006;2:431-447.
119. Paradiso G, Tripoli J, Galicchio S, et al. Epidemiological, clinical, and electrodiagnostic findings in childhood Guillain-Barre syndrome: a reappraisal. *Ann Neurol*. 1999;46:701-708.
120. Pareyson D, Scaiola V, Laurà M. Clinical and electrophysiological aspects of Charcot-Marie-Tooth disease. *Neuromolecular Med*. 2006;8:3-22.
121. Patel A, MacMahon S, Chalmers J, et al. Intensive blood glucose control and vascular outcomes in patients with type 2 diabetes. *N Engl J Med*. 2008;358:2560-2572.
122. Pickett JB. Localizing peroneal nerve lesions to the knee by motor conduction studies. *Arch Neurol*. 1998;41:192-195.
123. Plasma Exchange/Sandoglobulin Guillain-Barre Syndrome Trial Group. Randomised trial of plasma exchange, intravenous immunoglobulin, and combined treatments in Guillain-Barre syndrome. *Lancet*. 1997;349:225-230.
124. Postma TJ, Benard BA, Huijgens PC, et al. Long-term effects of vincristine on the peripheral nervous system. *J Neurooncol*. 1993;15(1):23-27.
125. Praet SFE, Jonkers RAM, Schep G, et al. Long-standing, insulin-treated type 2 diabetes patients with complications respond well to short-term resistance and interval exercise training. *Eur J Endocrinol*. 2008;158(2):163-172.
126. Pritchard J. What's new in Guillain-Barre syndrome? *Postgrad Med J*. 2008;84:532-538.
127. Quigley PA, Bulat T, Schulz B, et al. Exercise interventions, gait, and balance in older subjects with distal symmetric polyneuropathy: a three-group randomized trial. *Am J Phys Med Rehabil*. 2014;93:1-16.
128. Raja SN, Haythornthwaite A, Pappagallo M, et al. Opioids versus antidepressants in postherpetic neuralgia: a randomized, placebo-controlled trial. *Neurology*. 2002;59:1015-1021.
129. Ramdharry GM, Day BL, Reilly MM, Marsden JF. Hip flexor fatigue limits walking in Charcot Marie-Tooth disease. *Muscle Nerve*. 2009;40(1):103-111.
130. Rengachary SS, Burr D, Lucas S, Hassanein KM, Mohn MP, Matzke H. Suprascapular entrapment neuropathy: A clinical, anatomical, and comparative study: Part 2: Anatomical study. *Neurosurgery*. 1979;5(4):447-51.
131. Richardson JK. The clinical identification of peripheral neuropathy among older persons. *Arch Phys Med Rehabil*. 2002;83:205-209.
132. Richardson JK, Ching C, Hurvitz EA. The relationship between electromyographically documented peripheral neuropathy and falls. *J Am Geriatr Soc*. 1992;40:1008-1012.
133. Richardson JK, DeMott T, Allet L, et al. Hip strength: Ankle proprioceptive threshold ratio predicts falls and injury in diabetic neuropathy. *Muscle Nerve*. 2014;50(3):437-442.
134. Richardson JK, Green DF, Jamieson SC, et al. Gender, body mass index and age as risk factors for ulnar mononeuropathy at the elbow. *Muscle Nerve*. 2001;24:551-554.
135. Richardson JK, Ho S, Spiegelberg T, et al. The nature of the relationship between smoking and ulnar neuropathy at the elbow. *Am J Phys Med Rehabil*. 2009;88(9):711-718.
136. Richardson JK, Hurvitz EA. Peripheral neuropathy: a true risk factor for falls. *J Gerontol Ser A Biol Sci Med Sci*. 1995;50A:211-215.
137. Richardson JK, Sandman D, Vela S. A focused exercise regimen improves clinical measures of balance in patients with peripheral neuropathy. *Arch Phys Med Rehabil*. 2001;82:205-209.
138. Richardson JK, Thies SB, DeMott TK, et al. Interventions improve gait regularity in patients with peripheral neuropathy while walking on an irregular surface under low light. *J Am Geriatr Soc*. 2004;52:510-515.
139. Robinson LR, Binhammer P. Role of electrodiagnosis in nerve transfers for focal neuropathies and brachial plexopathies. *Muscle Nerve*. 2022;65(2):137-46.
140. Rodner CM, Tinsley BA, O'Malley MP. Pronator syndrome and interosseous nerve syndrome. *J Am Acad Orthop Surg*. 2013;21:268-275.
141. Said G. Infectious neuropathies. *Neurol Clin*. 2007;25:115-137.
142. Santoro L, Manganelli F, Briani C, et al. HCV Peripheral Nerve Study Group. Prevalence and characteristics of peripheral neuropathy in hepatitis C virus population. *J Neurol Neurosurg Psychiatry*. 2006;77(5):626-629.
143. Schneider DS, Reddy S. The lateral cutaneous nerve of the calf revisited. *PM&R*. 2010;2(6):579-80.
144. Schonberger LB, Bregman DJ, Sullivan-Bolyai JZ. Guillain-Barre syndrome following vaccination in the National Influenza Immunization Program, United States, 1976-7. *Am J Epidemiol*. 1979;110(2):105-123.
145. Scripture CD, Figg WD, Sparreboom A. Peripheral neuropathy induced by paclitaxel: recent insights and future perspectives. *Curr Neuropharmacol*. 2006;4(2):165-172.
146. Seddon HJ. Three types of nerve injury. *Brain*. 1943;66(4):237-288.

147. Seror P. Electrophysiological pattern of 53 cases of ulnar nerve lesion at the wrist. *Clin Neurophysiol.* 2013;43(2):95-103.
148. Seror P. Isolated sensory manifestations in neuralgic amyotrophy: report of eight cases. *Muscle Nerve.* 2004;29(1):134-138.
149. Seror P, Kuntz PP, Maisonnobe T, et al. Sensory nerve action potential abnormalities in neuralgic amyotrophy: a report of 18 cases. *J Clin Neuromuscul Dis.* 2002;4(2):45-49.
150. Seror P, Roren MA, Garofoli R, et al. Voluntary winging of the scapula: proposed diagnostic criteria. *Muscle Nerve.* 2020;62(5):611-616.
151. Shang P, Zhu M, Wang Y, et al. Axonal variants of Guillain-Barre syndrome: an update. *J Neurol.* 2021;268(7):2402-2419.
152. Shapiro BE, Preston DC. Entrapment and compressive neuropathies. *Med Clin N Am.* 2009;93:285-315.
153. Sillman JS, Niparko JK, Lee SS, et al. Prognostic value of evoked and standard electromyography in acute facial paralysis. *Otolaryngol Head Neck Surg.* 1992;107:377-381.
154. Simpson DM, Kitch D, Evans SR, et al. HIV neuropathy natural history cohort study: assessment measures and risk factors. *Neurology.* 2006;66:1679-1687.
155. Sindrup SH, Andersen G, Madsen C, et al. Tramadol relieves pain and allodynia in polyneuropathy: a randomized, double-blind, controlled trial. *Pain.* 1999;83:85-90.
156. Singleton J, Smith A. Neuropathy associated with prediabetes: what is new in 2007? *Curr Diab Rep.* 2007;7:420-424.
157. Son J, Ashton-Miller JA, Richardson JK. Frontal plane ankle proprioceptive thresholds and unipedal balance. *Muscle Nerve.* 2009;39(2):150-157.
158. Spinner RJ, Atkinson JL, Tiel RL. Peroneal intraneural ganglia: the importance of the articular branch. A unifying theory. *J Neurosurg.* 2003;99(2):330-343.
159. Stevens JC, Beard M, O'Fallon WM, et al. Conditions associated with carpal tunnel syndrome. *Mayo Clin Proc.* 1992;67(6):541-548.
160. Stiernstedt G, Gustafsson R, Karlsson M, et al. Clinical manifestations and diagnosis of neuroborreliosis. *Ann NY Acad Sci.* 1988;539:46-55.
161. Sunderland S. A classification of peripheral nerve injuries producing loss of function. *Brain.* 1951;74(4):491-516.
162. Sunderland S. *Nerves and Nerve Injuries.* 2nd ed. Churchill Livingstone; 1978:133-138.
163. Suranyi L. Median nerve compression by Struthers ligament. *J Neurol Neurosurg Psychiatry.* 1983;46(11):1047-1049.
164. Tai TW, Wu CY, Su FC, et al. Ultrasonography for diagnosing carpal tunnel syndrome: a meta-analysis of diagnostic test accuracy. *Ultrasound Med Biol.* 2012;38:1121-1128.
165. Takakura Y, Kumai T, Takoaka T, et al. Tarsal tunnel syndrome caused by coalition associated with a ganglion. *J Bone Joint Surg Br.* 1998;80:130-133.
166. Tesfaye S, Chaturvedi N, Eaton SEM, et al. Vascular risk factors and diabetic neuropathy. *N Engl J Med.* 2005;352:341-350.
167. Thaisethawatkul P, Logigia EL. Peripheral nervous system manifestations of Lyme borreliosis. *J Clin Neuromuscul Dis.* 2002;3:165-171.
168. Thomassen I, van Suijlekom JA, van de Gaag A, et al. Ultrasound-guided ilioinguinal/iliohypogastric nerve blocks for chronic pain after inguinal hernia repair. *Hernia.* 2009;17:329-332.
169. Thomson RM, Parry GJ. Neuropathies associated with excessive exposure to lead. *Muscle Nerve.* 2006;33(6):732-741.
170. Trojaborg W. Rate of recovery in motor and sensory fibres of the radial nerve: clinical and electrophysiological aspects. *J Neurol Neurosurg Psychiatry.* 1970;33:625-638.
171. Tsairis P, Dyck PJ, Mulder DW. Natural history of brachial plexus neuropathy: report on 99 patients. *Arch Neurol.* 1972;27:109-117.
172. Tung TH, Liu DZ, Mackinnon SE. Nerve transfer for elbow flexion in radiation-induced brachial plexopathy: a case report. *Hand (NY).* 2009;4(2):123-128.
173. Tzika M, Paraskevas GK, Kitsoulis P. The accessory deep peroneal nerve: a review of the literature. *Foot.* 2012;22(3):232-234.
174. Urguden M, Bilbasar H, Ozdemir H, et al. Tarsal tunnel syndrome—the effect of the associated features on outcome of surgery. *Int Orthop.* 2002;26(4):253-256.
175. van Alfen N, van Eijk JJ, Ennik T, et al. Incidence of neuralgic amyotrophy (Parsonage-Turner syndrome) in a primary care setting—a prospective cohort study. *PLoS One.* 2015;10(5):e0128361.
176. van Alfen N, van Engelen BG. The clinical spectrum of neuralgic amyotrophy in 246 cases. *Brain.* 2006;129(Pt 2):438-450.
177. Van den Bergh PYK, van Doorn PA, Hadden RDM, et al. European Academy of Neurology/Peripheral Nerve Society guideline on diagnosis and treatment of chronic inflammatory demyelinating polyradiculoneuropathy: report of a joint task force—second revision. *J Peripher Nerv Syst.* 2021;26(3):242-268.
178. Van Eijk JJ, Groothuis JT, Van Alfen N. Neuralgic amyotrophy: an update on diagnosis, pathophysiology, and treatment. *Muscle Nerve.* 2016;53(3):337-350.
179. van Eijk JJ, van Alfen N, Berrevoets M, et al. Evaluation of prednisolone treatment in the acute phase of neuralgic amyotrophy: an observational study. *J Neurol Neurosurg Psychiatry.* 2009;80(10):1120-1124.
180. Venkatesh S, Kothari MJ, Preston DC. The limitations of the dorsal ulnar cutaneous sensory response in patients with ulnar neuropathy at the elbow. *Muscle Nerve.* 1995;18(3):345-347.
181. Verboon C, van Doorn PA, Jacobs BC. Treatment dilemmas in Guillain-Barré syndrome. *J Neurol Neurosurg Psychiatry.* 2017;88:346-352.
182. Vileikyte L, Peyrot M, Gonzalez JS, et al. Predictors of depressive symptoms in person with diabetic peripheral neuropathy: a longitudinal study. *Diabetologia.* 2009;52:1265-1273.
183. Warner MA, Warner DO, Harper CM, et al. Lower extremity neuropathies associated with lithotomy positions. *Anesthesiology.* 2000;93(4):938-942.
184. Warner WC. Neuromuscular disorders. In: Canale ST, Beaty JH, eds. *Campbell's Operative Orthopaedics.* 11th ed. Mosby; 2009.
185. Watson CPN, Vernich L, Chipman M, et al. Nortriptyline versus amitriptyline in postherpetic neuralgia: a randomized trial. *Neurology.* 1998;51:1166-1171.
186. Wedel DJ, Horlocker TT. Nerve blocks. In: Miller RD, ed. *Miller's Anesthesia.* 7th ed. Elsevier; 2010.
187. Weickhardt A, Wells K, Messersmith W. Oxaliplatin-induced neuropathy in colorectal cancer. *J Oncol.* 2011;2011:201593.
188. Werner CO. Paralysis of the posterior interosseous nerve caused by tumour: brief report. *J Bone Joint Surg Br.* 1987;69:670-671.
189. Wilbourn AJ. Thoracic outlet syndrome is over diagnosed. *Muscle Nerve.* 1999;22:130-136.
190. Wilmhurst JM. Diagnosis and management of pediatric peripheral neuropathies in resource-poor settings. *Future Neurol.* 2013;8(2):133-148.
191. Winer JB, Hughes RA. Identification of patients at risk of arrhythmia in the Guillain-Barré syndrome. *Q J Med.* 1988;68:735-739.
192. World Health Organization. Global leprosy situation. *Wkly Epidemiol Rec.* 2010;85:337-348.
193. Yoon MS, Obermann M, Dockweiler C, et al. Sensory neuropathy in patients with cryoglobulin negative hepatitis-C infection. *J Neurol.* 2011;258:80-88.